

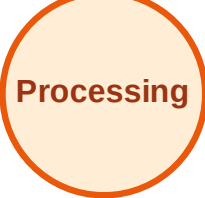
BFS Traversal

Step 1: Start at E

Legend



Complete



Processing

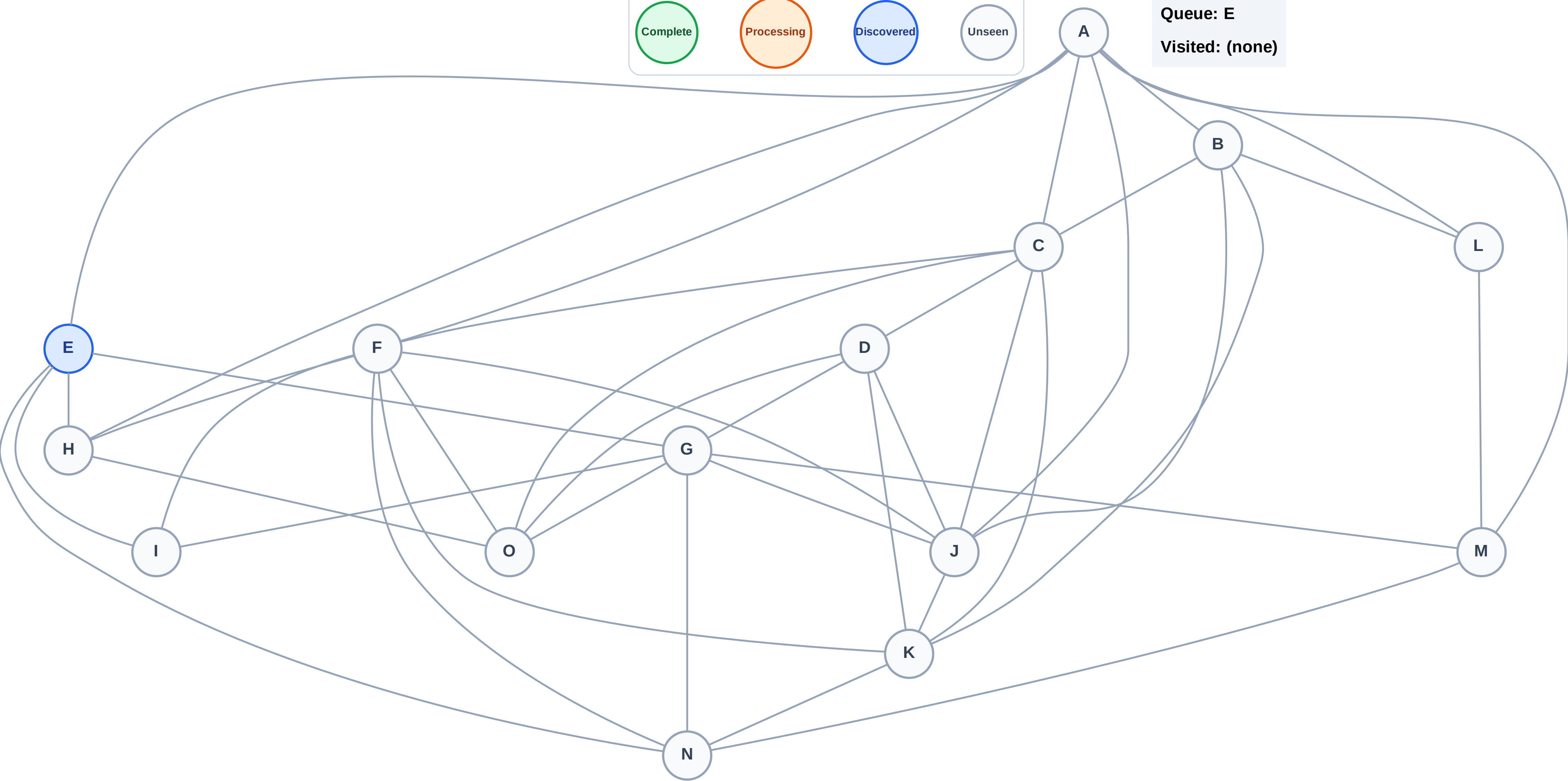


Discovered



Unseen

Queue: E
Visited: (none)



BFS Traversal
Step 2: Process node E

Legend

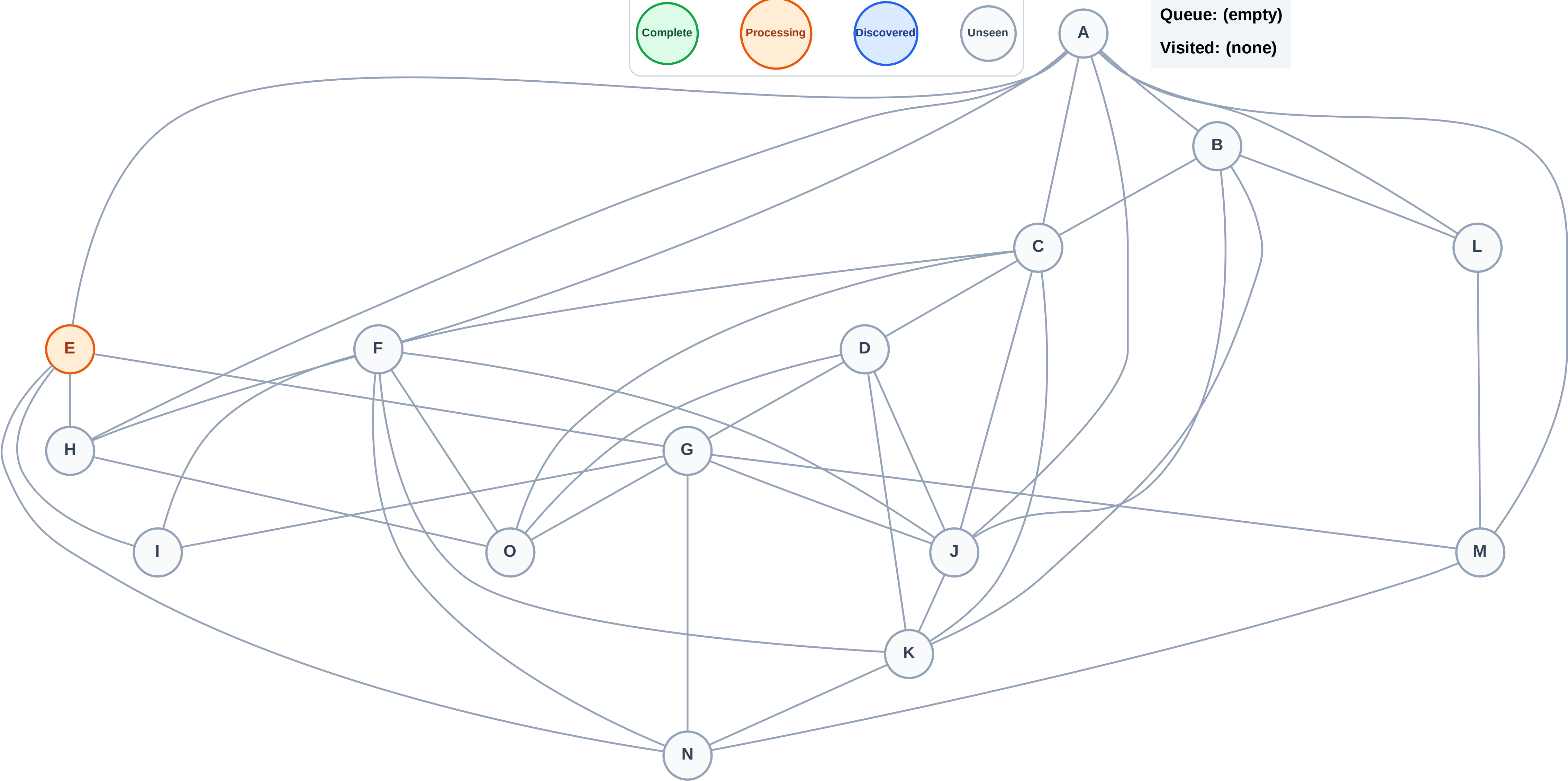
Complete

Processing

Discovered

Unseen

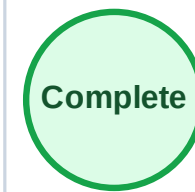
Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover A, G, H, I, N from E

Legend



Complete



Processing

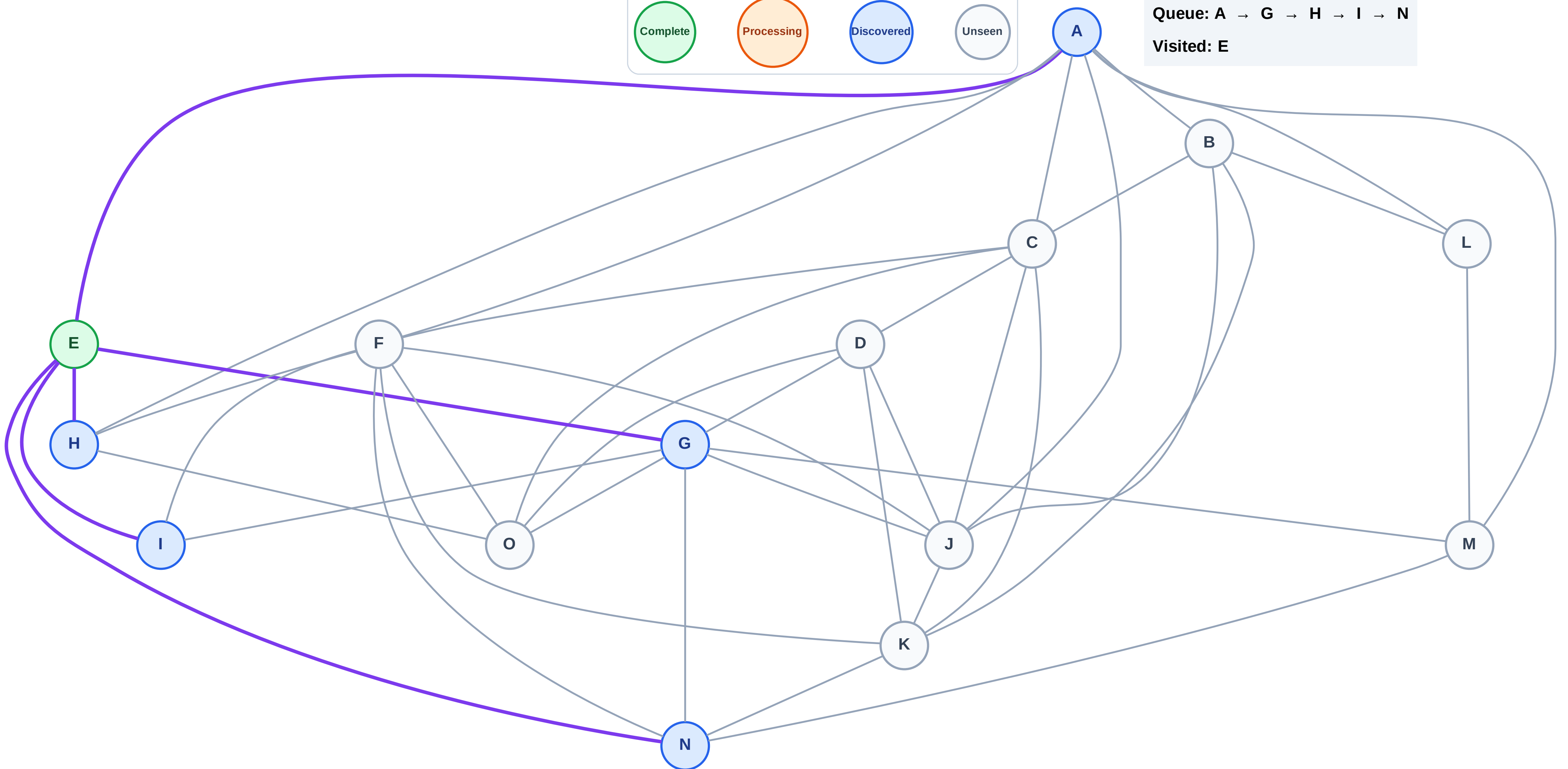


Discovered



Unseen

Queue: A → G → H → I → N
Visited: E



BFS Traversal

Step 4: Process node A

Legend

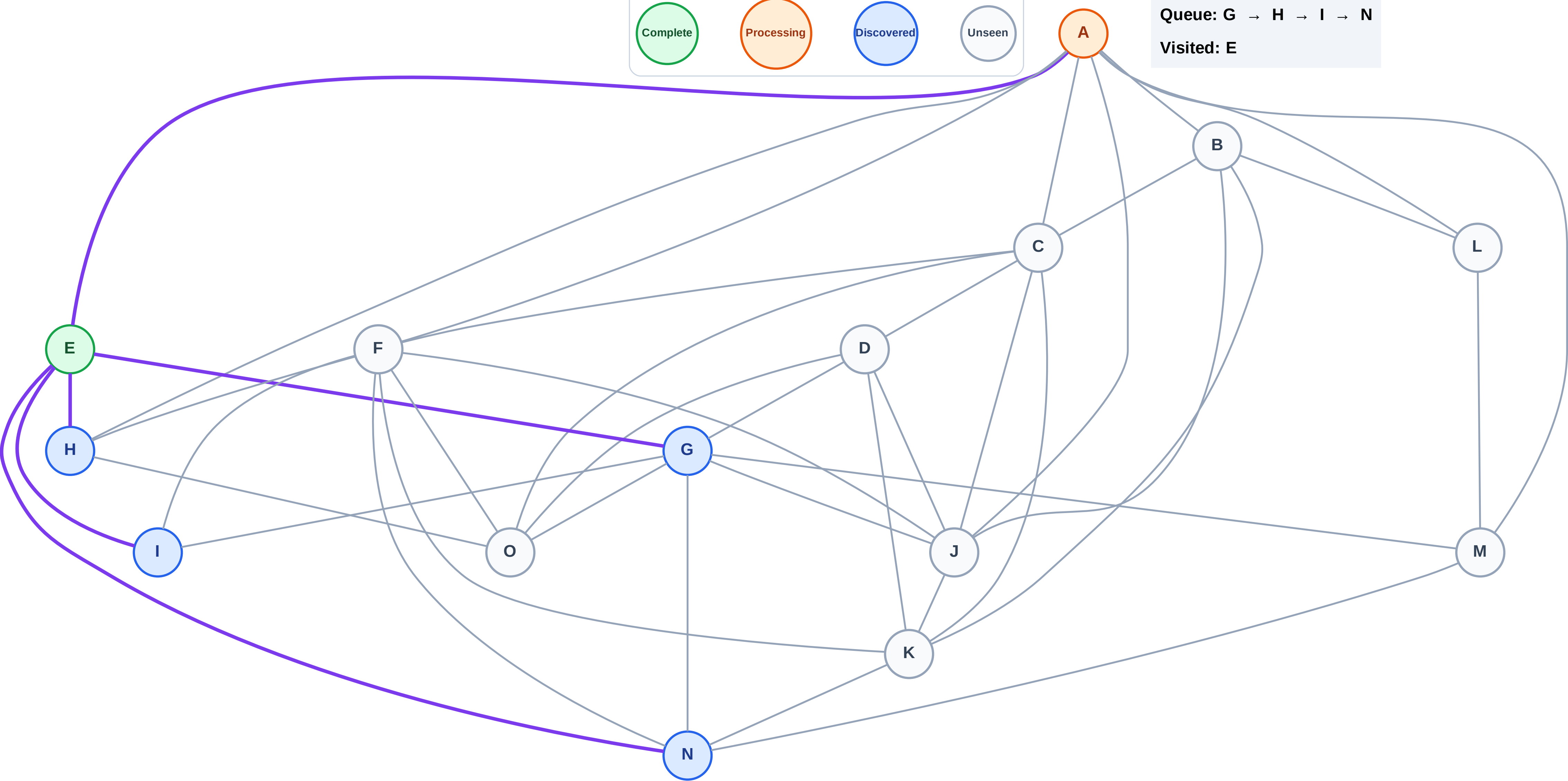
Complete

Processing

Discovered

Unseen

Queue: G → H → I → N
Visited: E



BFS Traversal

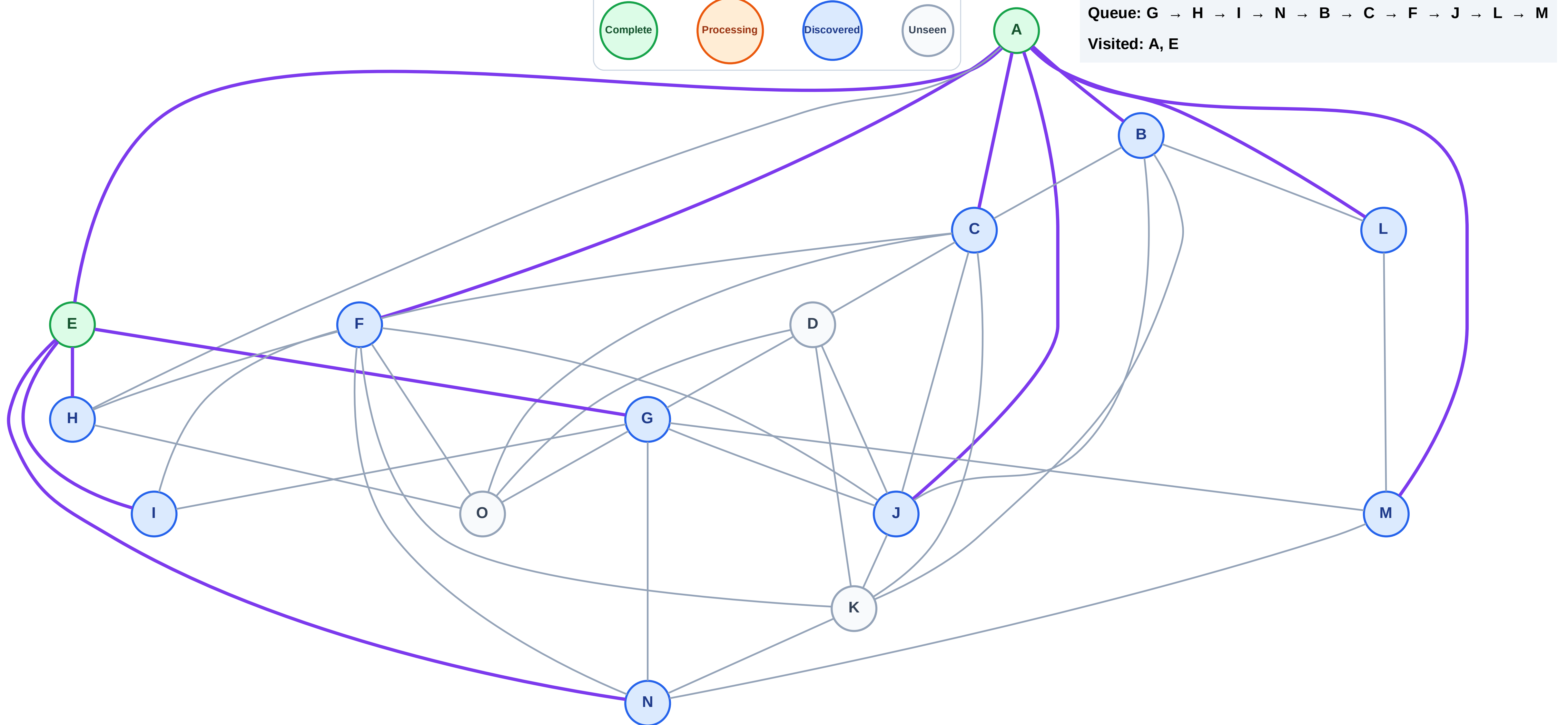
Step 5: Discover B, C, F, J, L, M from A

Legend



Queue: G → H → I → N → B → C → F → J → L → M

Visited: A, E



BFS Traversal
Step 6: Process node G

Legend



Complete



Processing



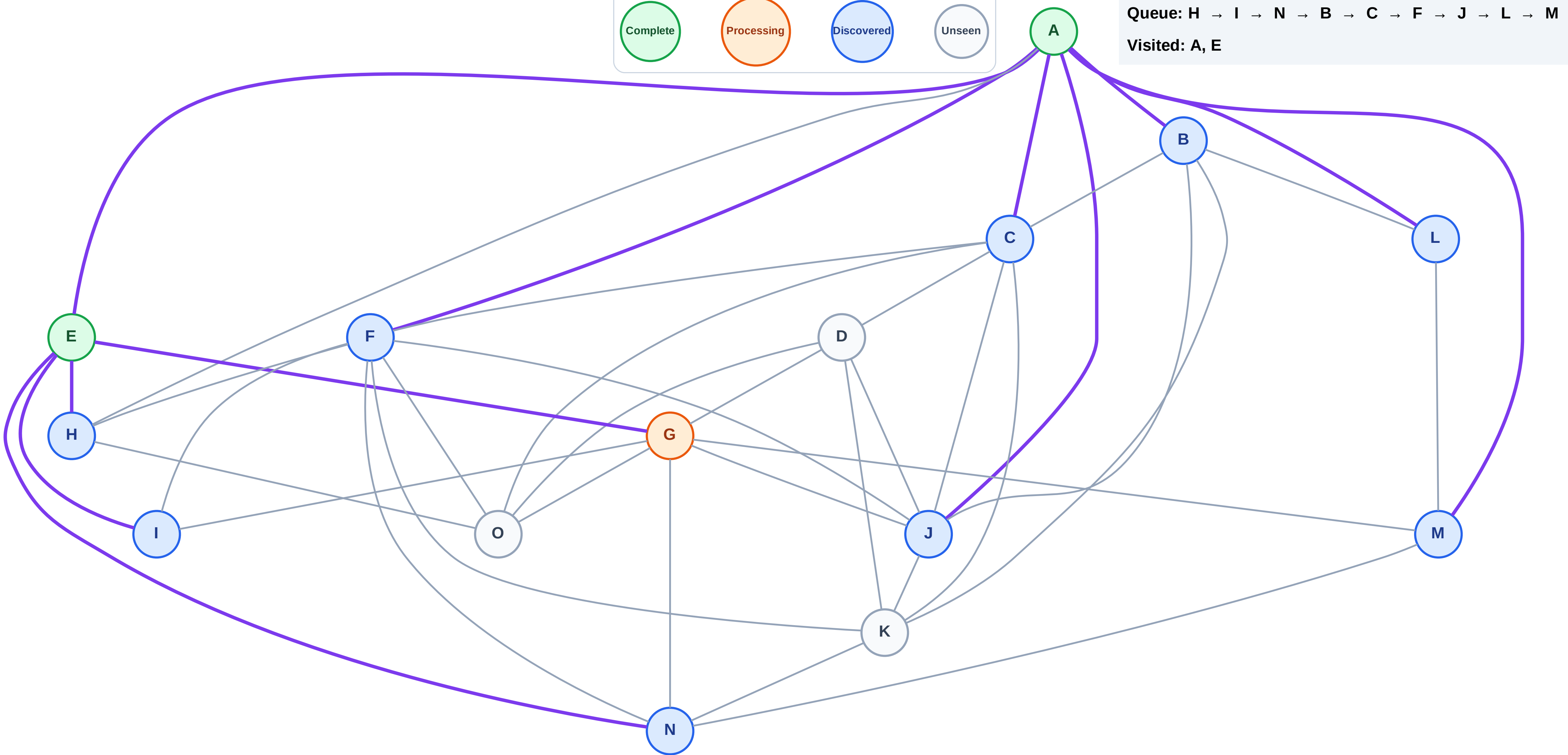
Discovered



Unseen

Queue: H → I → N → B → C → F → J → L → M

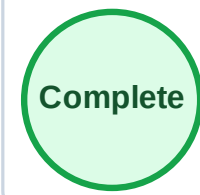
Visited: A, E



BFS Traversal

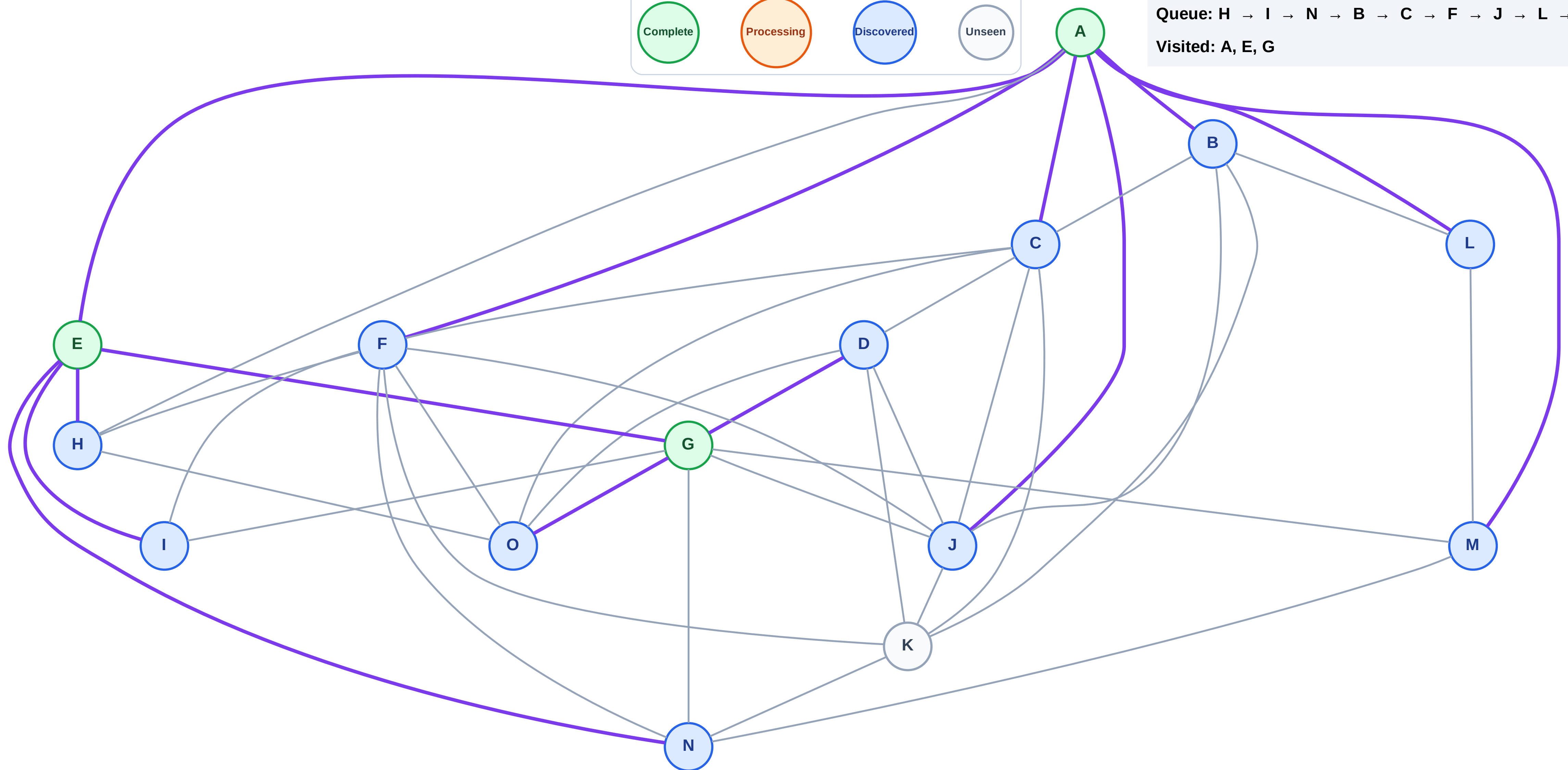
Step 7: Discover D, O from G

Legend



Queue: H → I → N → B → C → F → J → L → M → D → O

Visited: A, E, G



Step 8: Process node H

Step 8: Process node H

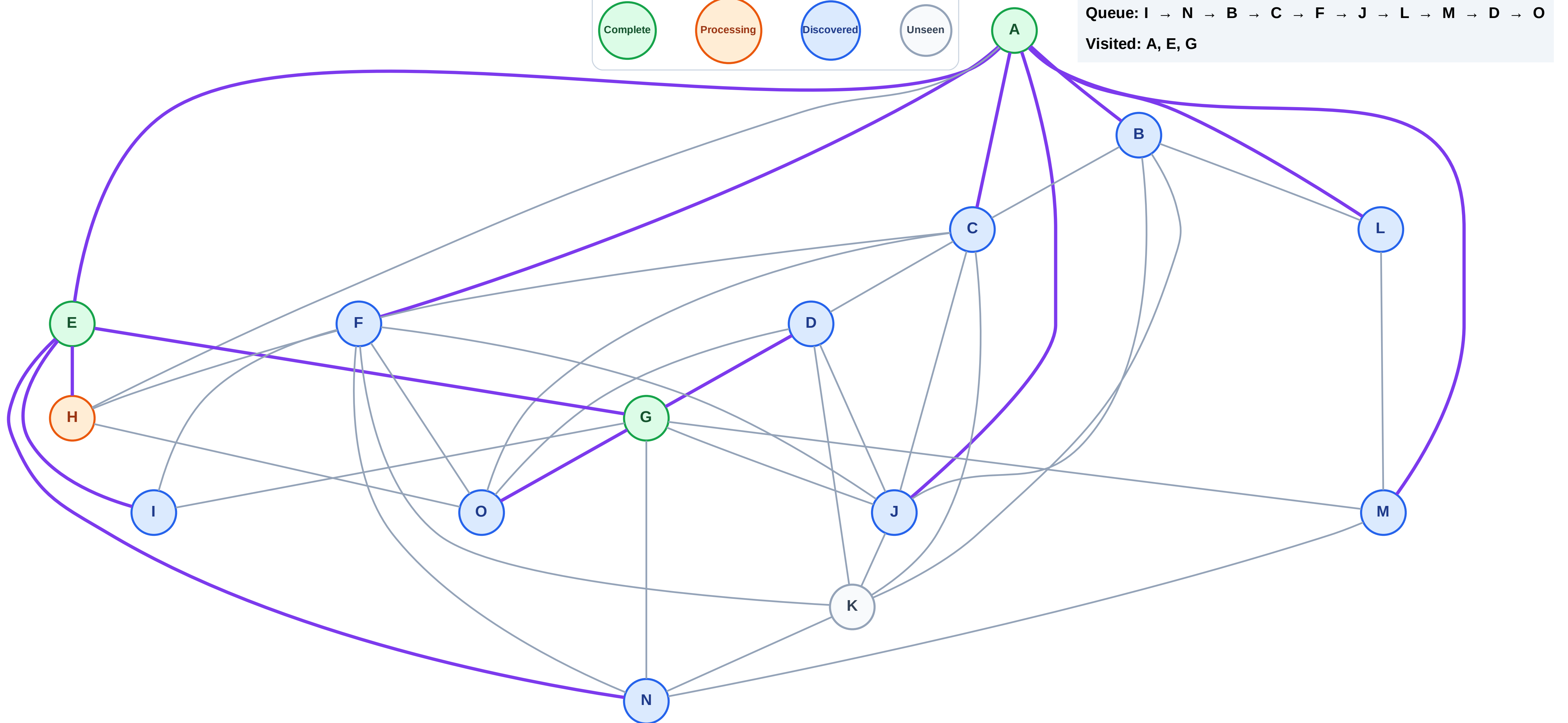
Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Unseen

Visited: A, E, G



BFS Traversal

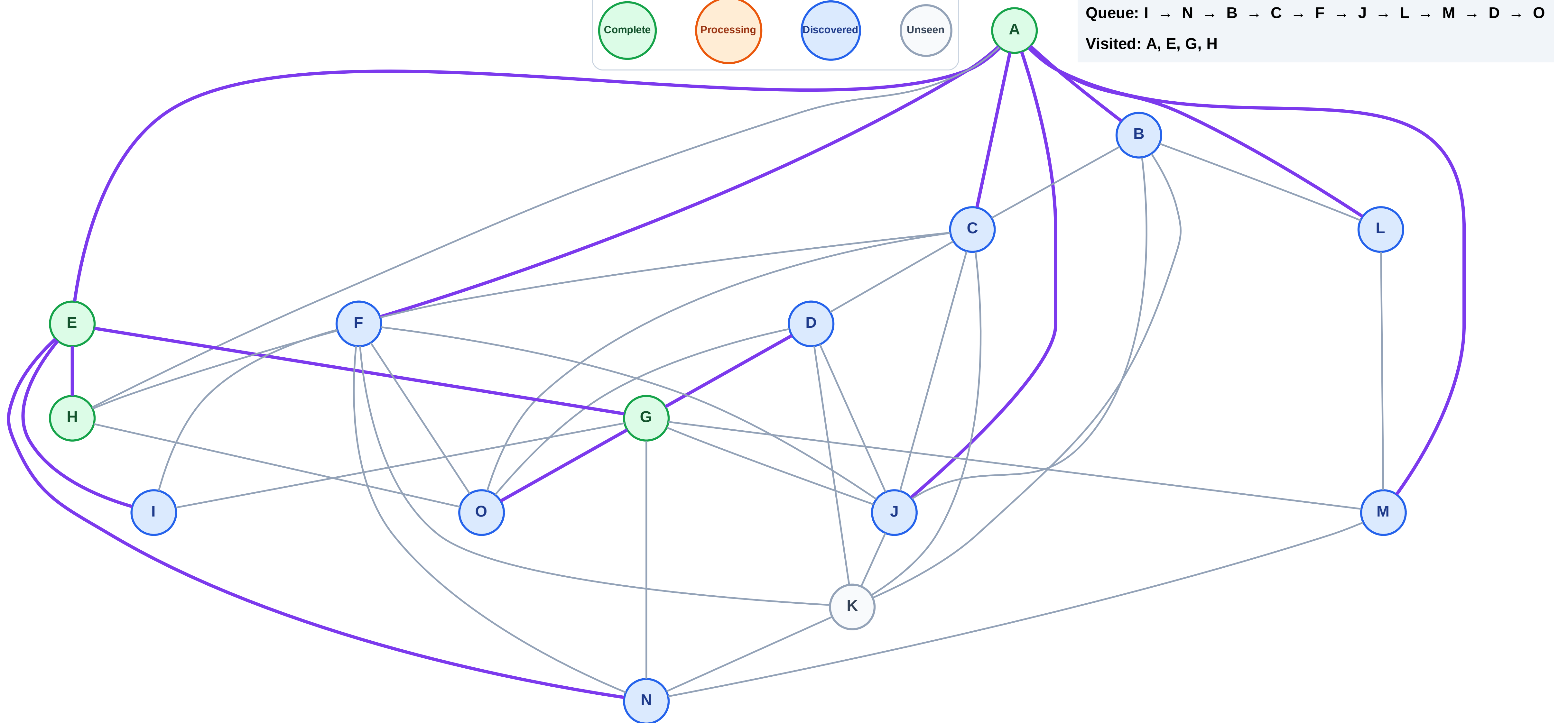
Step 9: No new neighbors from H

Legend



Queue: I → N → B → C → F → J → L → M → D → O

Visited: A, E, G, H



BFS Traversal
Step 10: Process node I

Legend

Complete

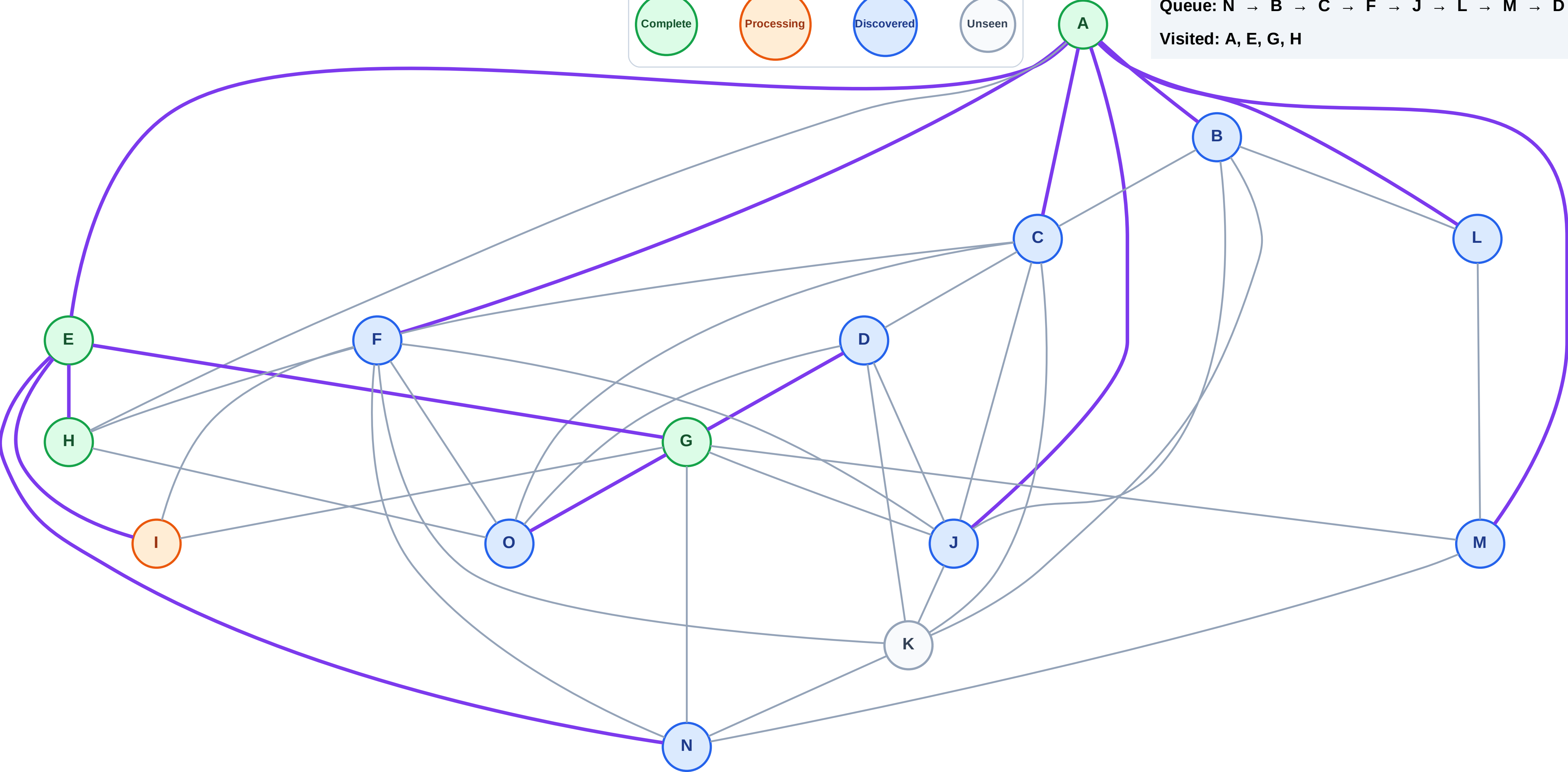
Processing

Discovered

Unseen

Queue: N → B → C → F → J → L → M → D → O

Visited: A, E, G, H



Step 11: No new neighbors from I

Step 11: No new neighbors from I



BFS Traversal

Step 12: Process node N

Legend



Complete



Processing

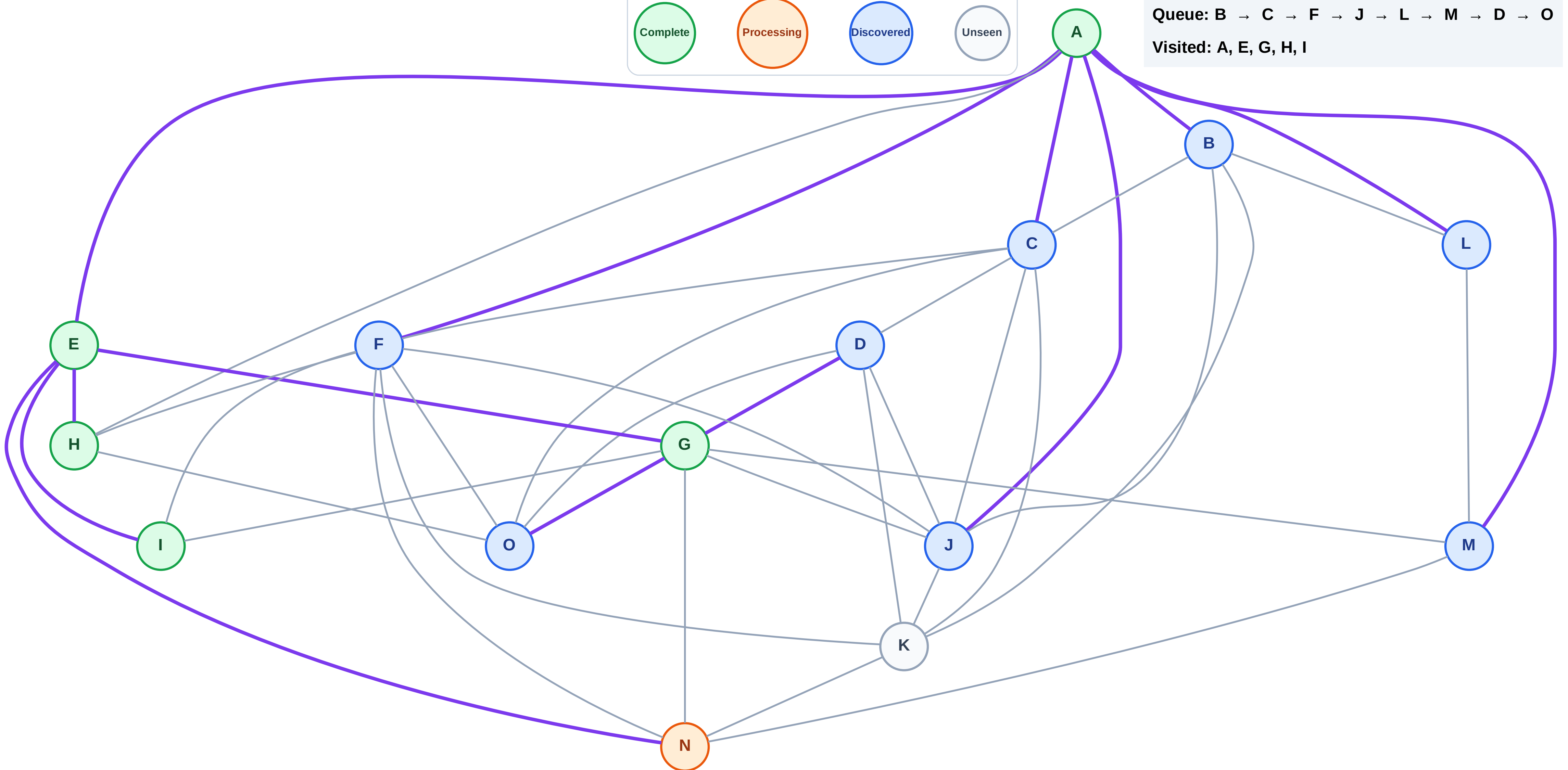


Discovered



Unseen

Queue: B → C → F → J → L → M → D → O
Visited: A, E, G, H, I



BFS Traversal

Step 13: Discover K from N

Legend



Complete



Processing



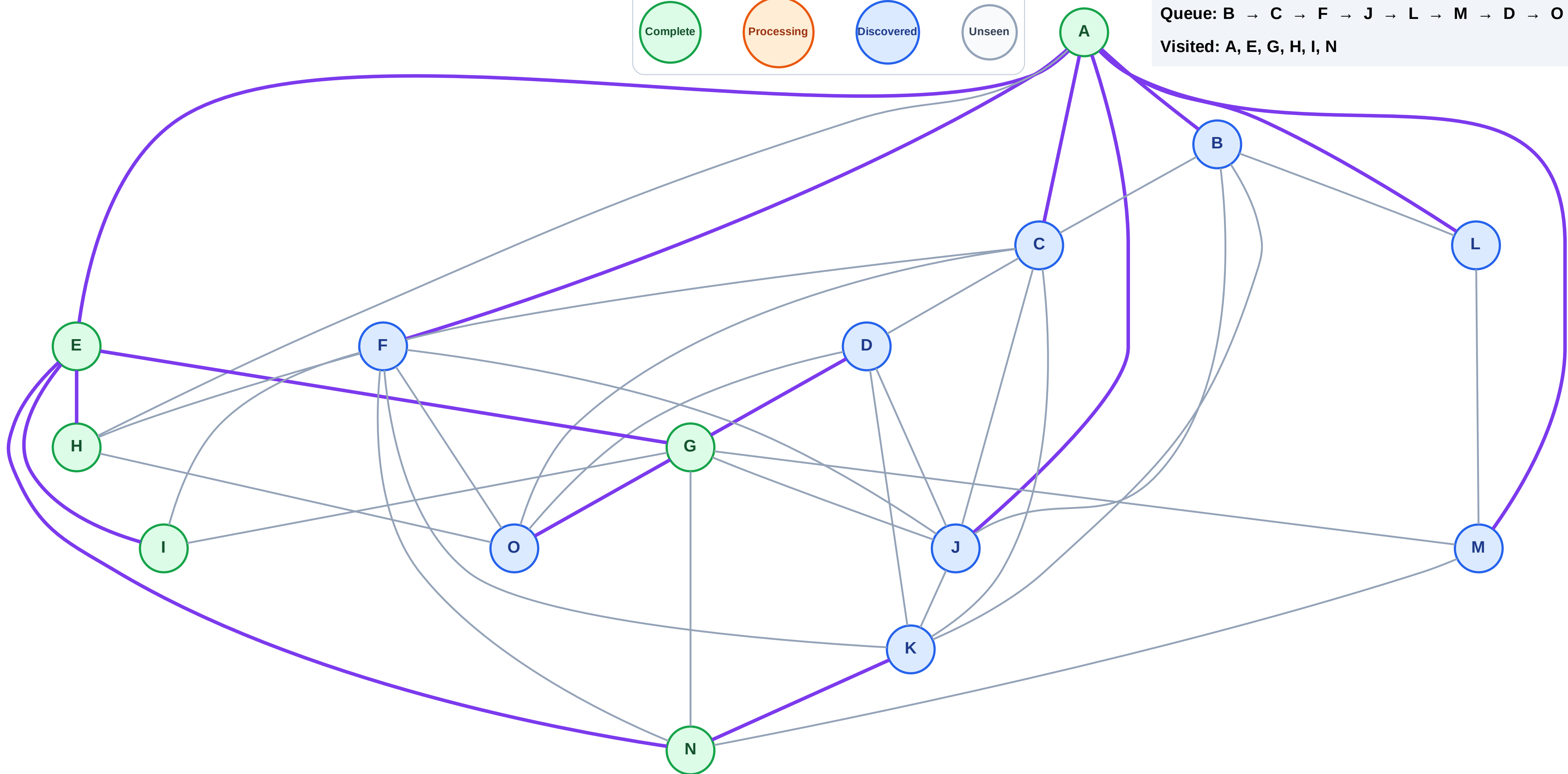
Discovered



Unseen

Queue: B → C → F → J → L → M → D → O → K

Visited: A, E, G, H, I, N



BFS Traversal

Step 14: Process node B

Legend



Complete



Processing



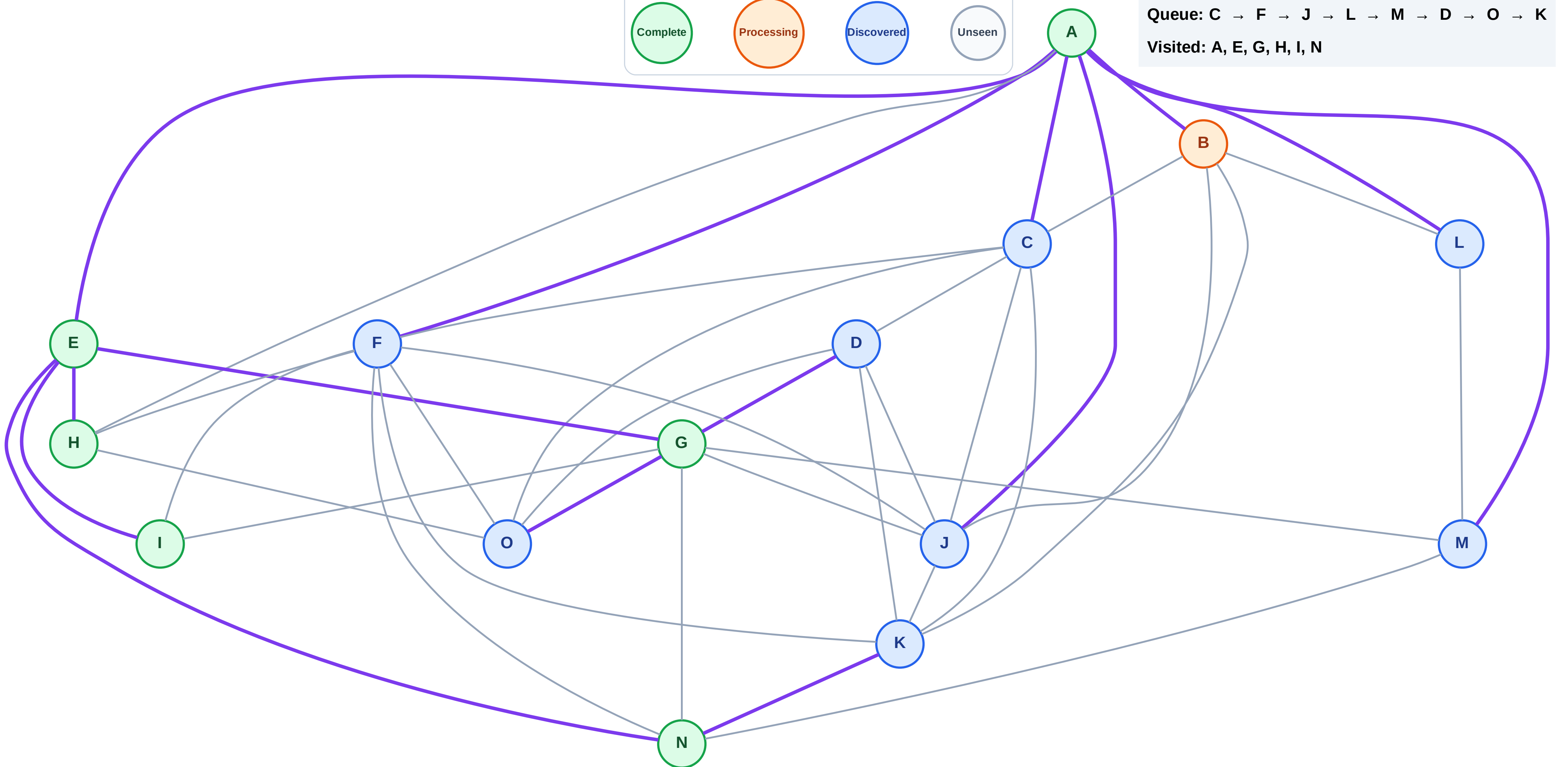
Discovered



Unseen

Queue: C → F → J → L → M → D → O → K

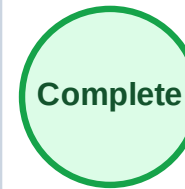
Visited: A, E, G, H, I, N



BFS Traversal

Step 15: No new neighbors from B

Legend



Complete



Processing

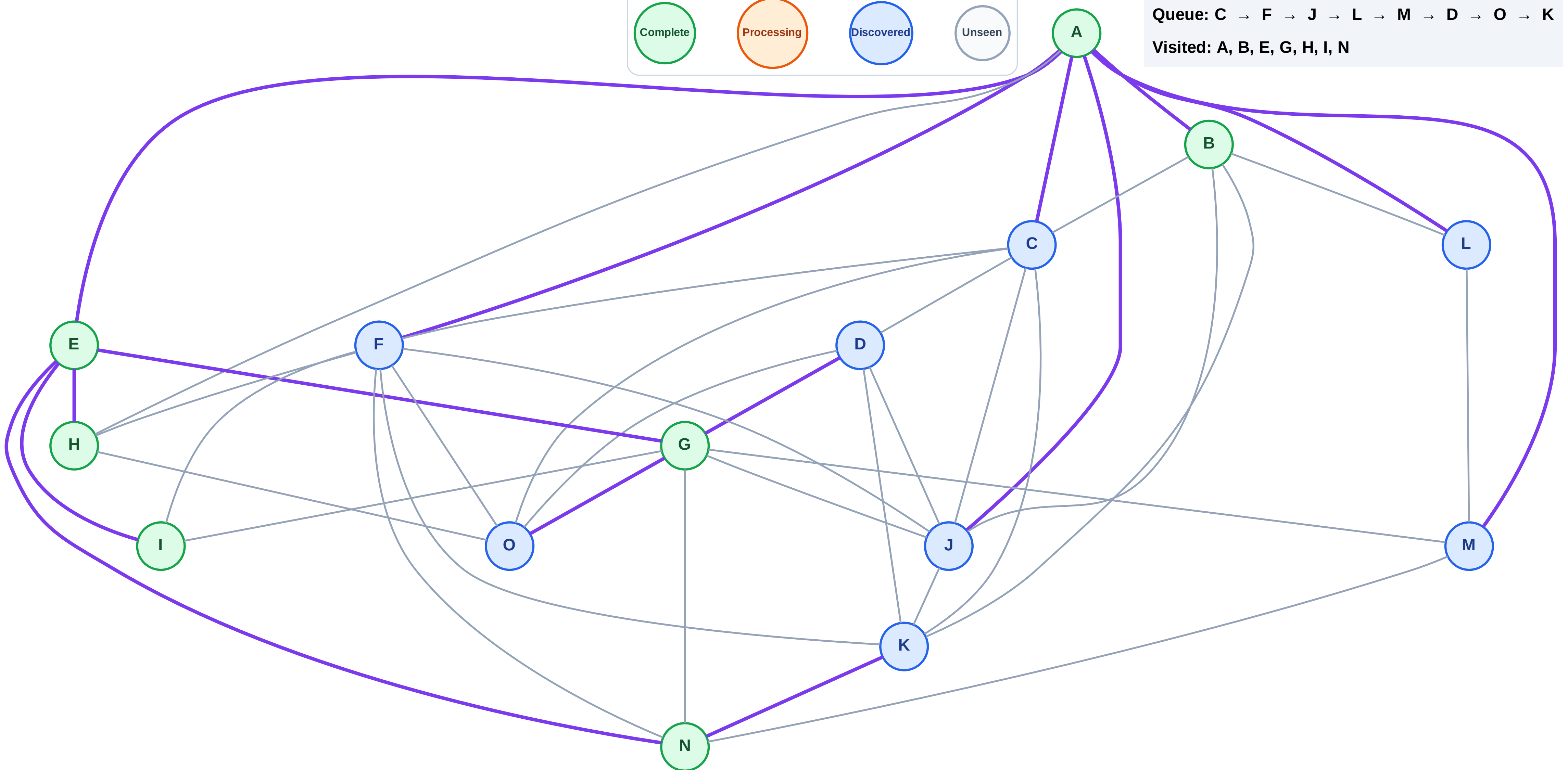


Discovered



Unseen

Queue: C → F → J → L → M → D → O → K
Visited: A, B, E, G, H, I, N



BFS Traversal

Step 16: Process node C

Legend



Complete



Processing

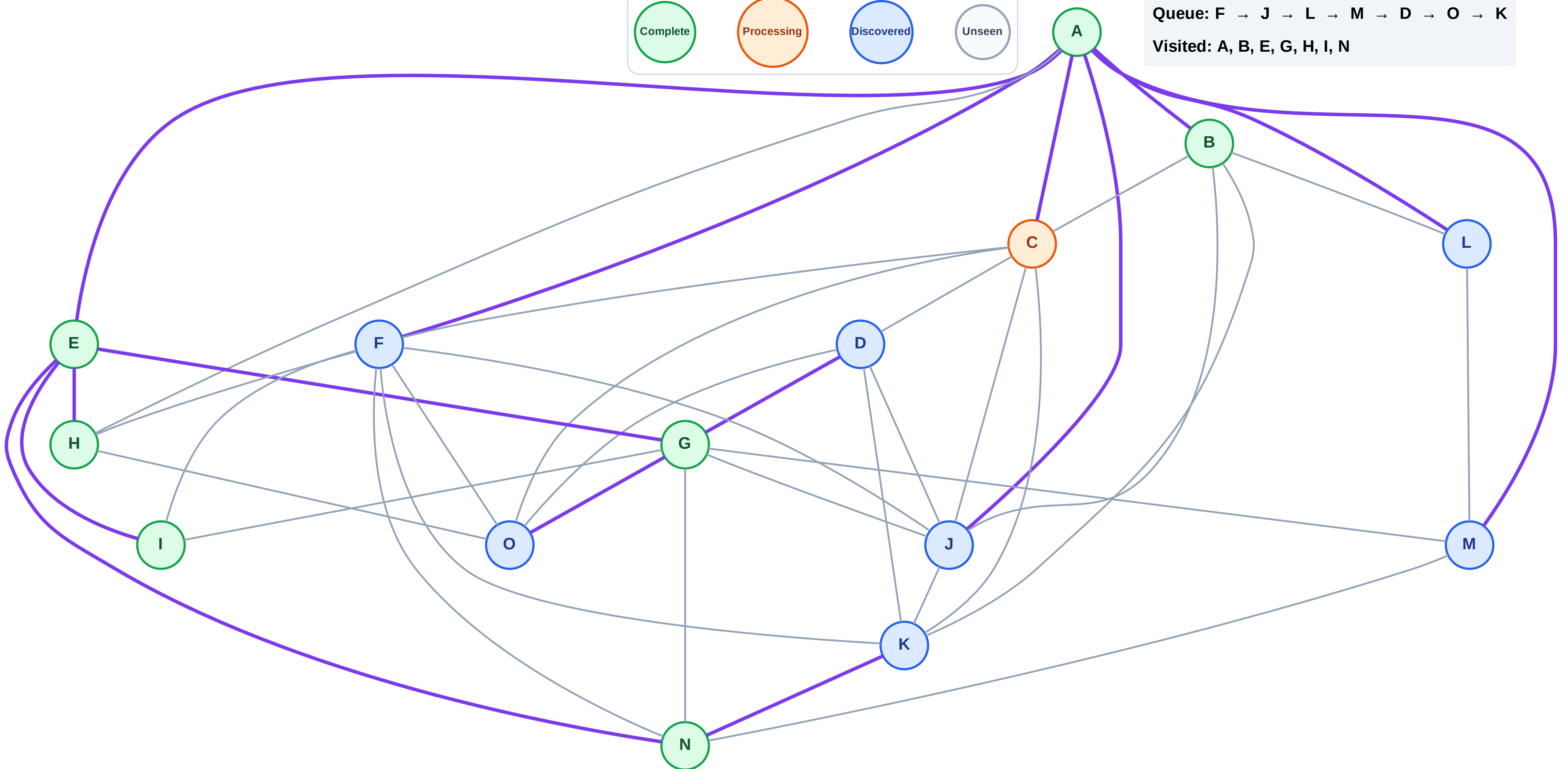


Discovered



Unseen

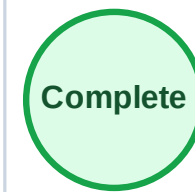
Queue: F → J → L → M → D → O → K
Visited: A, B, E, G, H, I, N



BFS Traversal

Step 17: No new neighbors from C

Legend



Complete



Processing

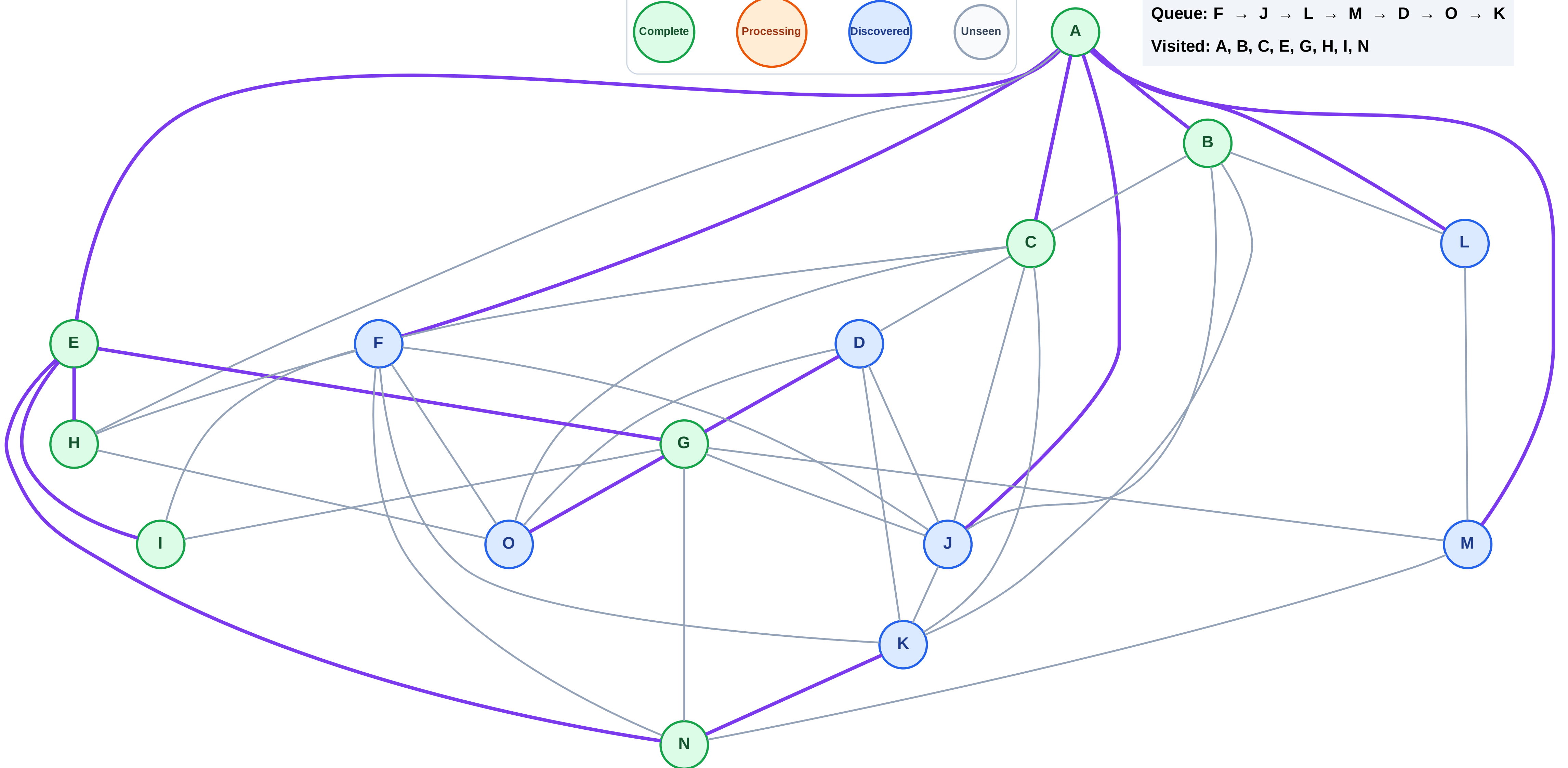


Discovered



Unseen

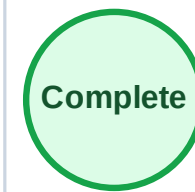
Queue: F → J → L → M → D → O → K
Visited: A, B, C, E, G, H, I, N



BFS Traversal

Step 18: Process node F

Legend



Complete



Processing



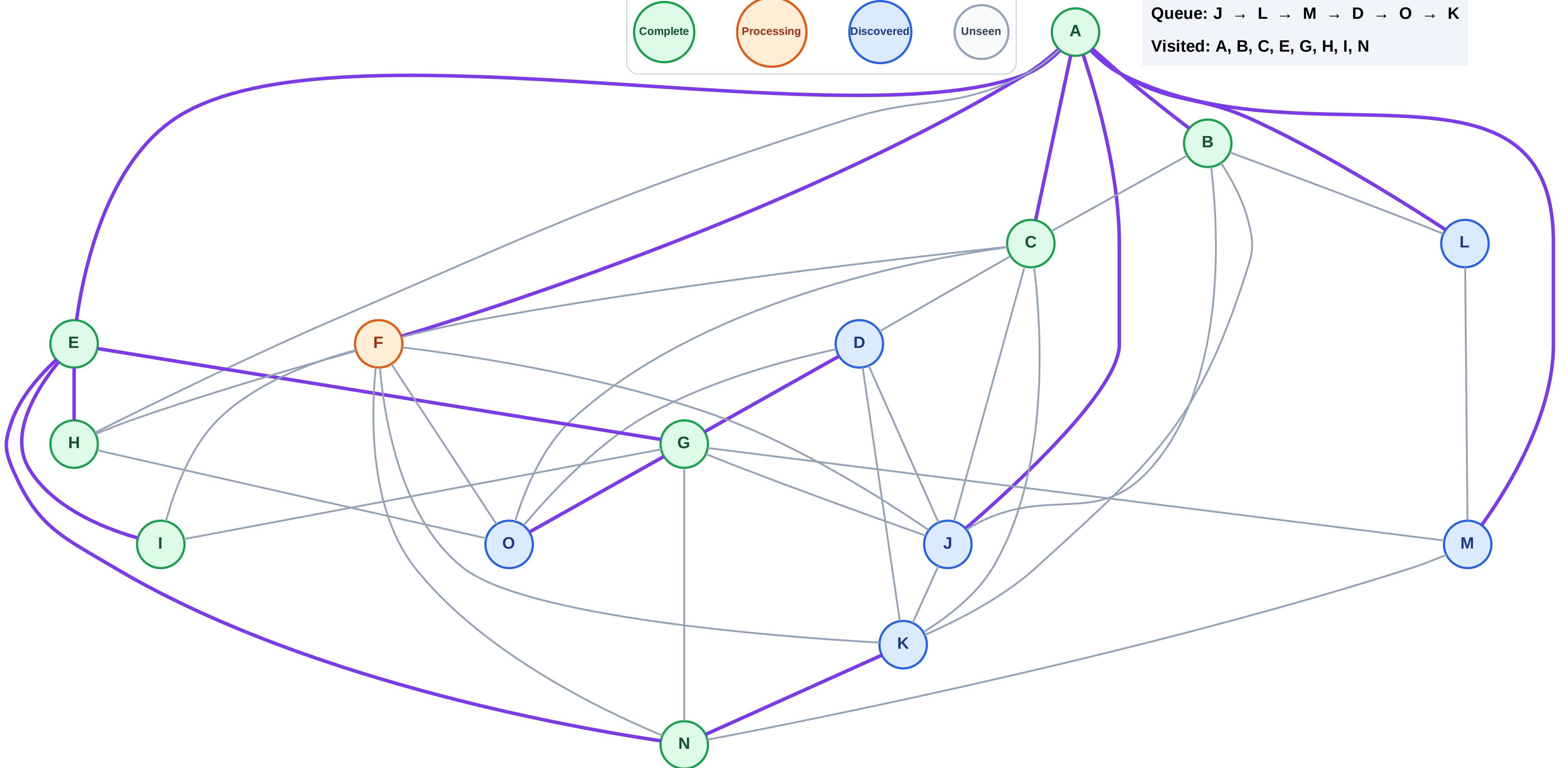
Discovered



Unseen

Queue: J → L → M → D → O → K

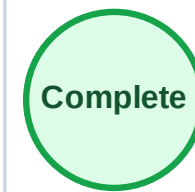
Visited: A, B, C, E, G, H, I, N



BFS Traversal

Step 19: No new neighbors from F

Legend



Complete



Processing



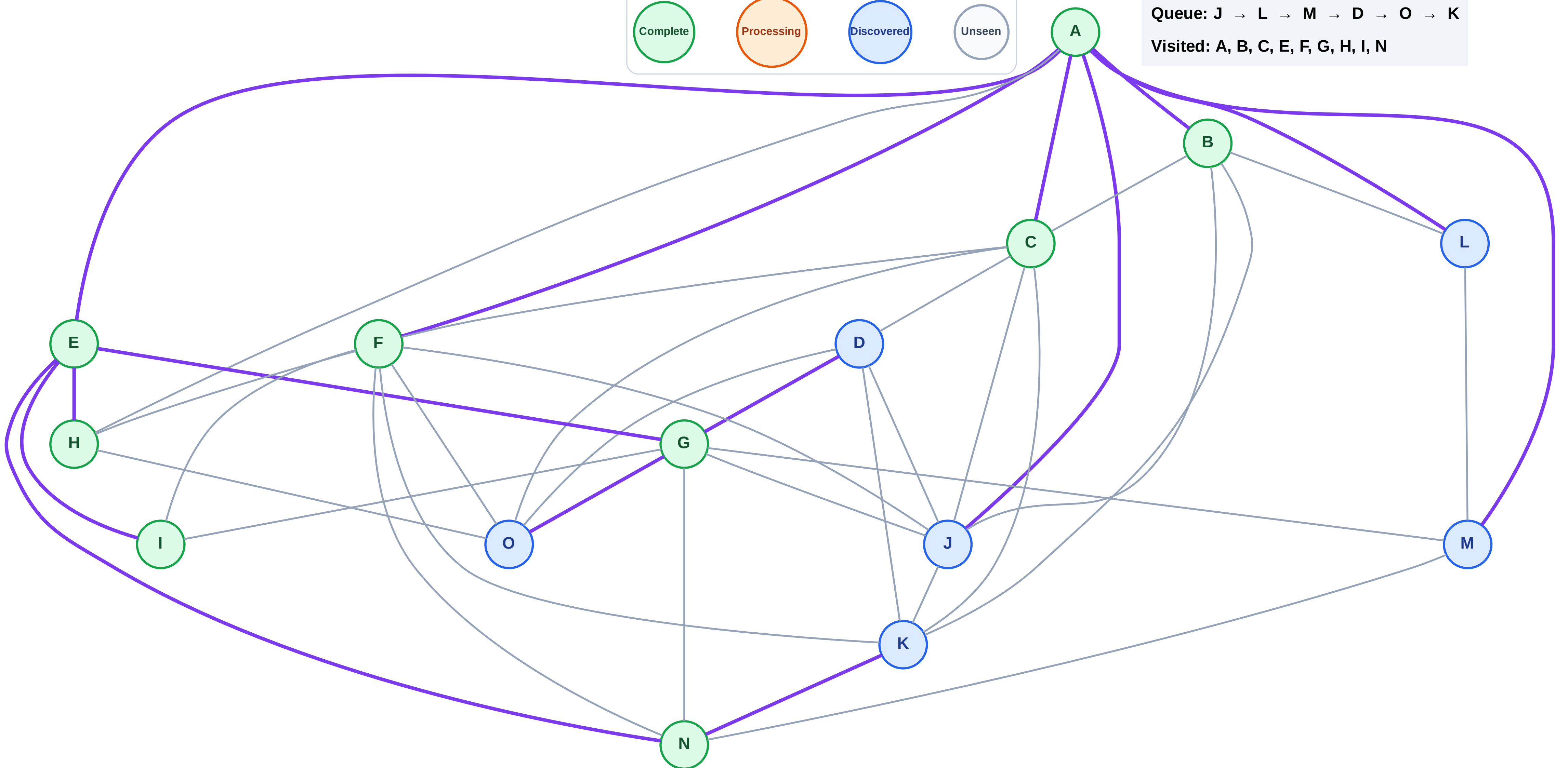
Discovered



Unseen

Queue: J → L → M → D → O → K

Visited: A, B, C, E, F, G, H, I, N



BFS Traversal

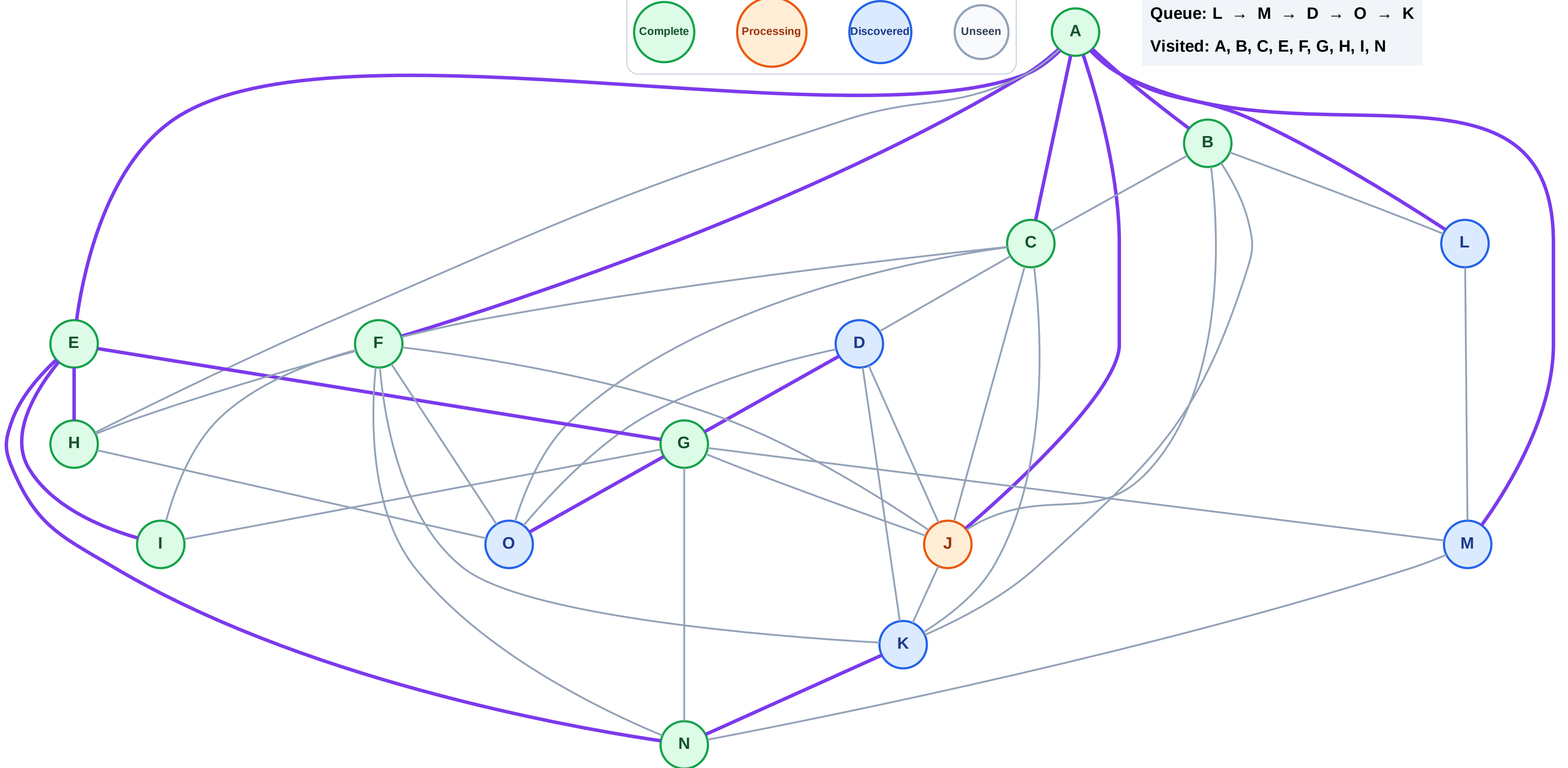
Step 20: Process node J

Legend



Queue: L → M → D → O → K

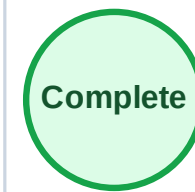
Visited: A, B, C, E, F, G, H, I, N



BFS Traversal

Step 21: No new neighbors from J

Legend



Complete



Processing

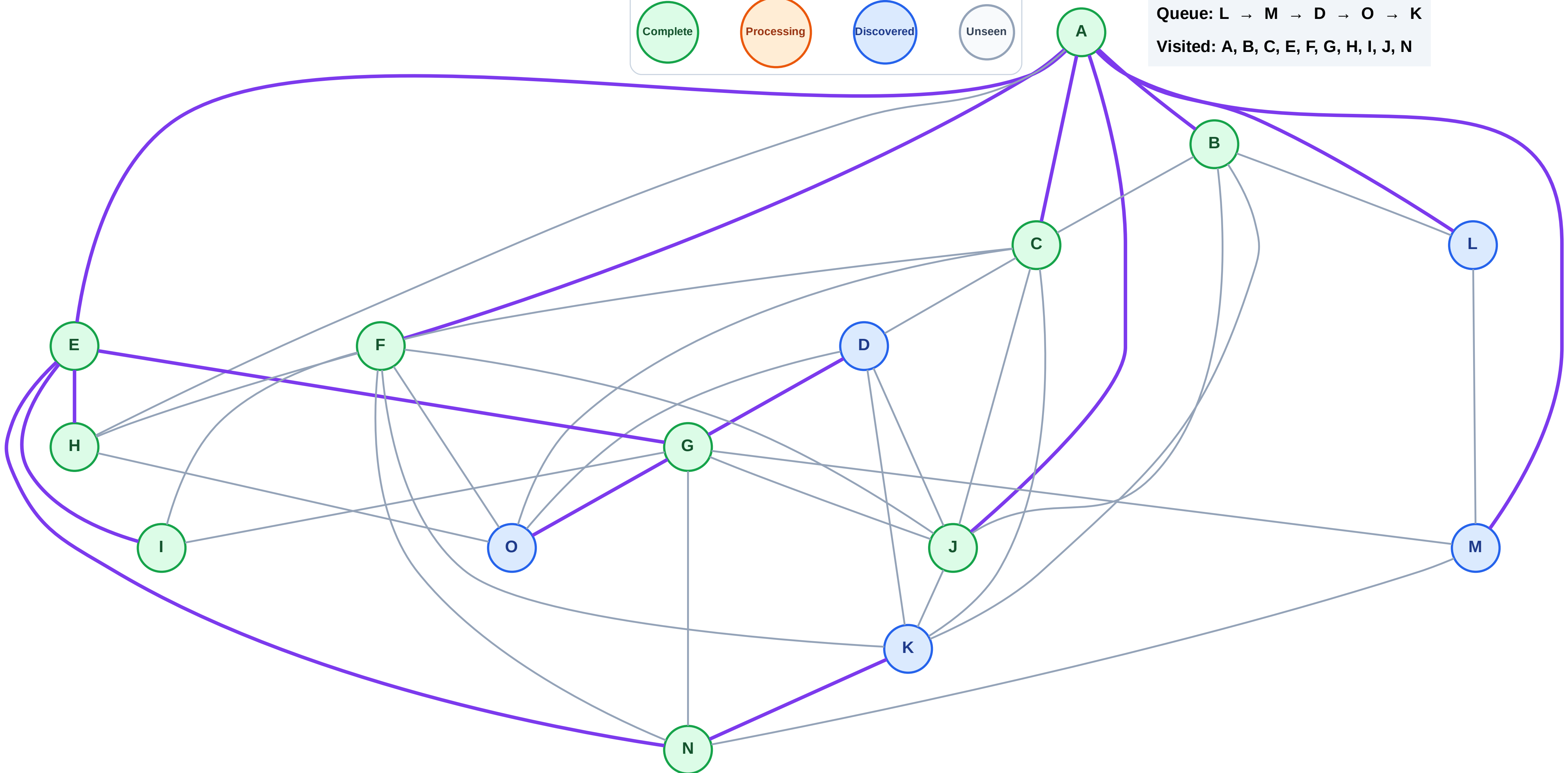


Discovered



Unseen

Queue: L → M → D → O → K
Visited: A, B, C, E, F, G, H, I, J, N



BFS Traversal

Step 22: Process node L

Legend

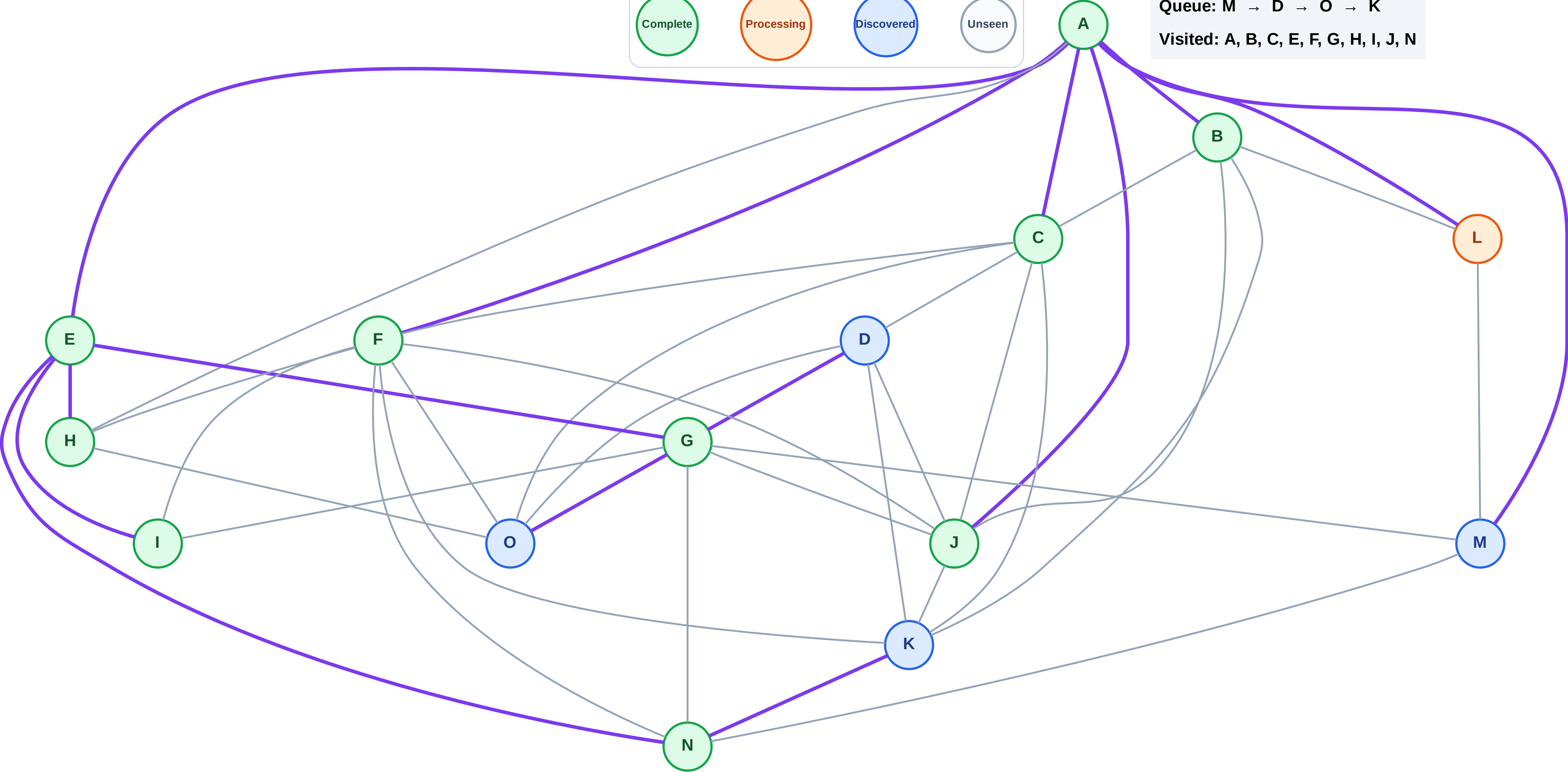
Complete

Processing

Discovered

Unseen

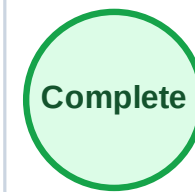
Queue: M → D → O → K
Visited: A, B, C, E, F, G, H, I, J, N



BFS Traversal

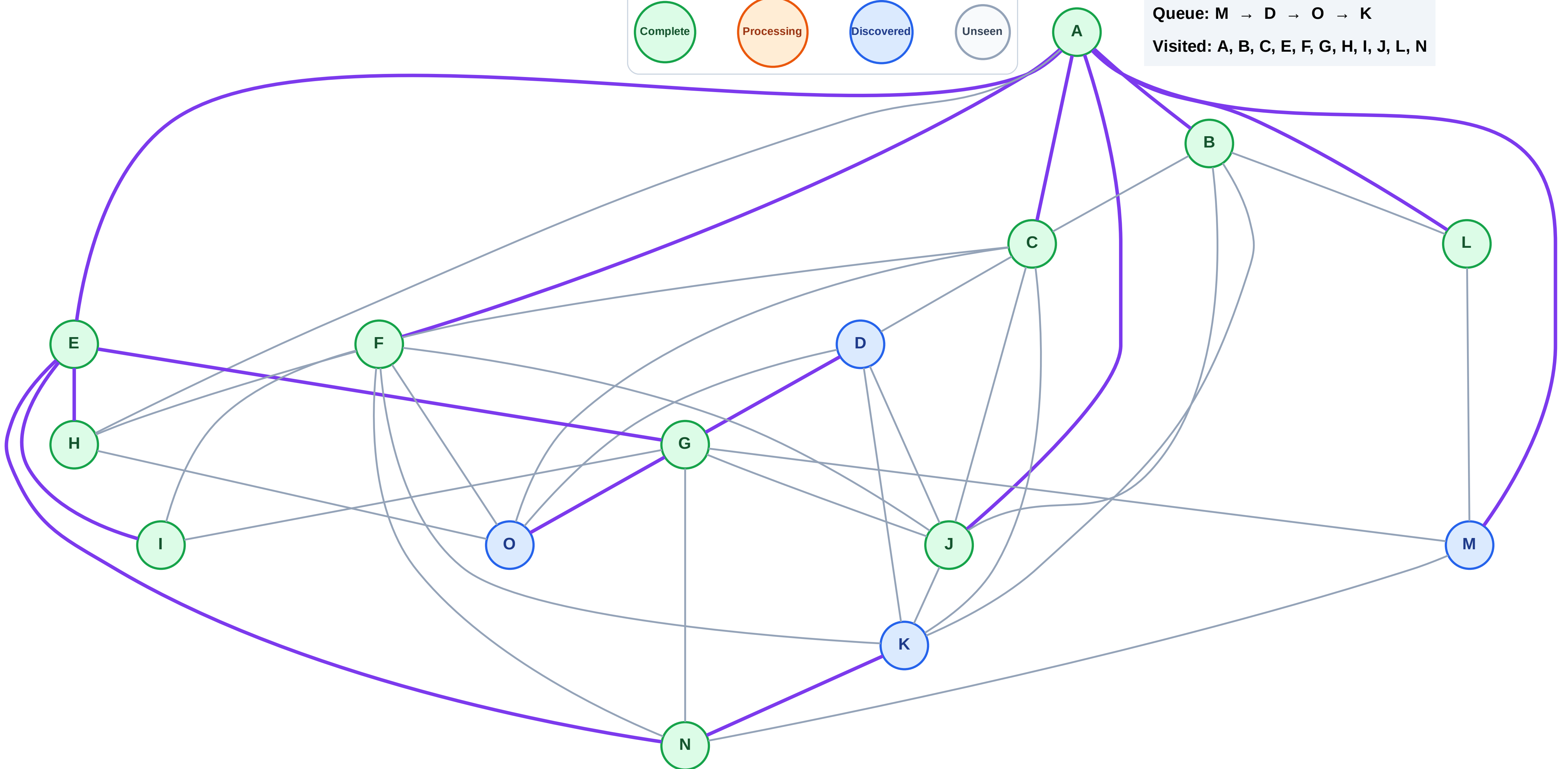
Step 23: No new neighbors from L

Legend



Queue: M → D → O → K

Visited: A, B, C, E, F, G, H, I, J, L, N



BFS Traversal

Step 24: Process node M

Legend



Complete



Processing



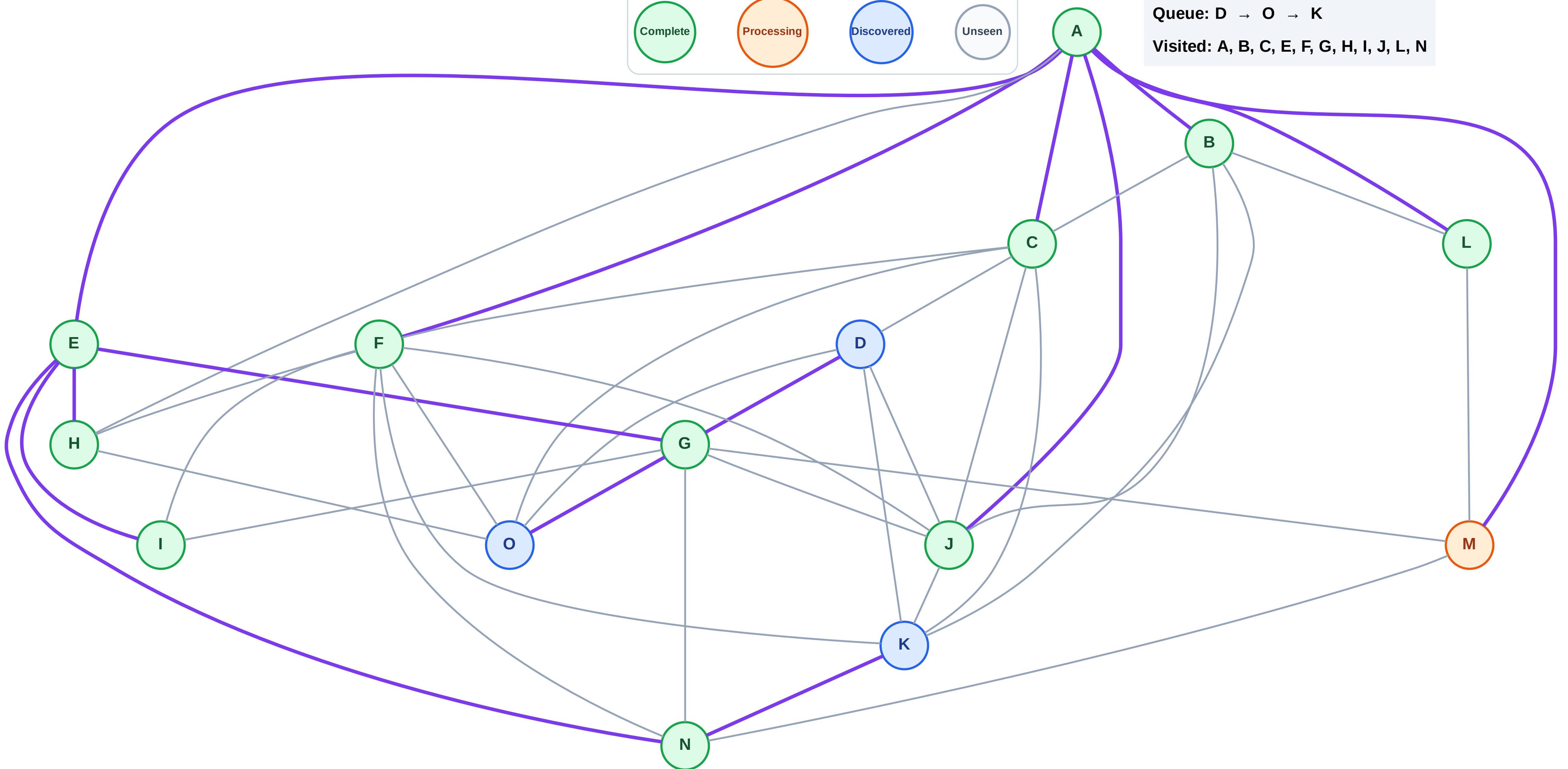
Discovered



Unseen

Queue: D → O → K

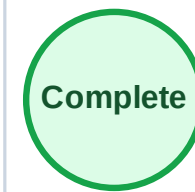
Visited: A, B, C, E, F, G, H, I, J, L, N



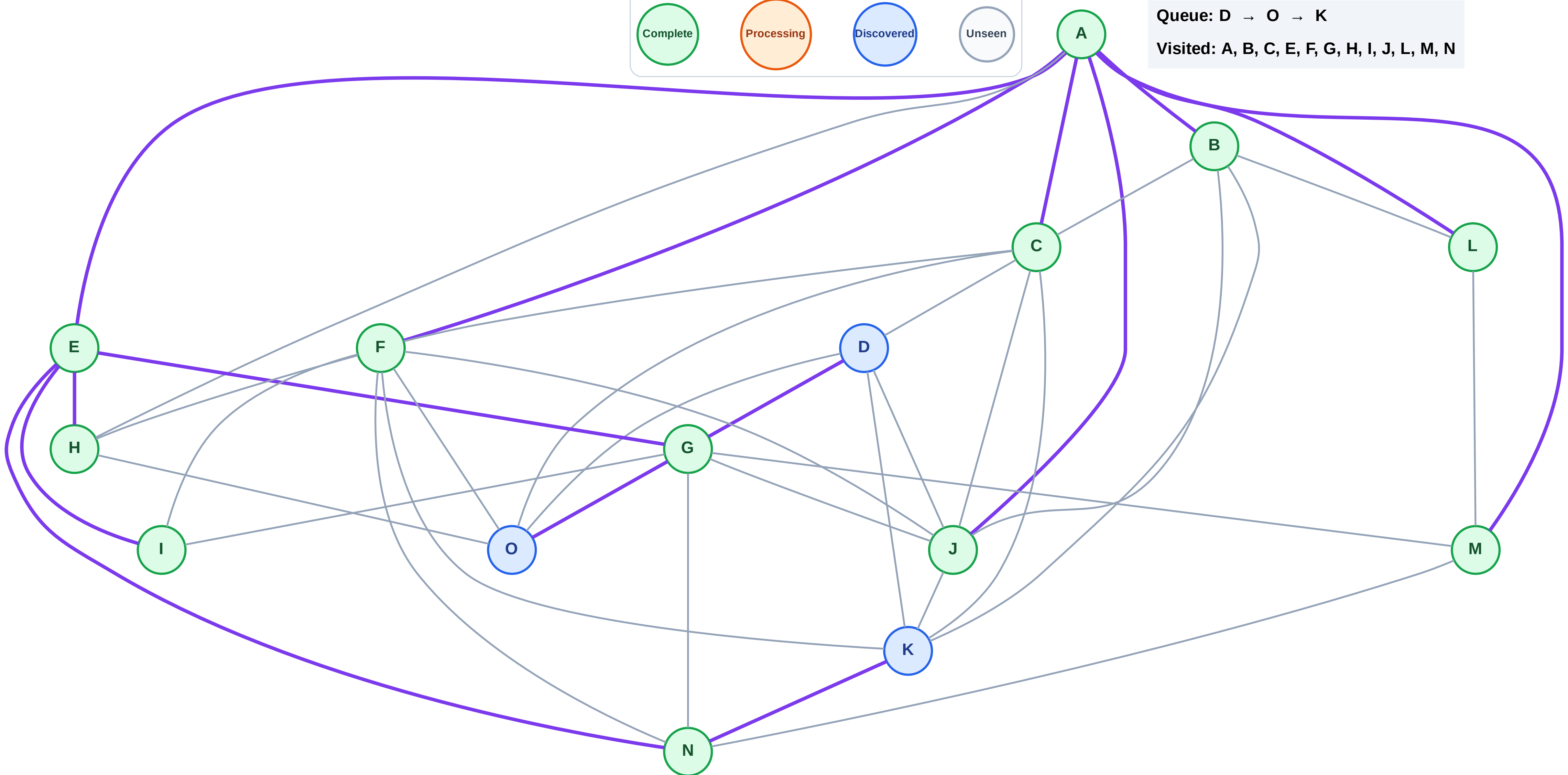
BFS Traversal

Step 25: No new neighbors from M

Legend



Queue: D → O → K
Visited: A, B, C, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 26: Process node D

Legend



Complete



Processing



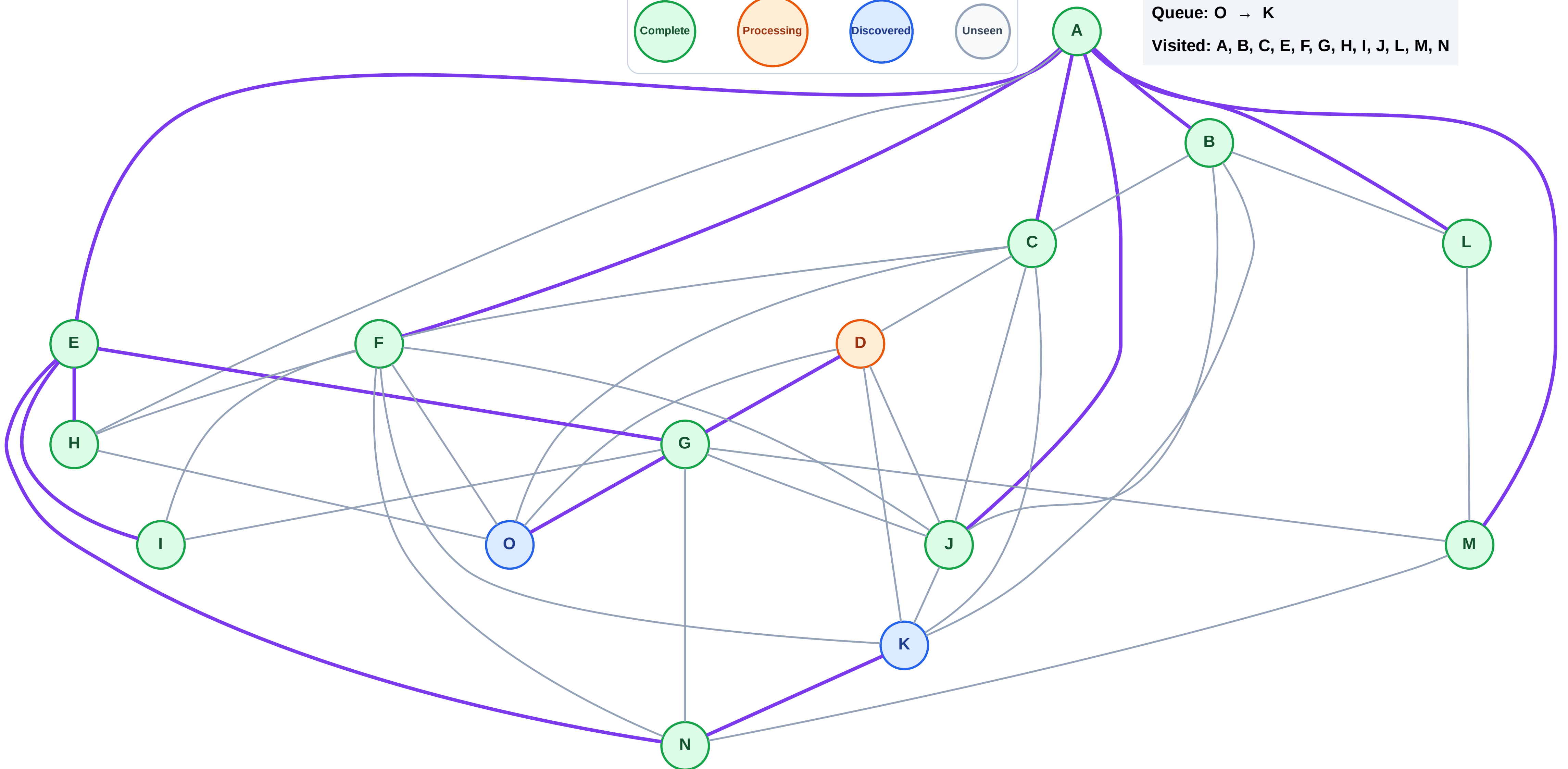
Discovered



Unseen

Queue: O → K

Visited: A, B, C, E, F, G, H, I, J, L, M, N



BFS Traversal

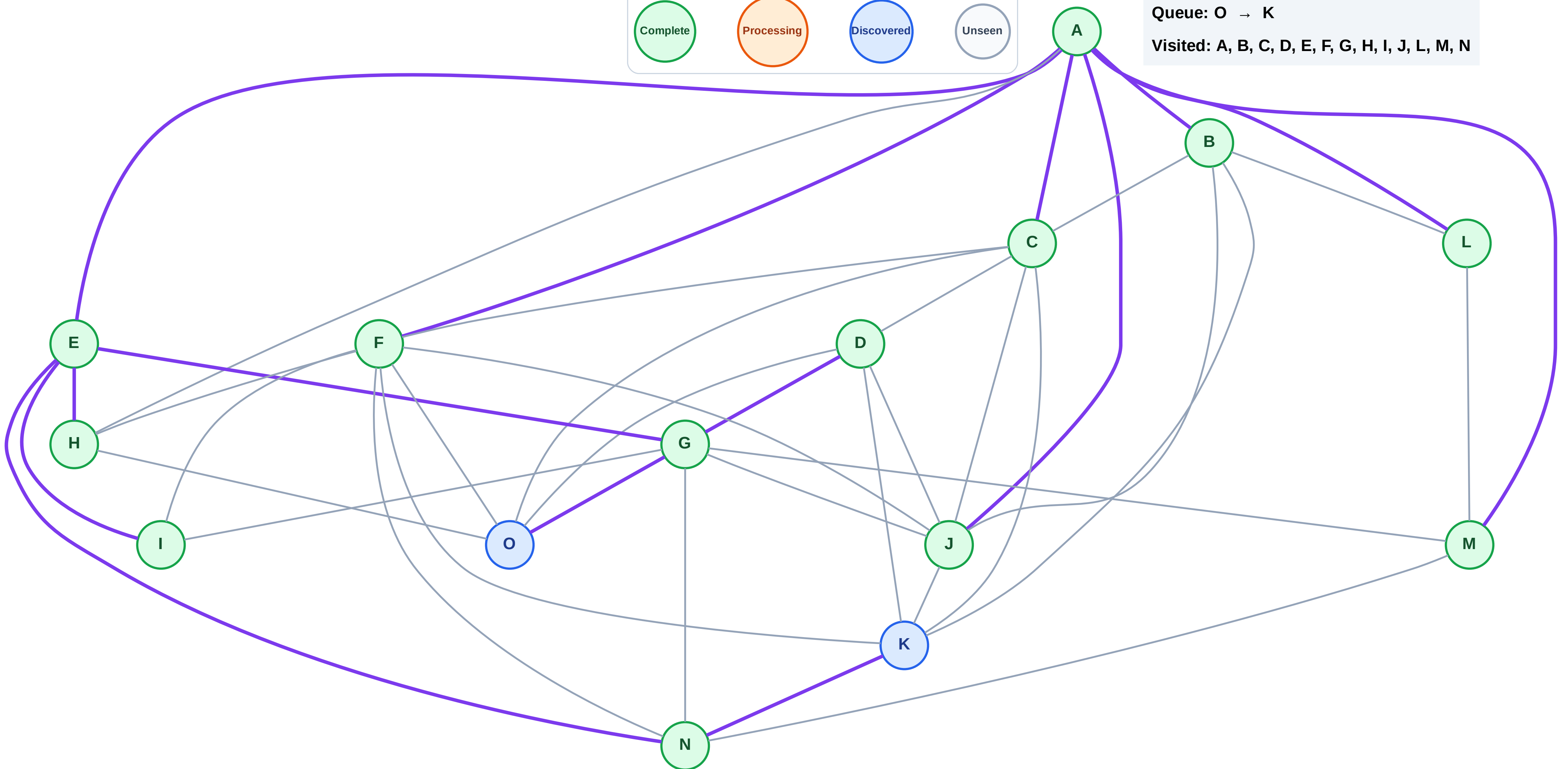
Step 27: No new neighbors from D

Legend



Queue: O → K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 28: Process node O

Legend



Complete



Processing

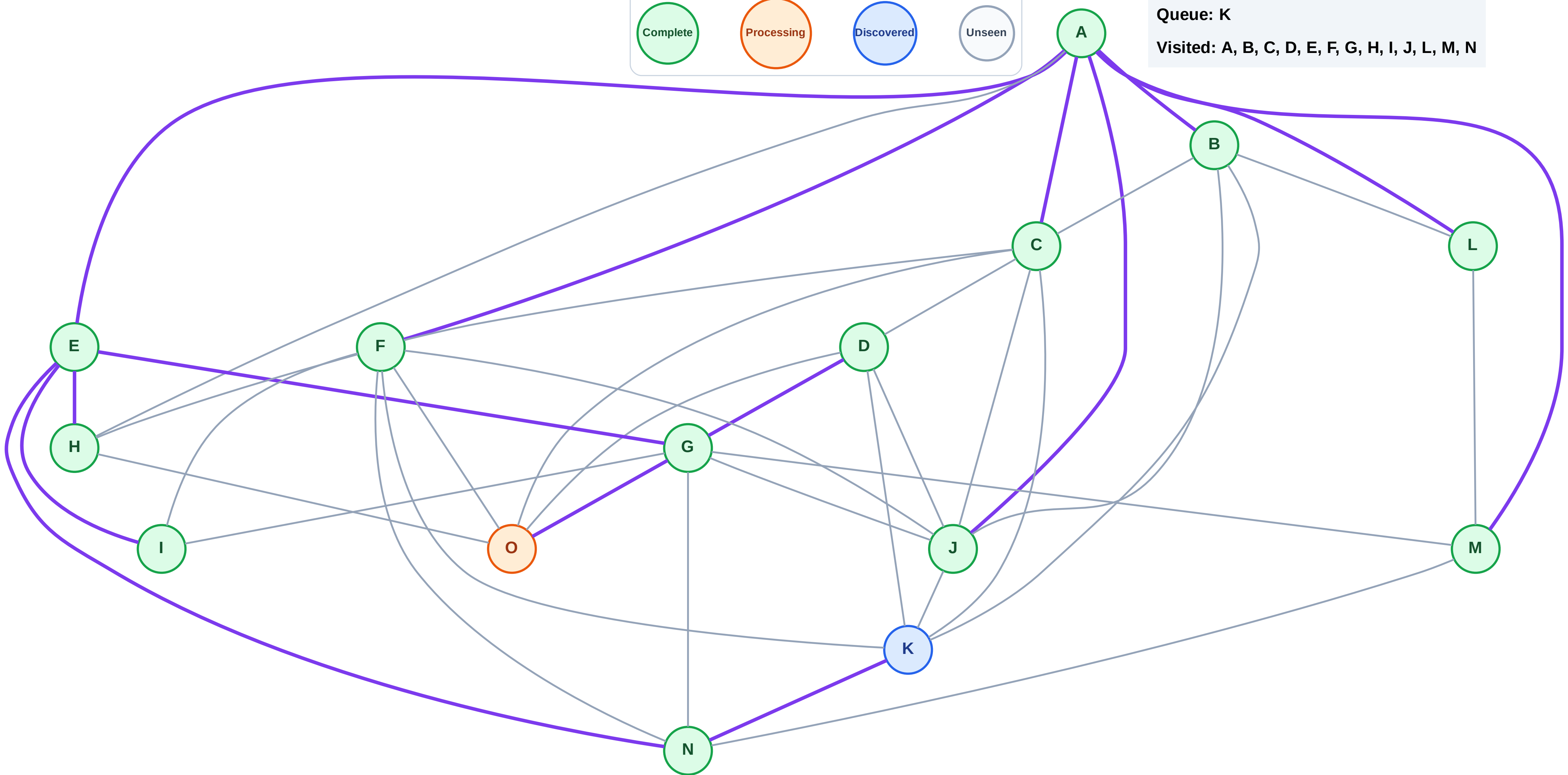


Discovered



Unseen

Queue: K
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 29: No new neighbors from O

Legend



Complete



Processing



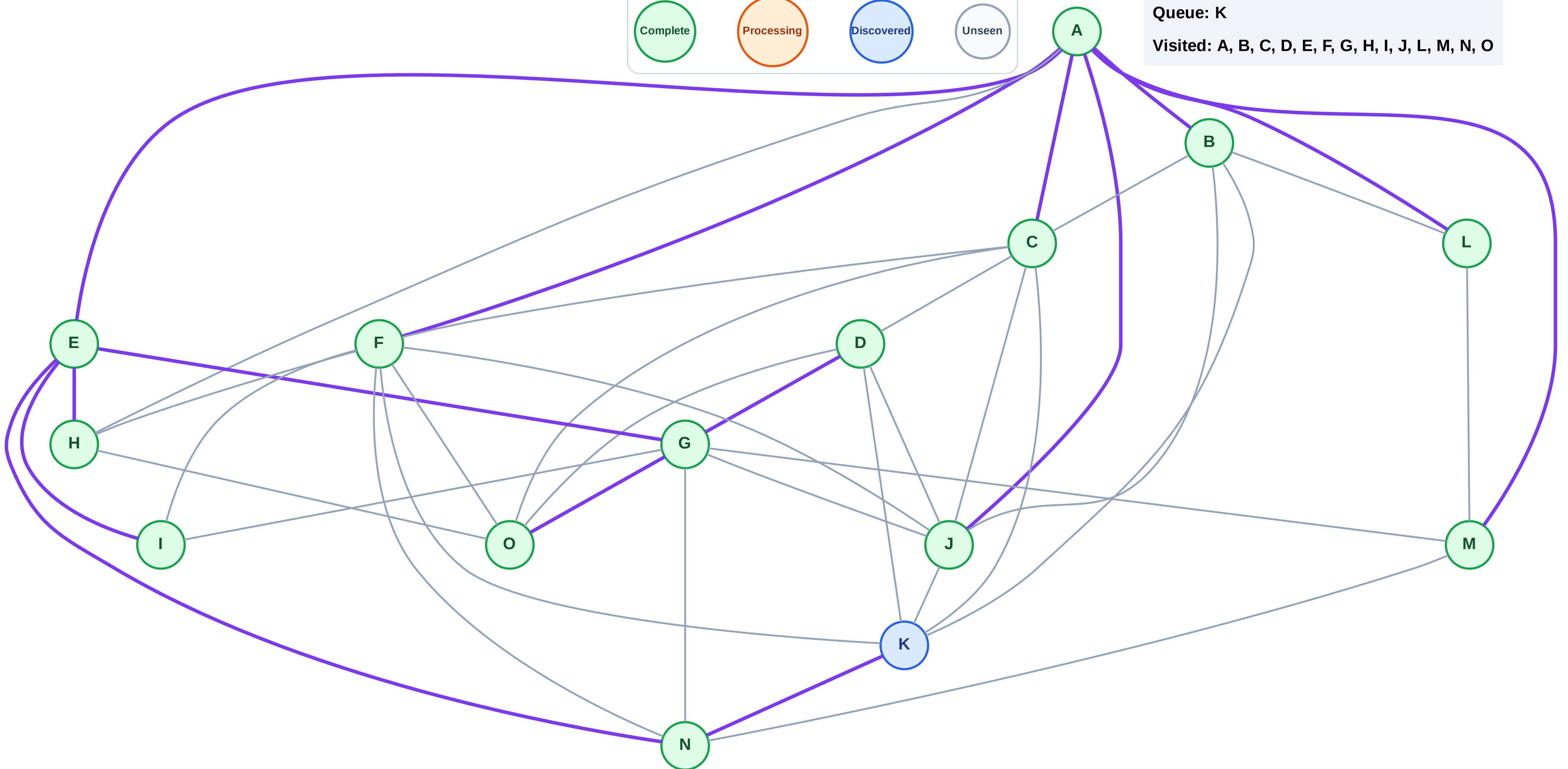
Discovered



Unseen

Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

Step 30: Process node K

Legend



Complete



Processing

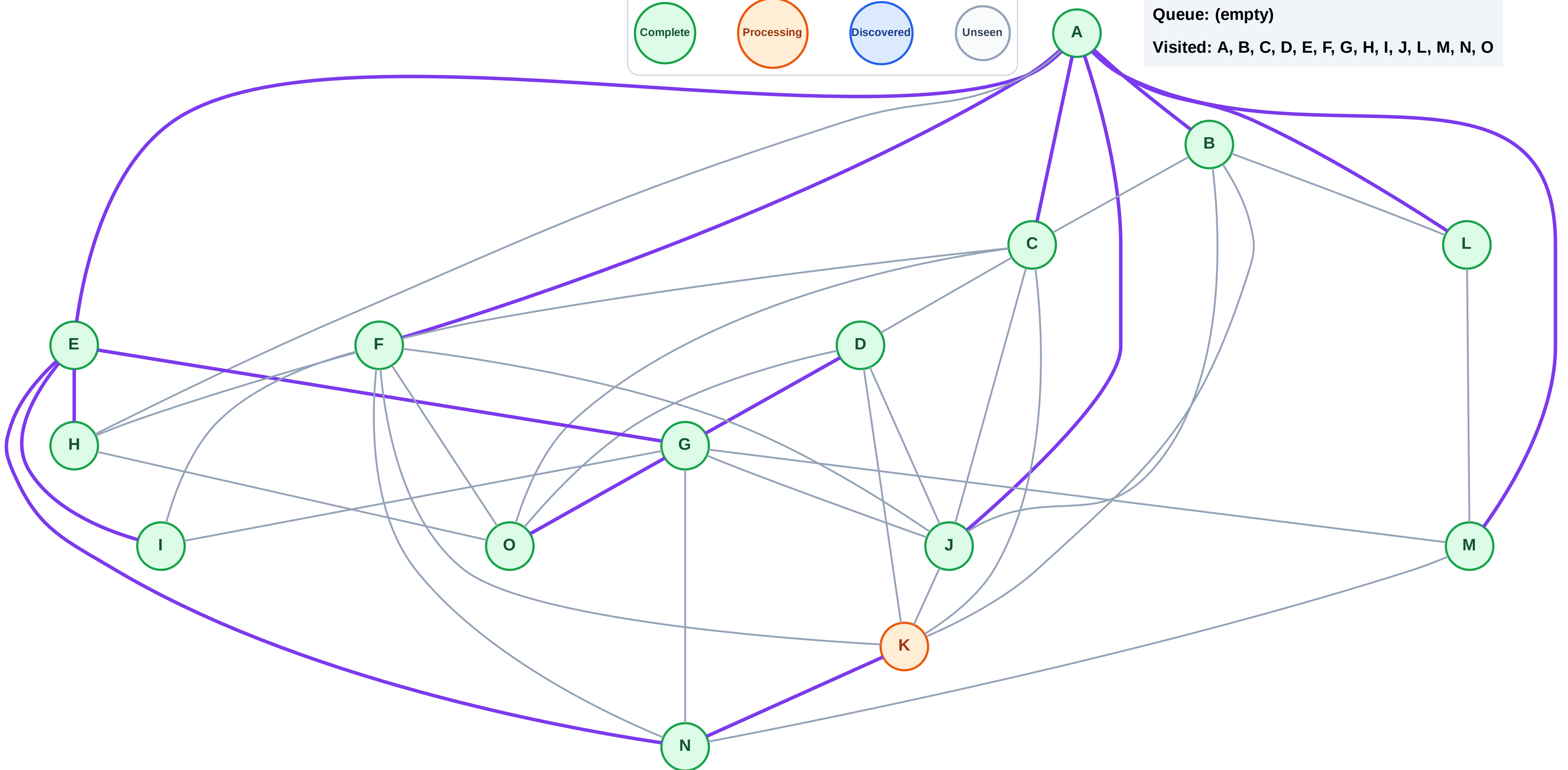


Discovered



Unseen

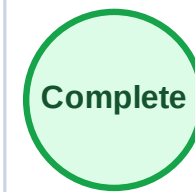
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

Step 31: No new neighbors from K

Legend



Complete



Processing



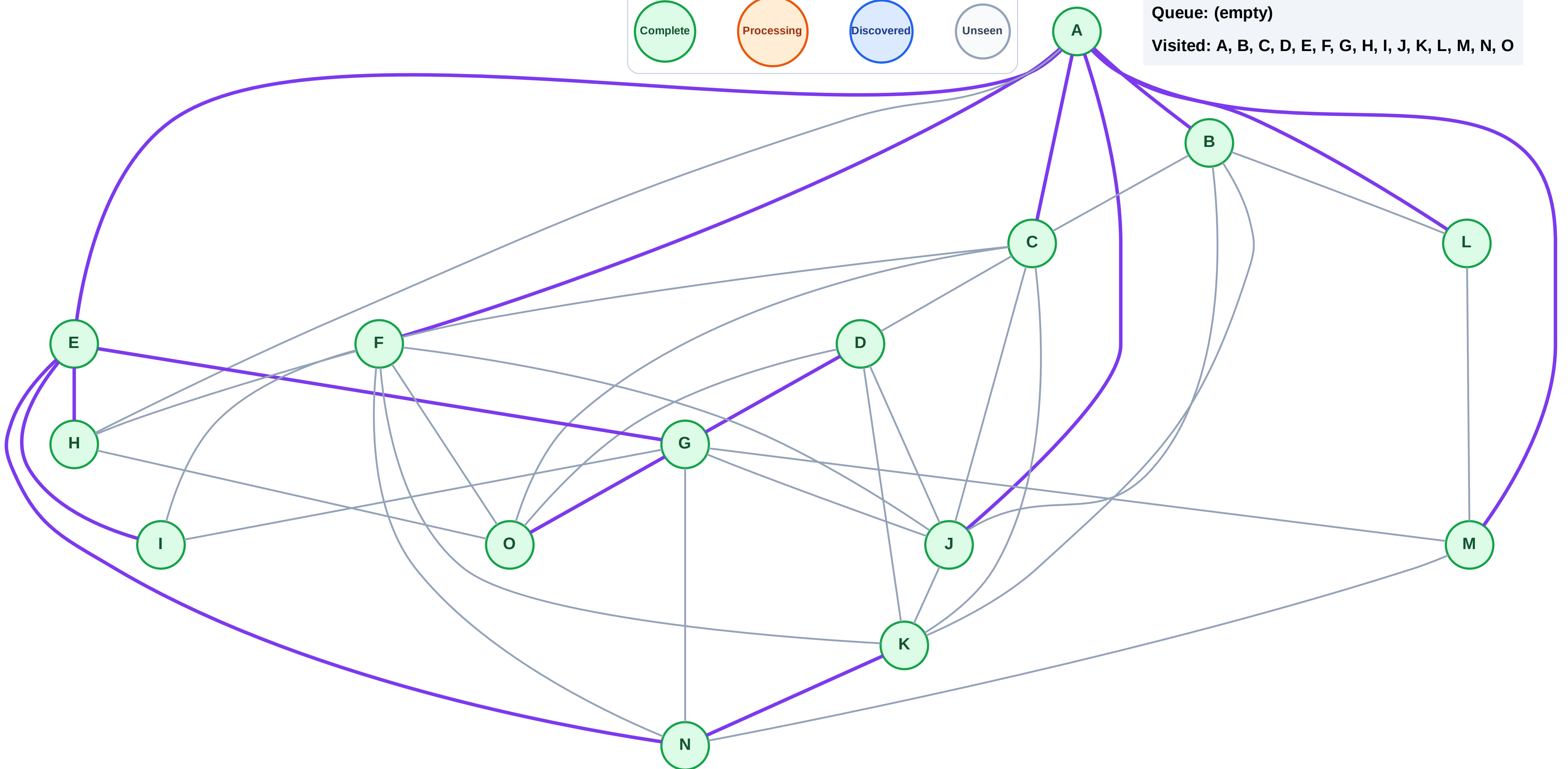
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



DFS Traversal

Step 1: Start at E

Legend

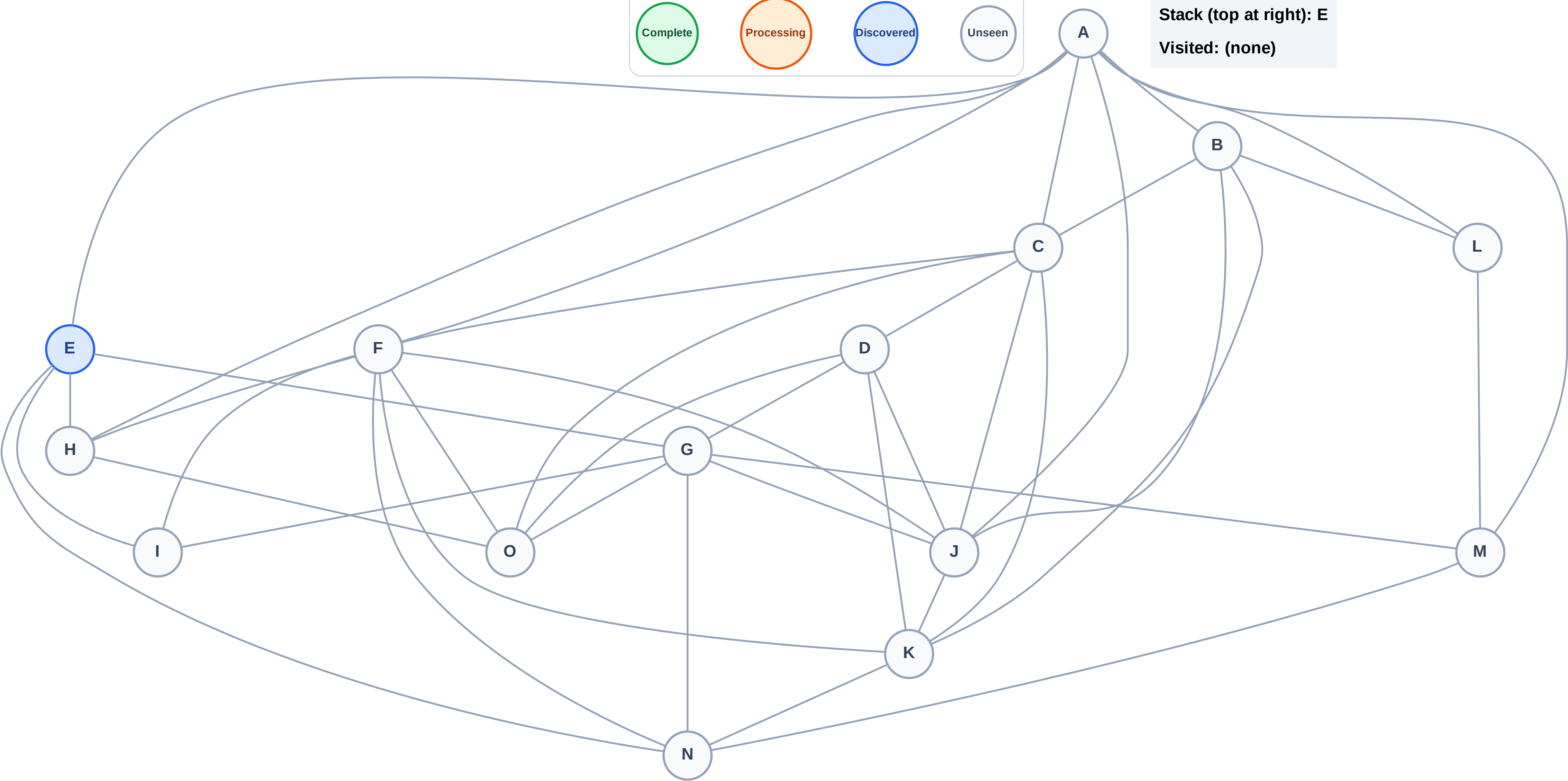
Complete

Processing

Discovered

Unseen

Stack (top at right): E
Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

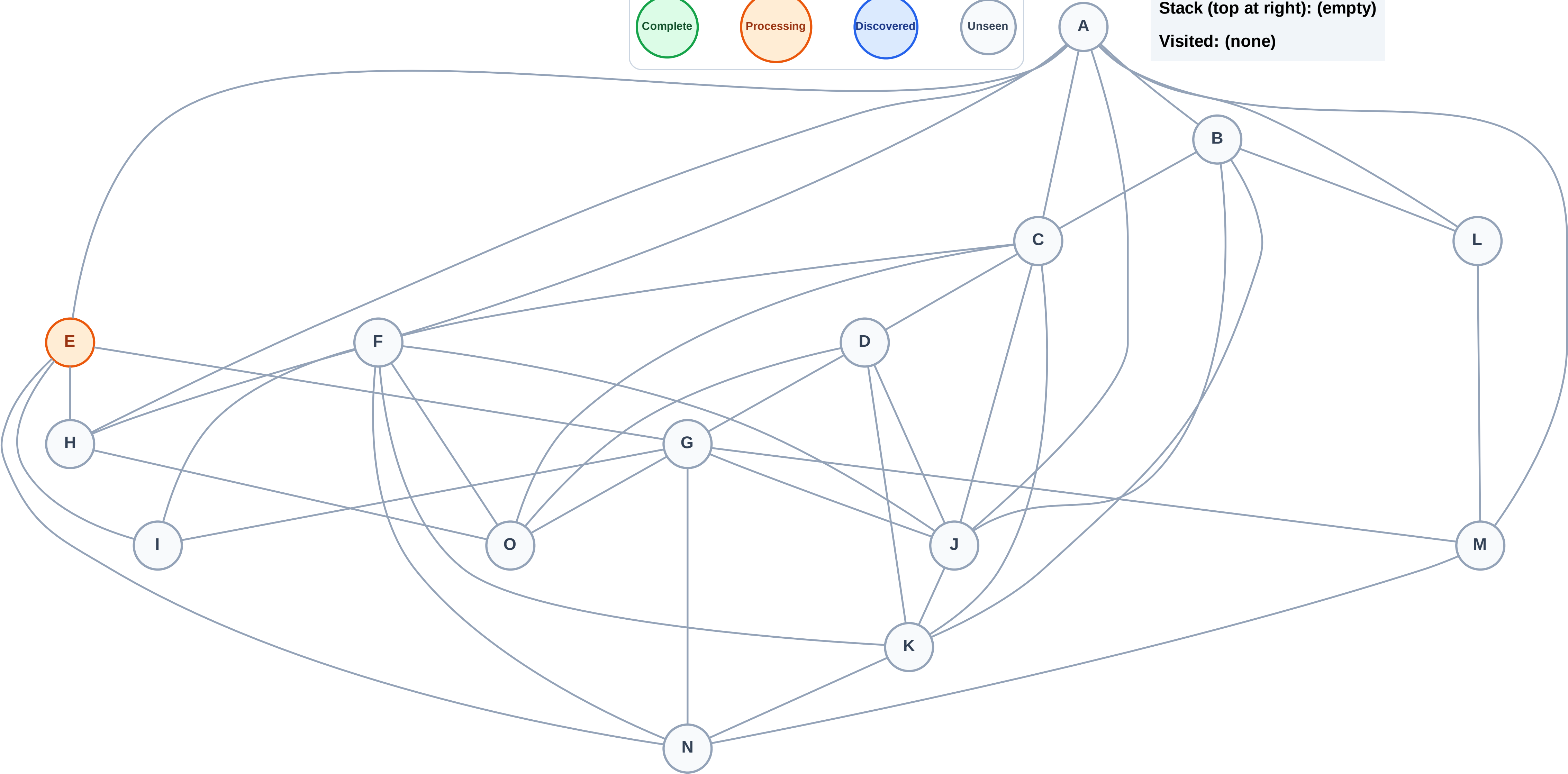
Complete

Processing

Discovered

Unseen

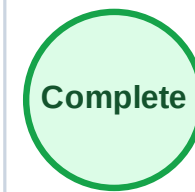
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

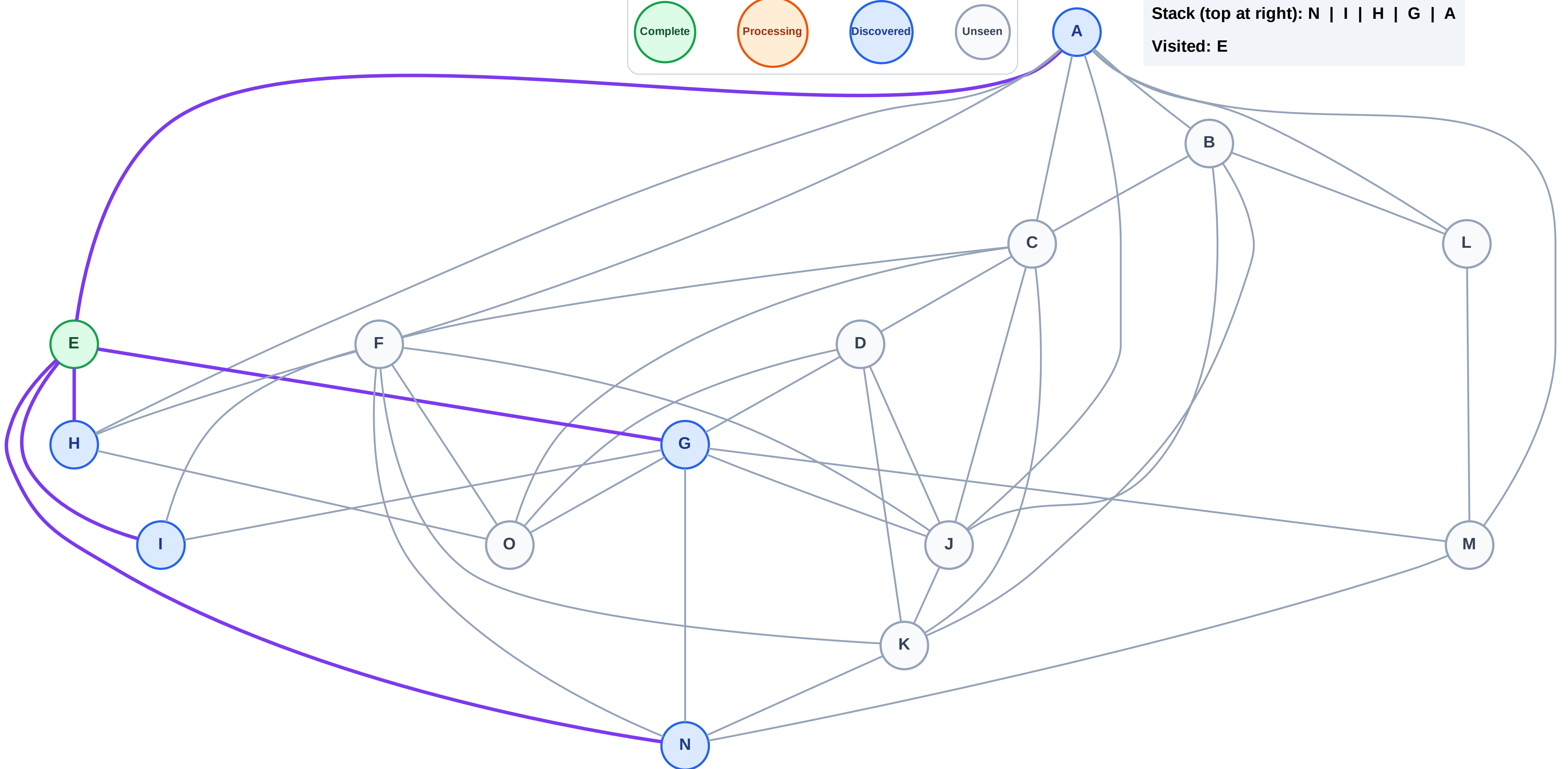
Step 3: Discover N, I, H, G, A from E

Legend



Stack (top at right): N | I | H | G | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

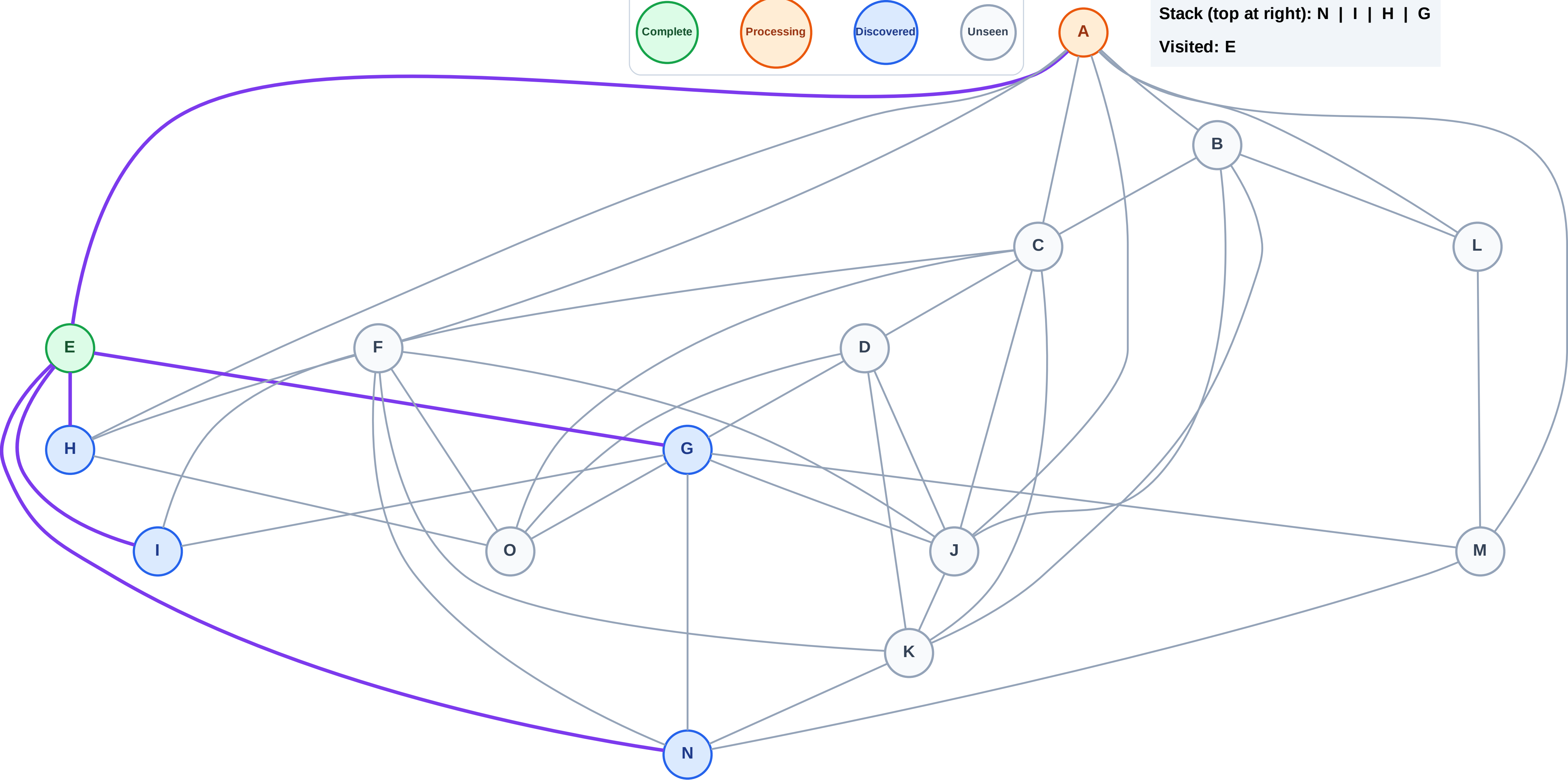
Processing

Discovered

Unseen

Stack (top at right): N | I | H | G

Visited: E



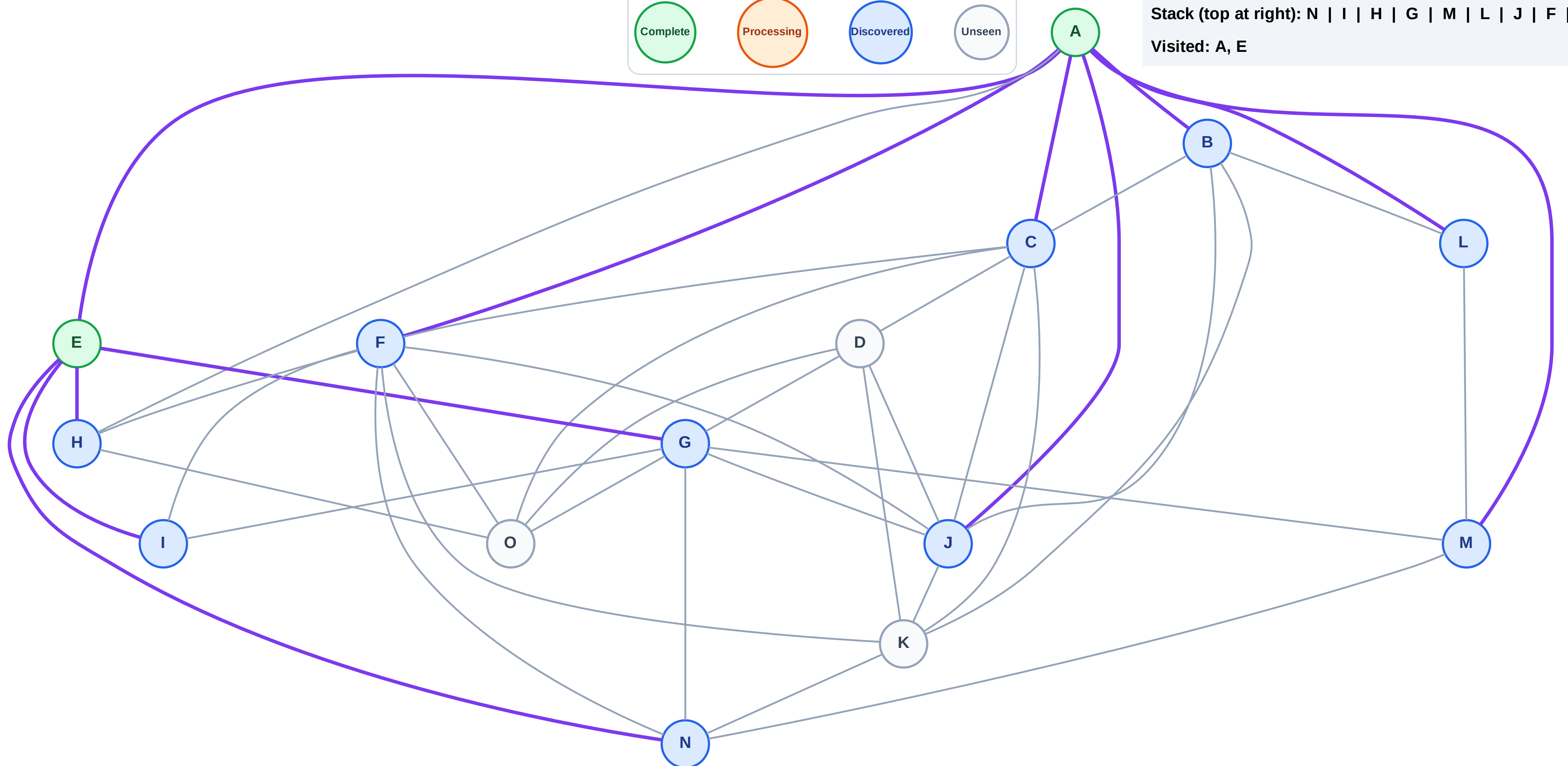
DFS Traversal

Step 5: Discover M, L, J, F, C, B from A

Legend



Stack (top at right): N | I | H | G | M | L | J | F | C | B
Visited: A, E



DFS Traversal
Step 6: Process node B

Legend

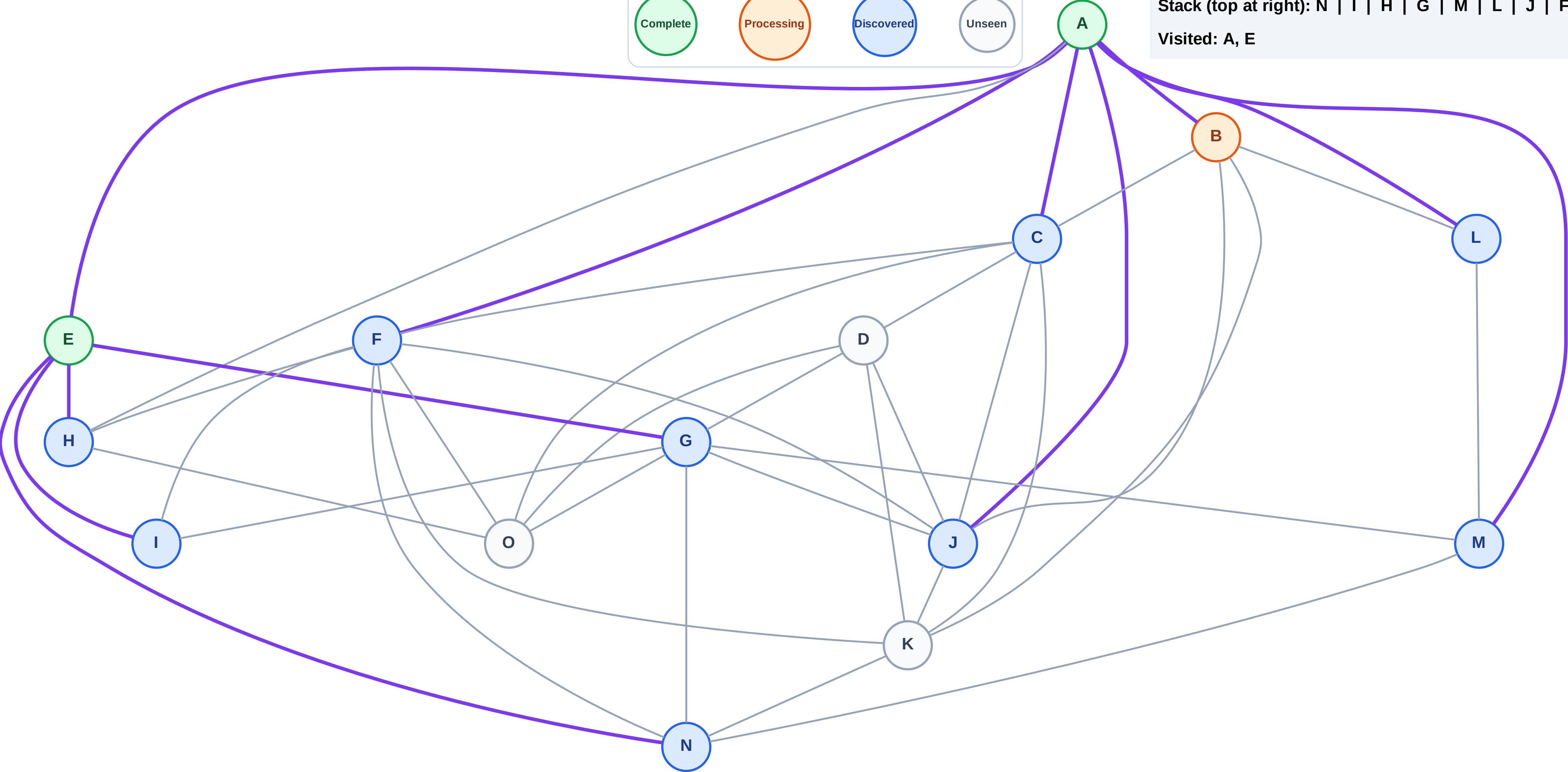
Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | H | G | M | L | J | F | C
Visited: A, E



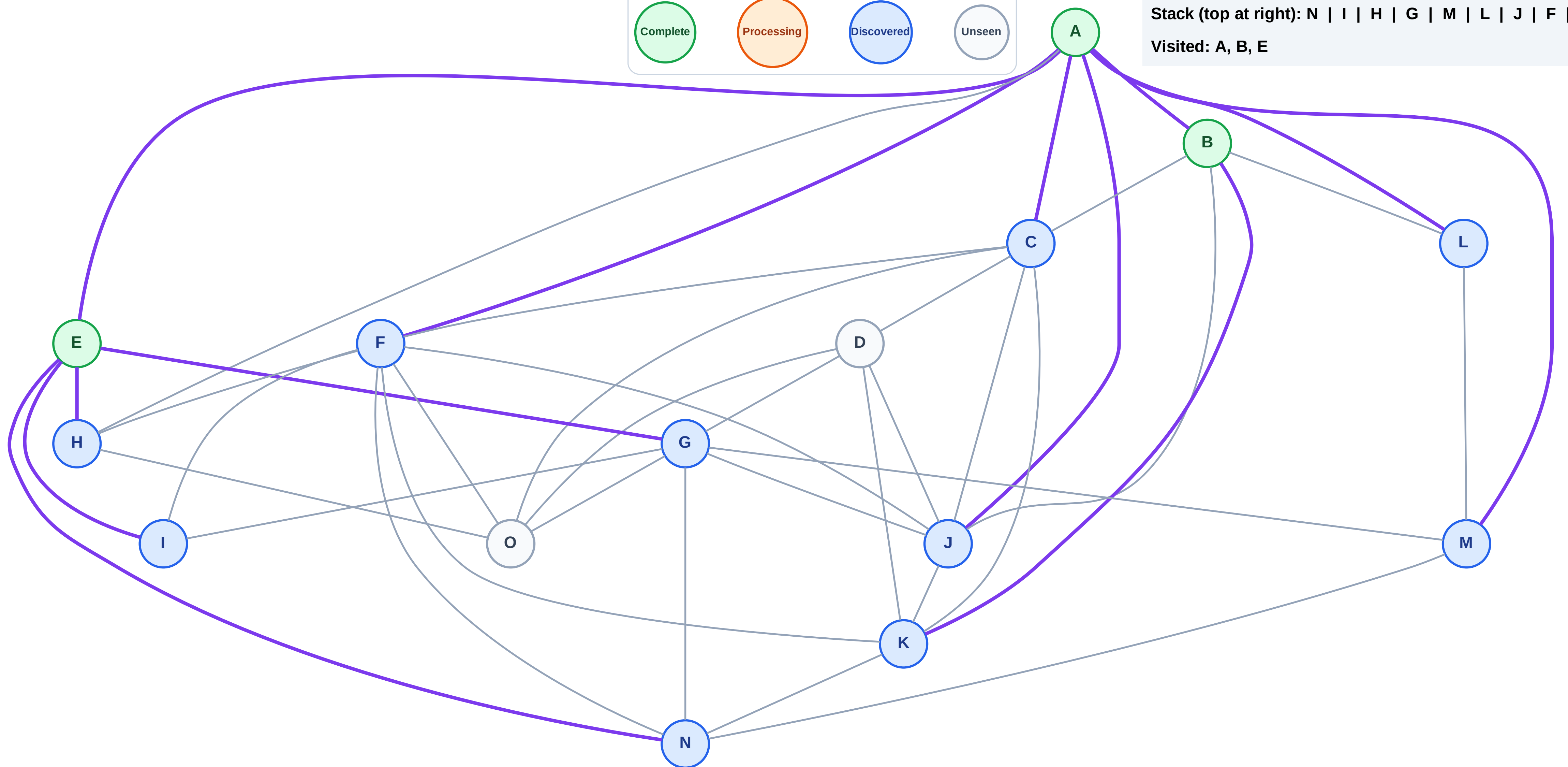
DFS Traversal

Step 7: Discover K from B

Legend



Stack (top at right): N | I | H | G | M | L | J | F | C | K
Visited: A, B, E



DFS Traversal
Step 8: Process node K

Legend

Complete

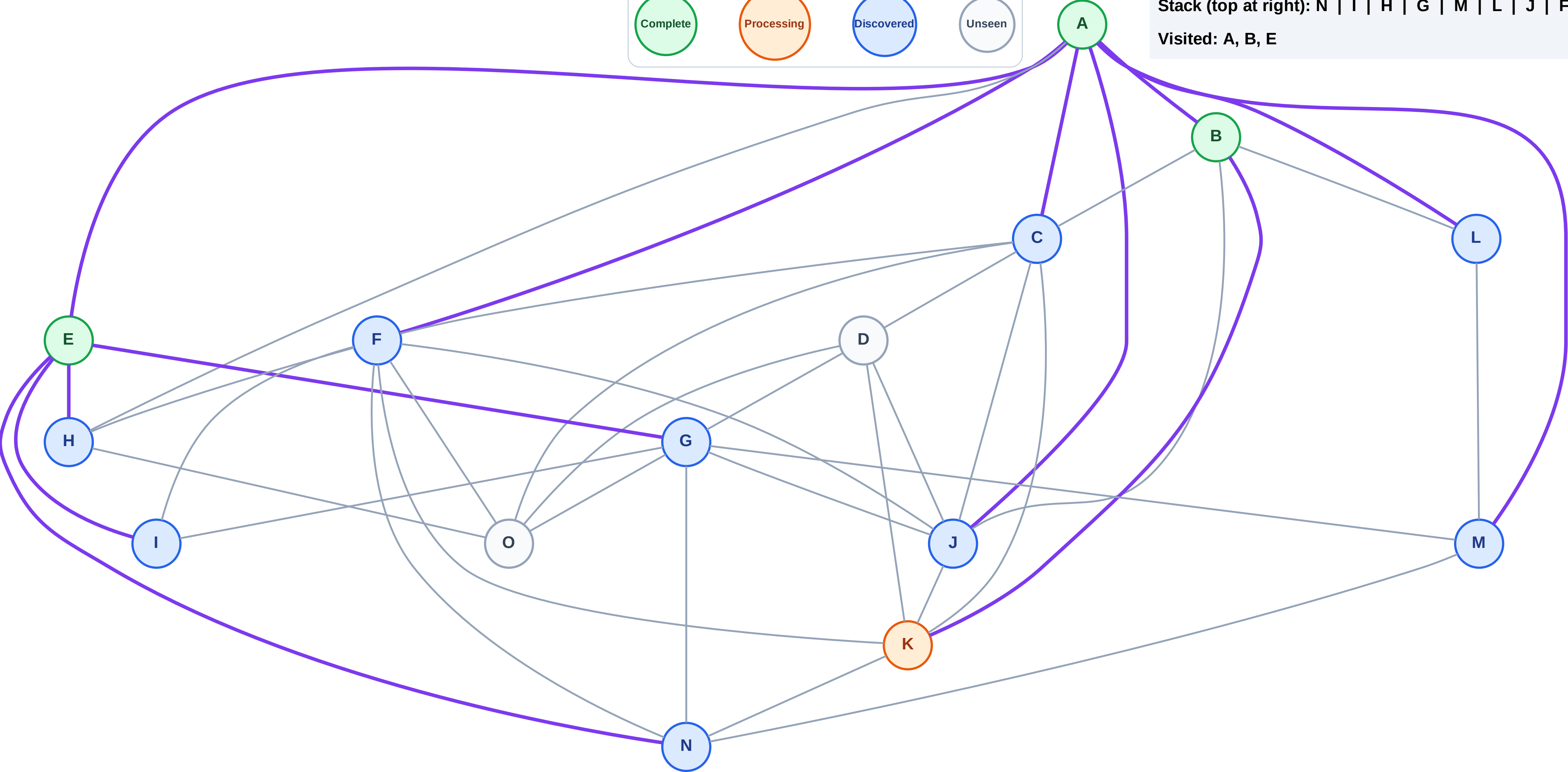
Processing

Discovered

Unseen

Stack (top at right): N | I | H | G | M | L | J | F | C

Visited: A, B, E



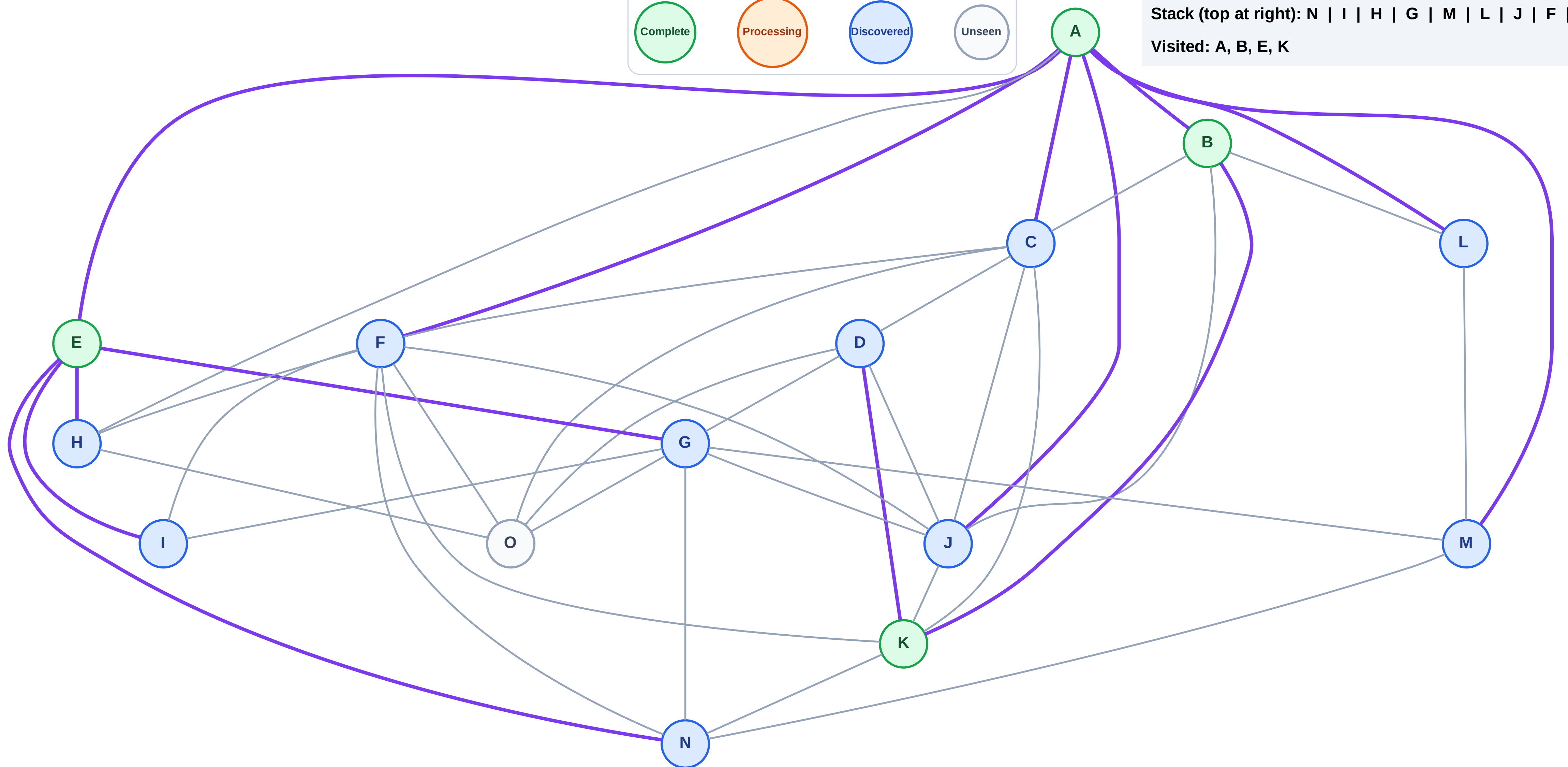
DFS Traversal

Step 9: Discover D from K

Legend



Stack (top at right): N | I | H | G | M | L | J | F | C | D
Visited: A, B, E, K



DFS Traversal
Step 10: Process node D

Legend

Complete

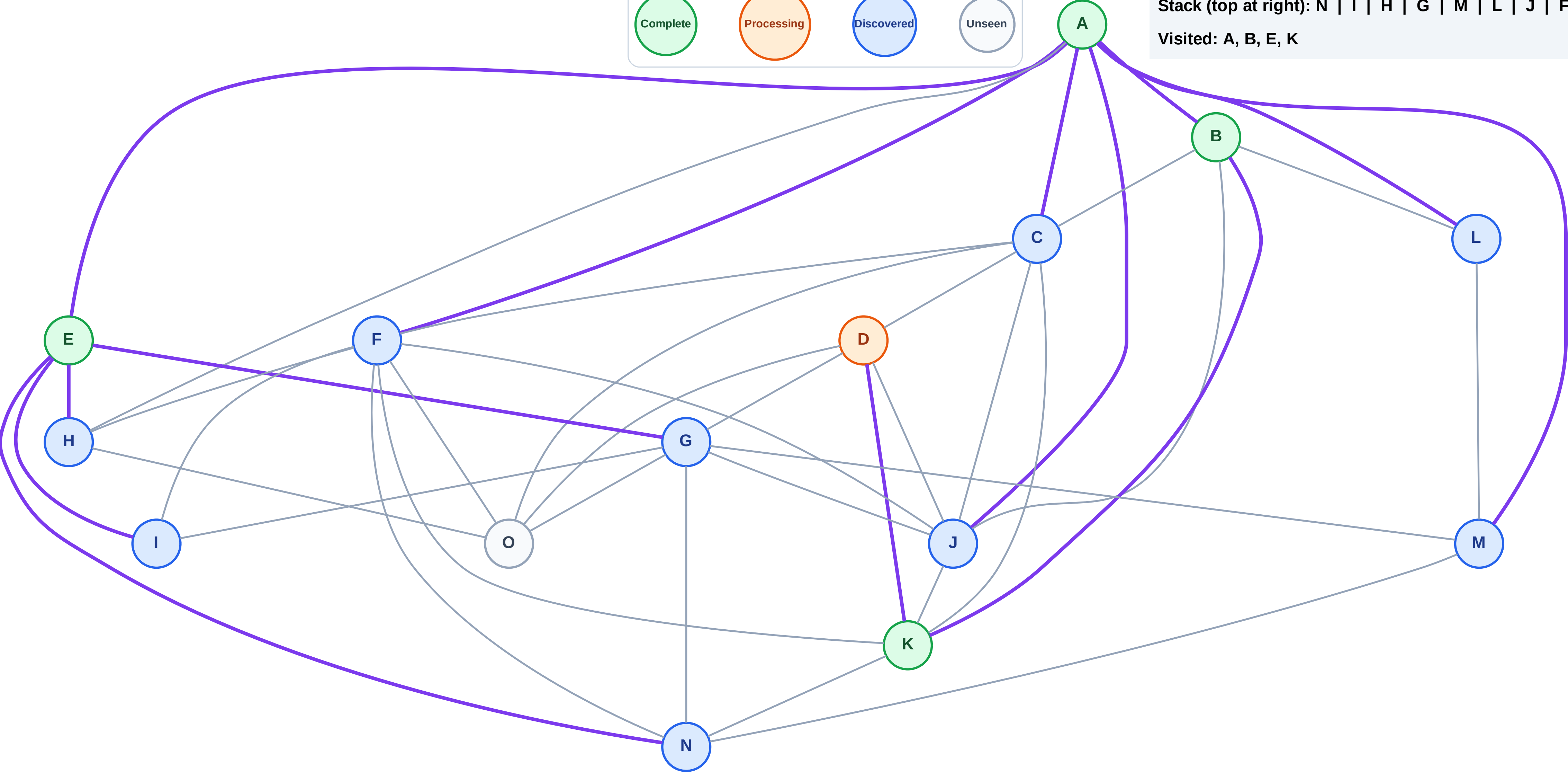
Processing

Discovered

Unseen

Stack (top at right): N | I | H | G | M | L | J | F | C

Visited: A, B, E, K



DFS Traversal

Step 11: Discover O from D

Legend



Complete



Processing



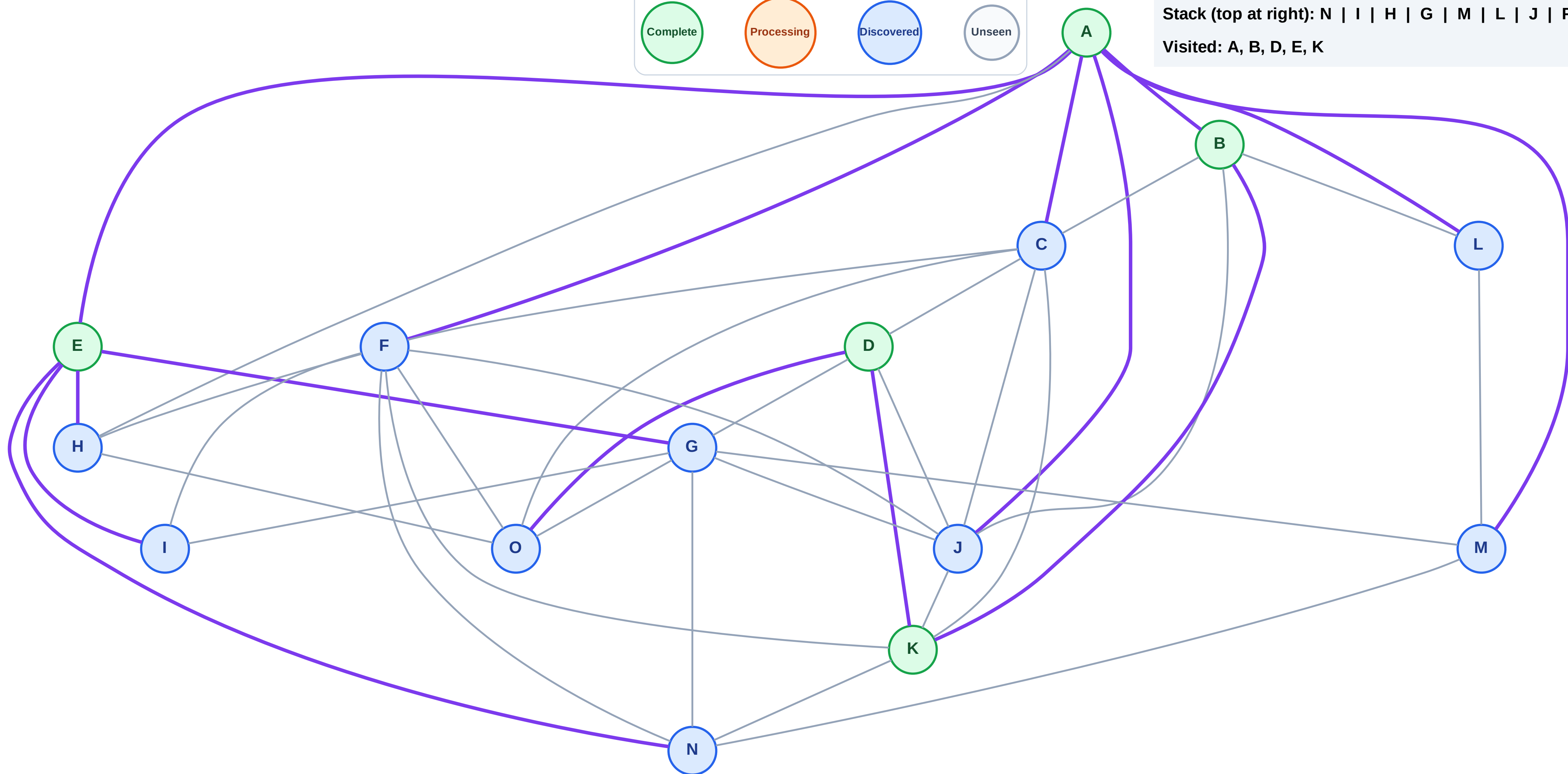
Discovered



Unseen

Stack (top at right): N | I | H | G | M | L | J | F | C | O

Visited: A, B, D, E, K



DFS Traversal

Step 12: Process node O

Legend

Complete

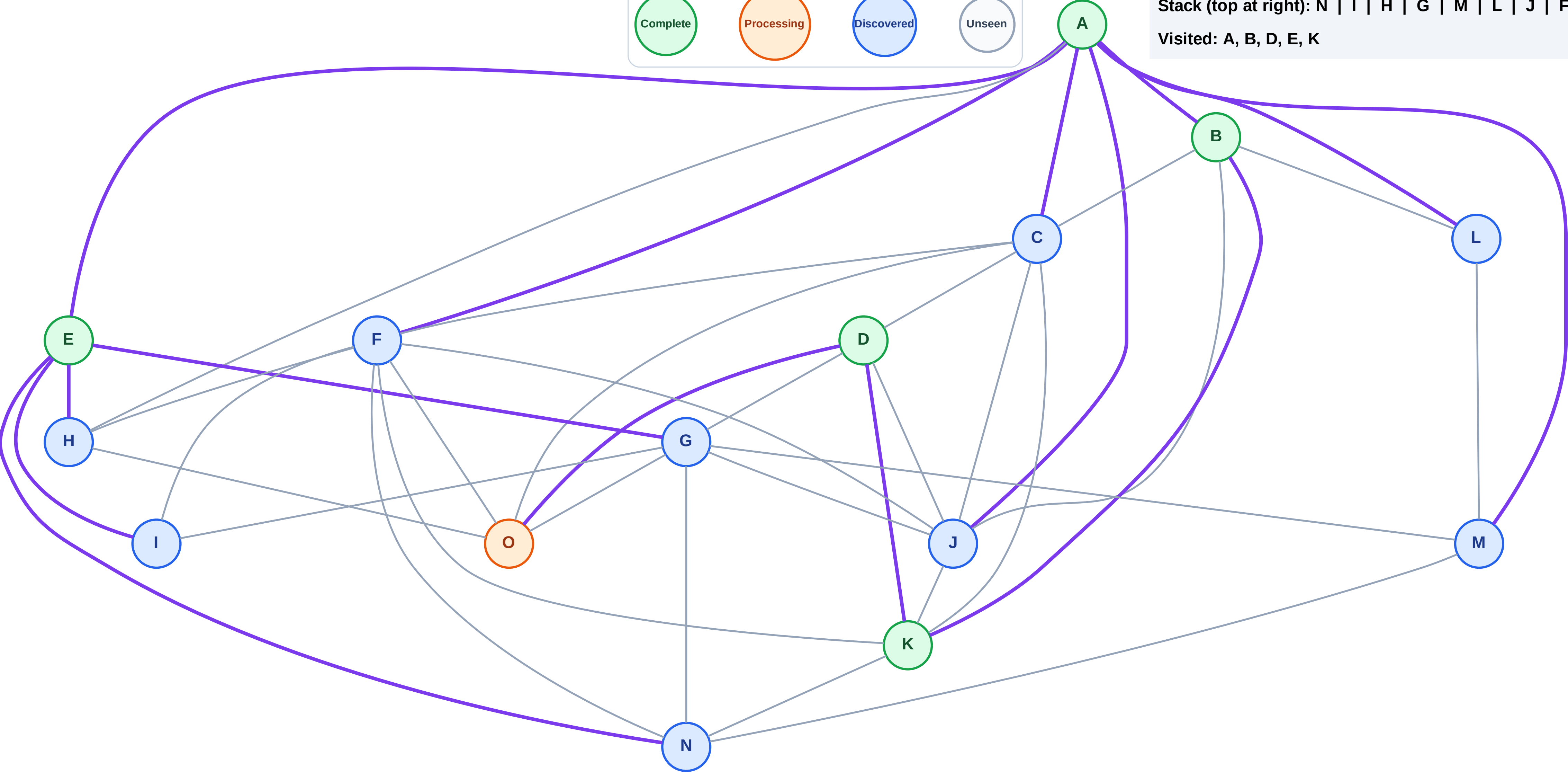
Processing

Discovered

Unseen

Stack (top at right): N | I | H | G | M | L | J | F | C

Visited: A, B, D, E, K



DFS Traversal

Step 13: No new neighbors from O

Legend



Complete



Processing



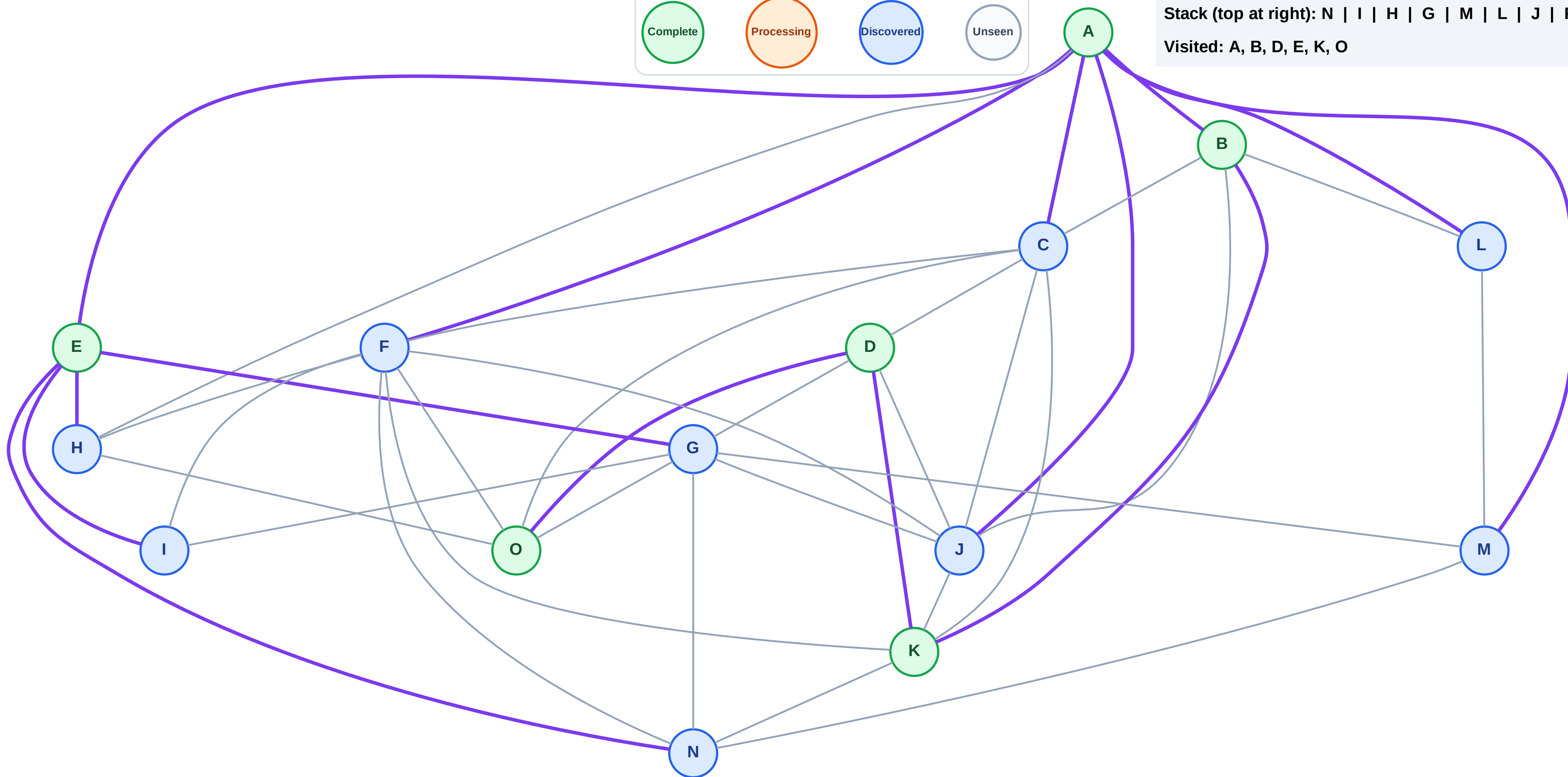
Discovered



Unseen

Stack (top at right): N | I | H | G | M | L | J | F | C

Visited: A, B, D, E, K, O



DFS Traversal

Step 14: Process node C

Legend

Complete

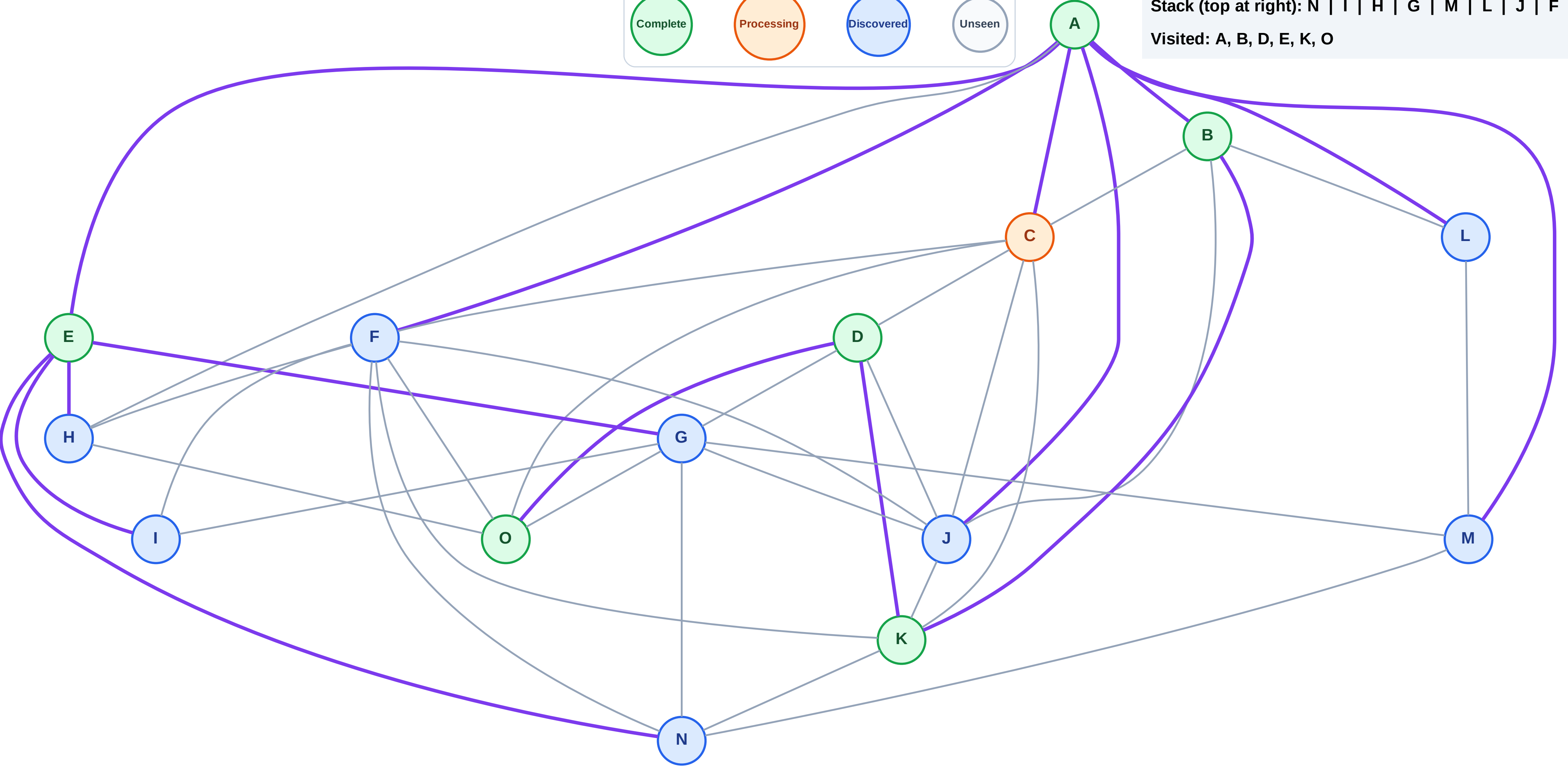
Processing

Discovered

Unseen

Stack (top at right): N | I | H | G | M | L | J | F

Visited: A, B, D, E, K, O



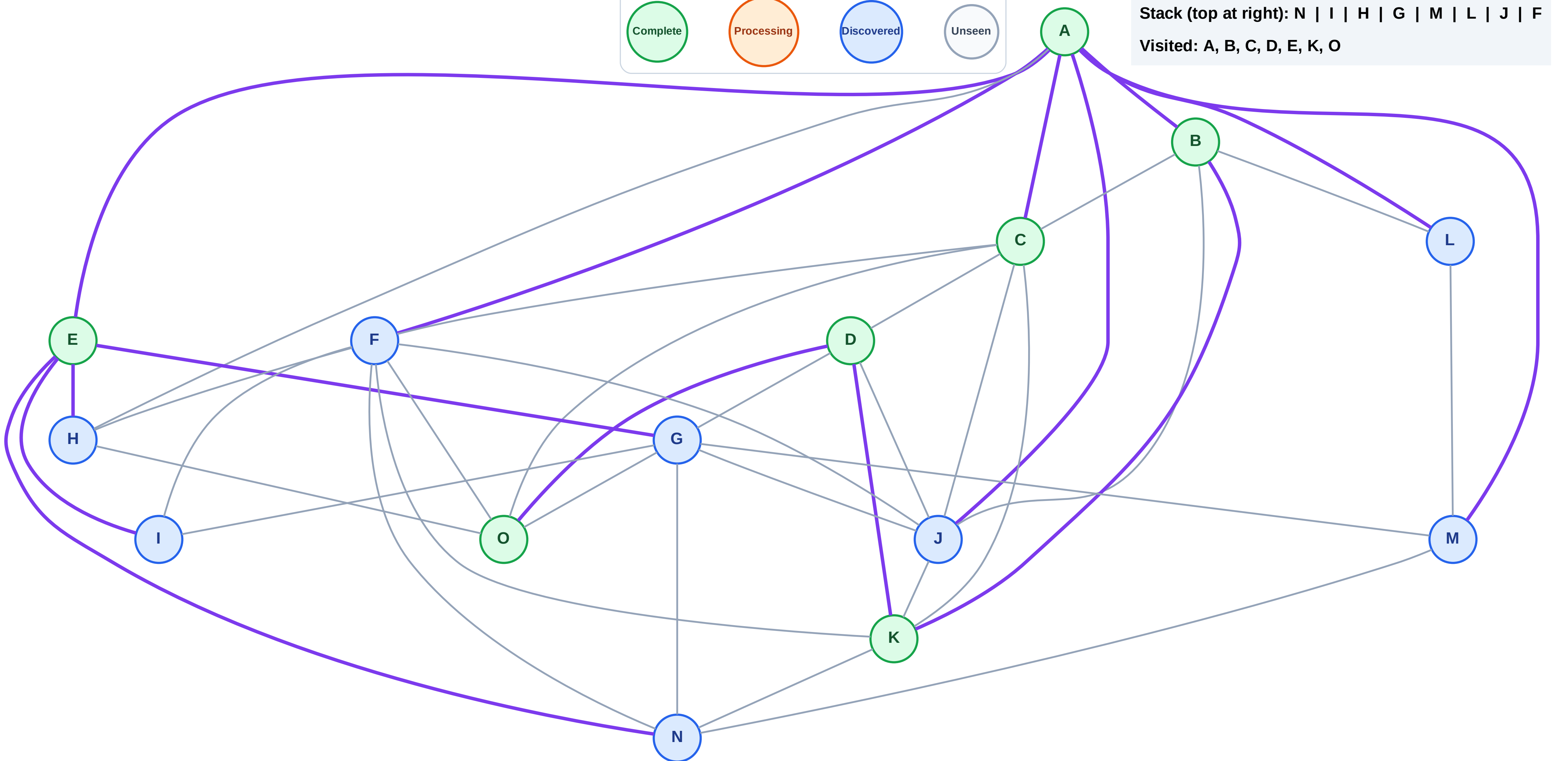
DFS Traversal

Step 15: No new neighbors from C

Legend



Stack (top at right): N | I | H | G | M | L | J | F
Visited: A, B, C, D, E, K, O



DFS Traversal

Step 16: Process node F

Legend



Complete



Processing



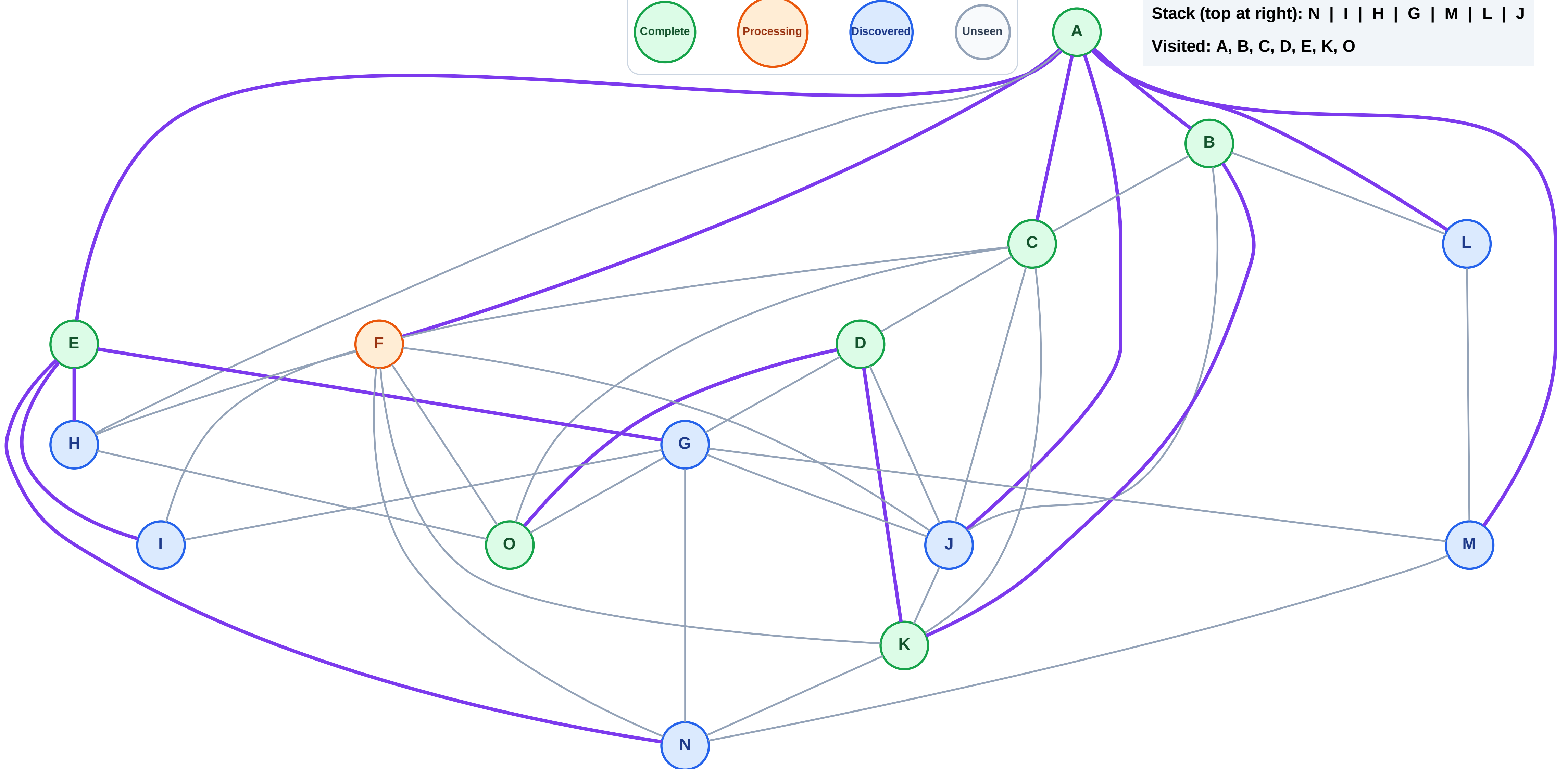
Discovered



Unseen

Stack (top at right): N | I | H | G | M | L | J

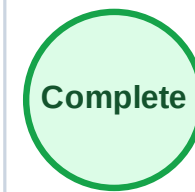
Visited: A, B, C, D, E, K, O



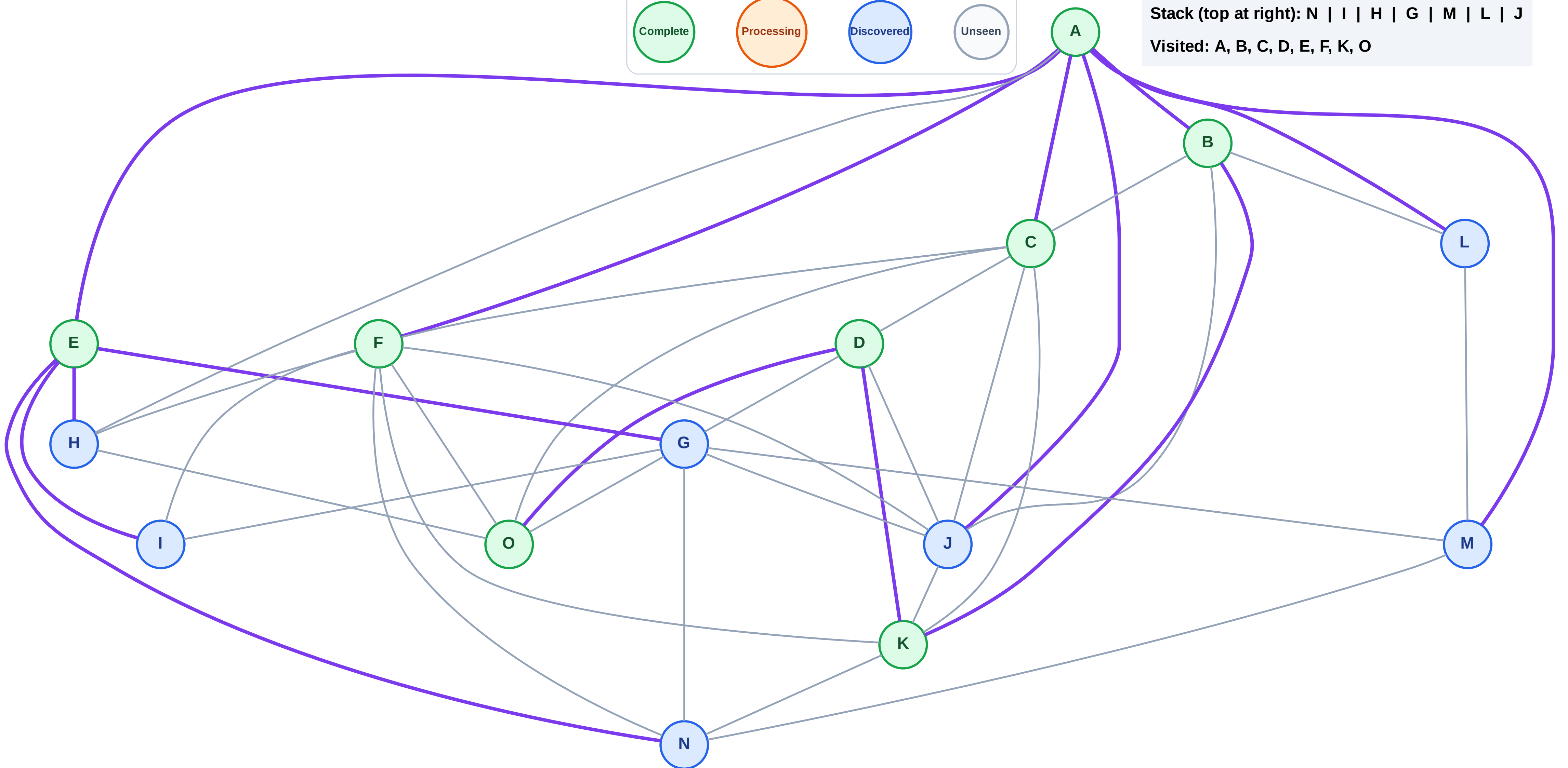
DFS Traversal

Step 17: No new neighbors from F

Legend



Stack (top at right): N | I | H | G | M | L | J
Visited: A, B, C, D, E, F, K, O



DFS Traversal

Step 18: Process node J

Legend

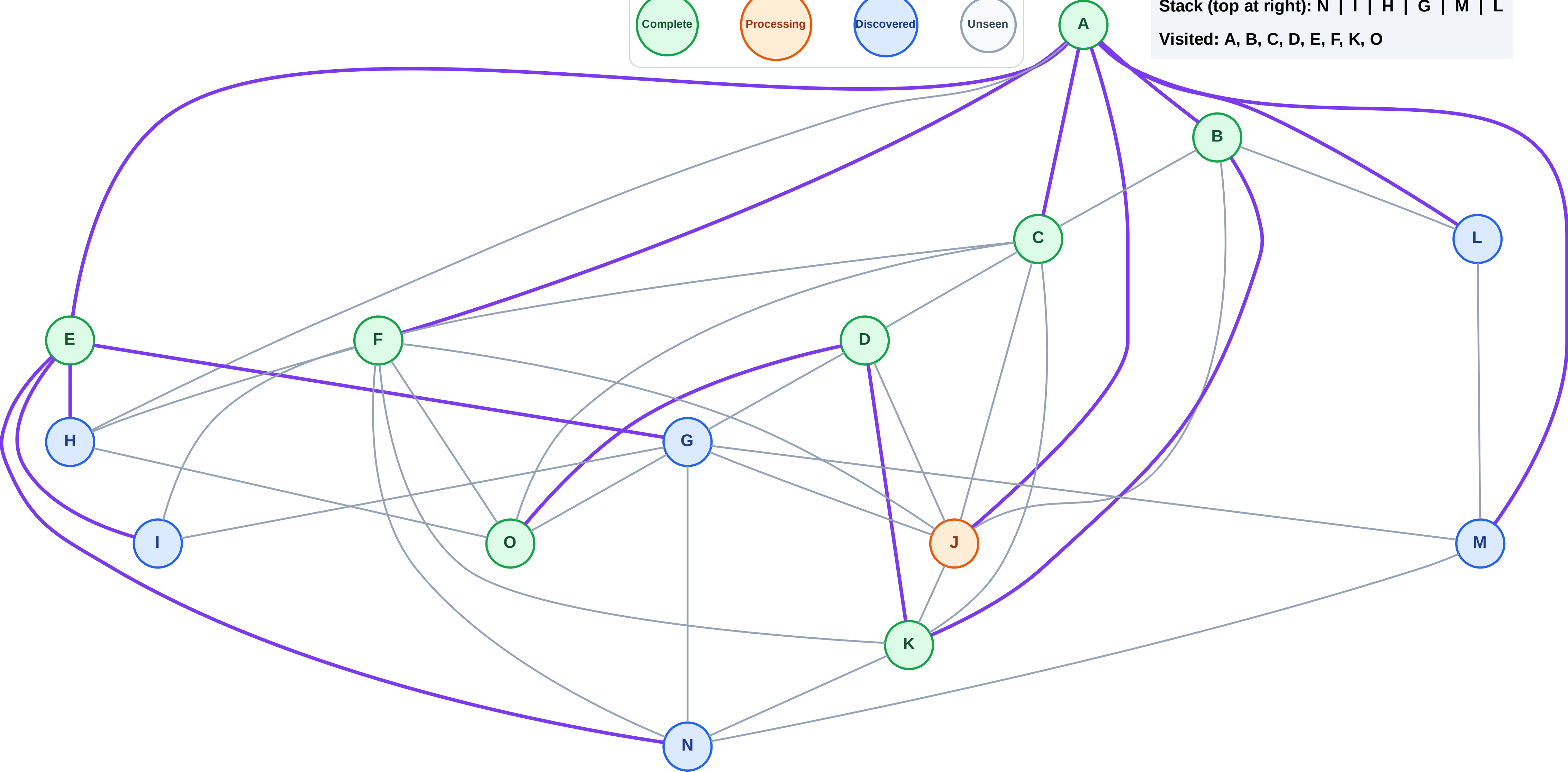
Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | H | G | M | L
Visited: A, B, C, D, E, F, K, O



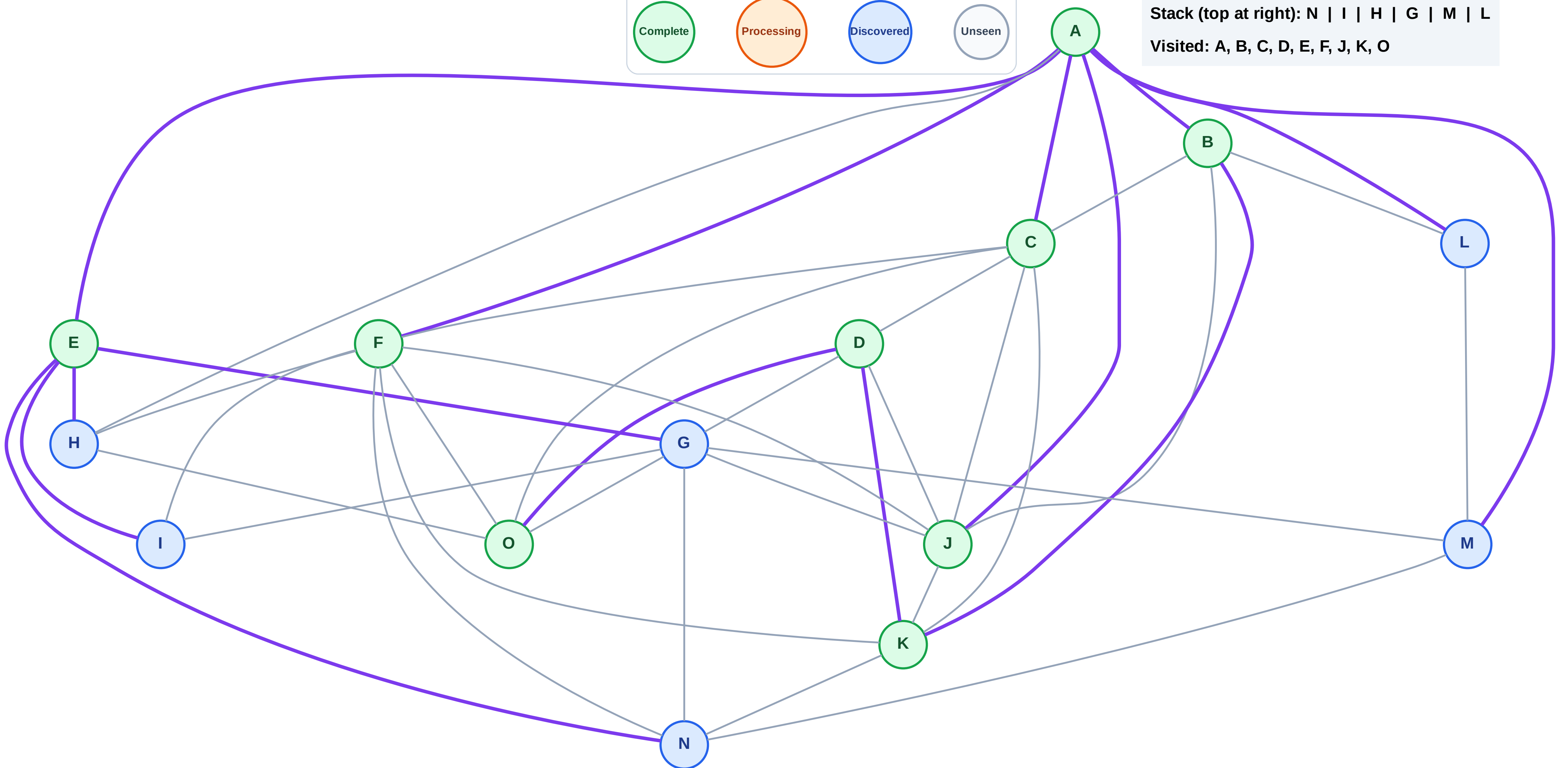
DFS Traversal

Step 19: No new neighbors from J

Legend



Stack (top at right): N | I | H | G | M | L
Visited: A, B, C, D, E, F, J, K, O



DFS Traversal

Step 20: Process node L

Legend

Complete

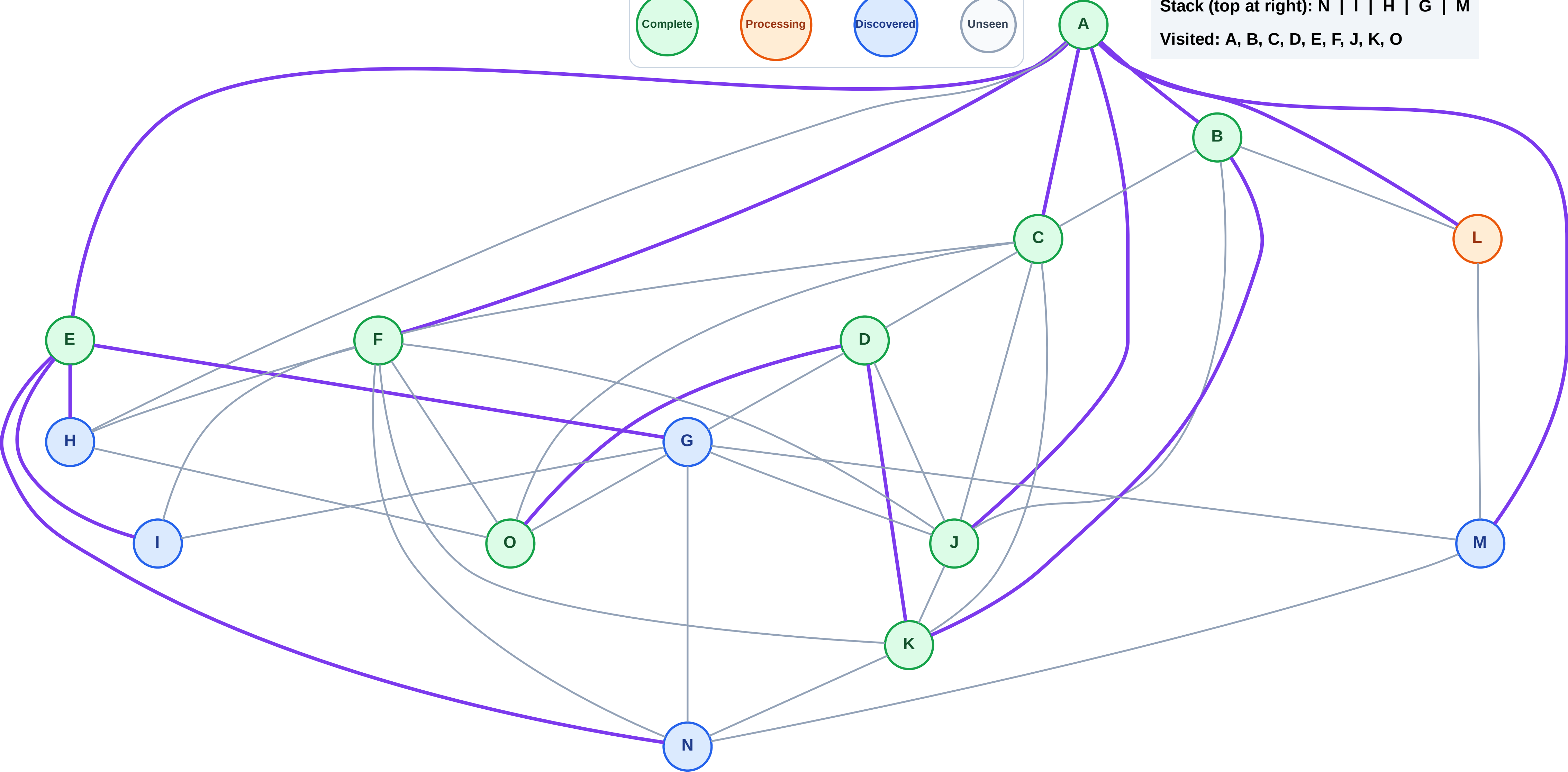
Processing

Discovered

Unseen

Stack (top at right): N | I | H | G | M

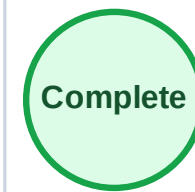
Visited: A, B, C, D, E, F, J, K, O



DFS Traversal

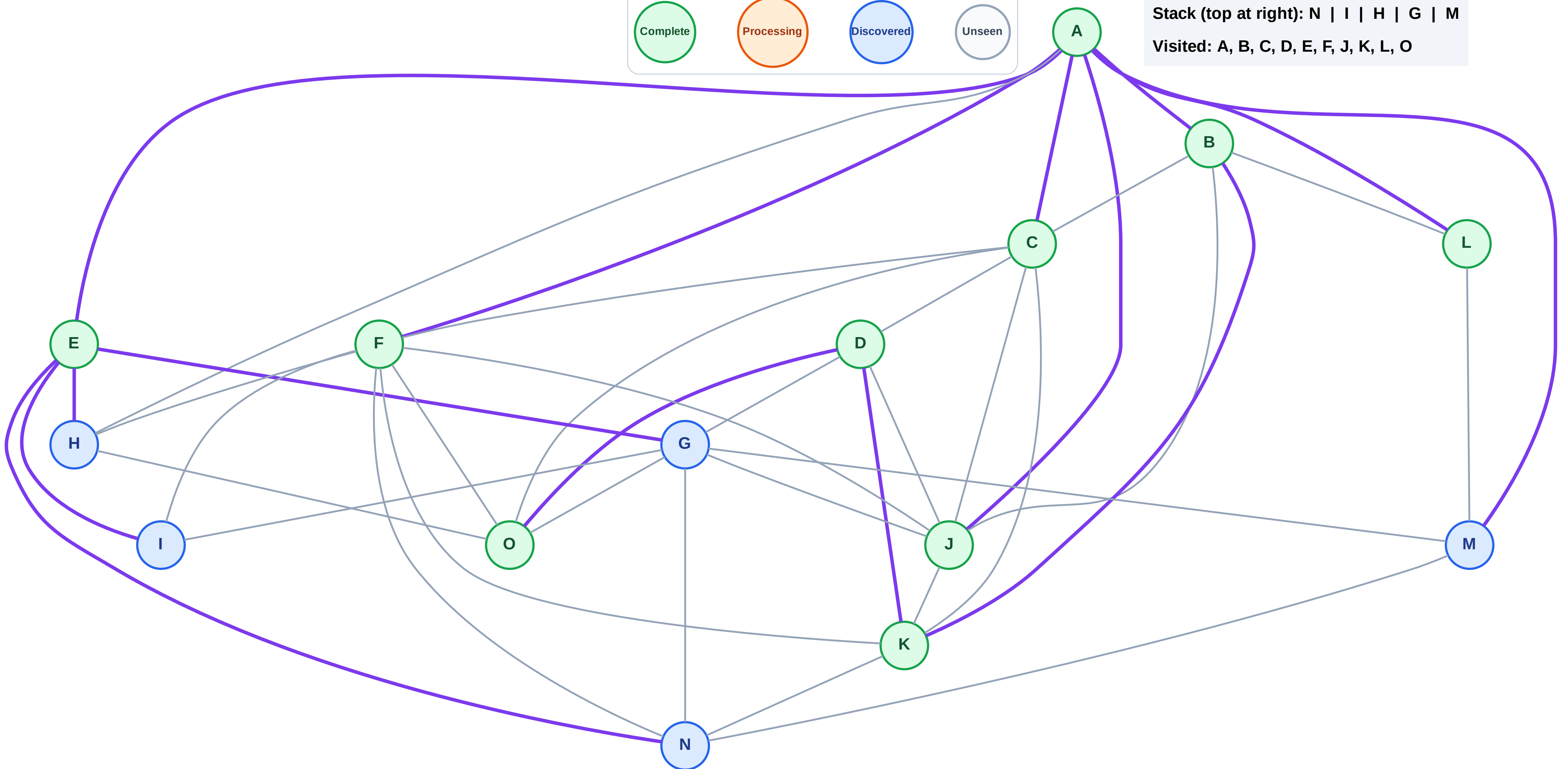
Step 21: No new neighbors from L

Legend



Stack (top at right): N | I | H | G | M

Visited: A, B, C, D, E, F, J, K, L, O



DFS Traversal

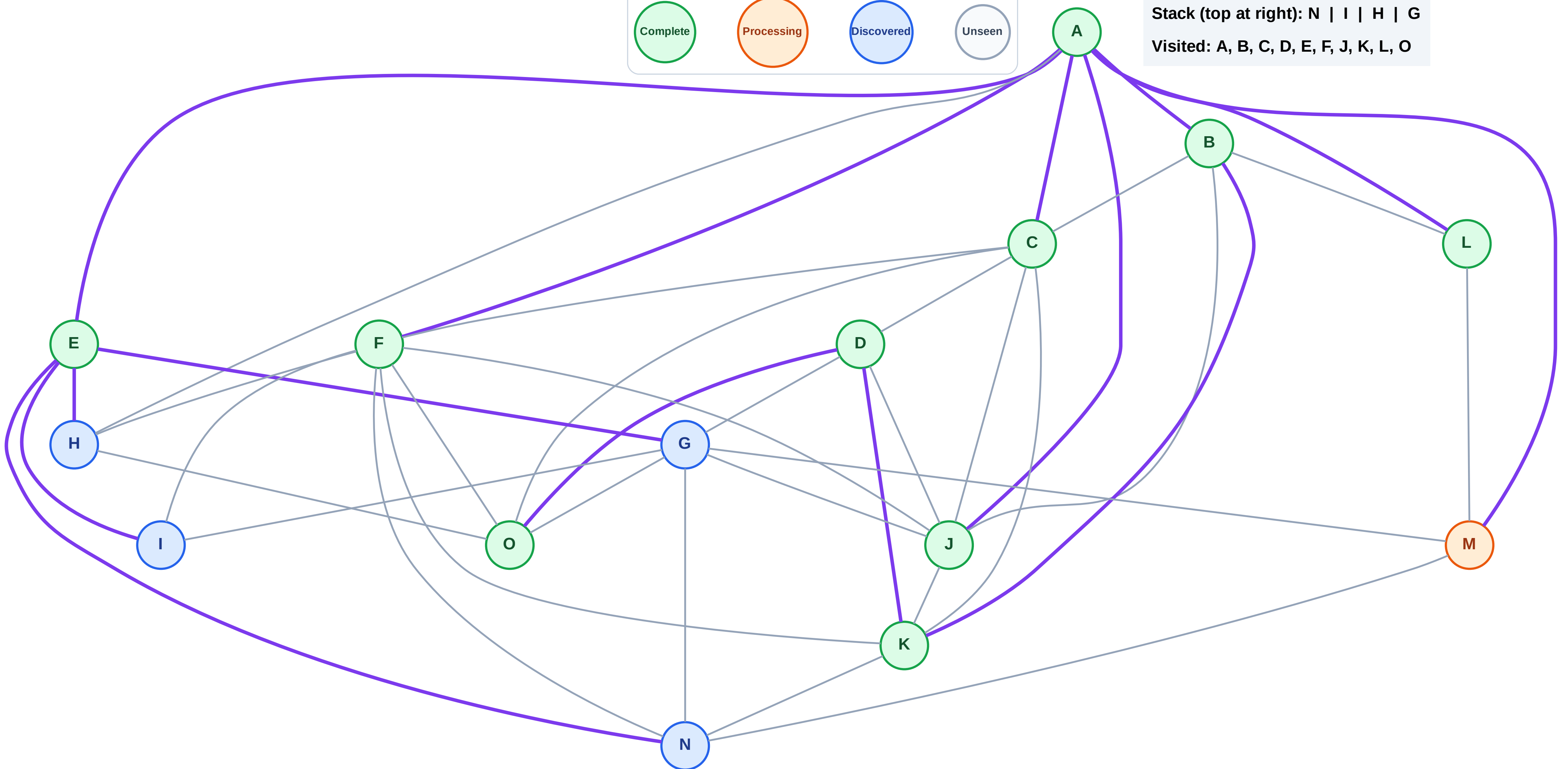
Step 22: Process node M

Legend



Stack (top at right): N | I | H | G

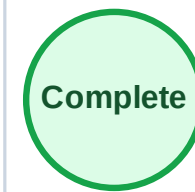
Visited: A, B, C, D, E, F, J, K, L, O



DFS Traversal

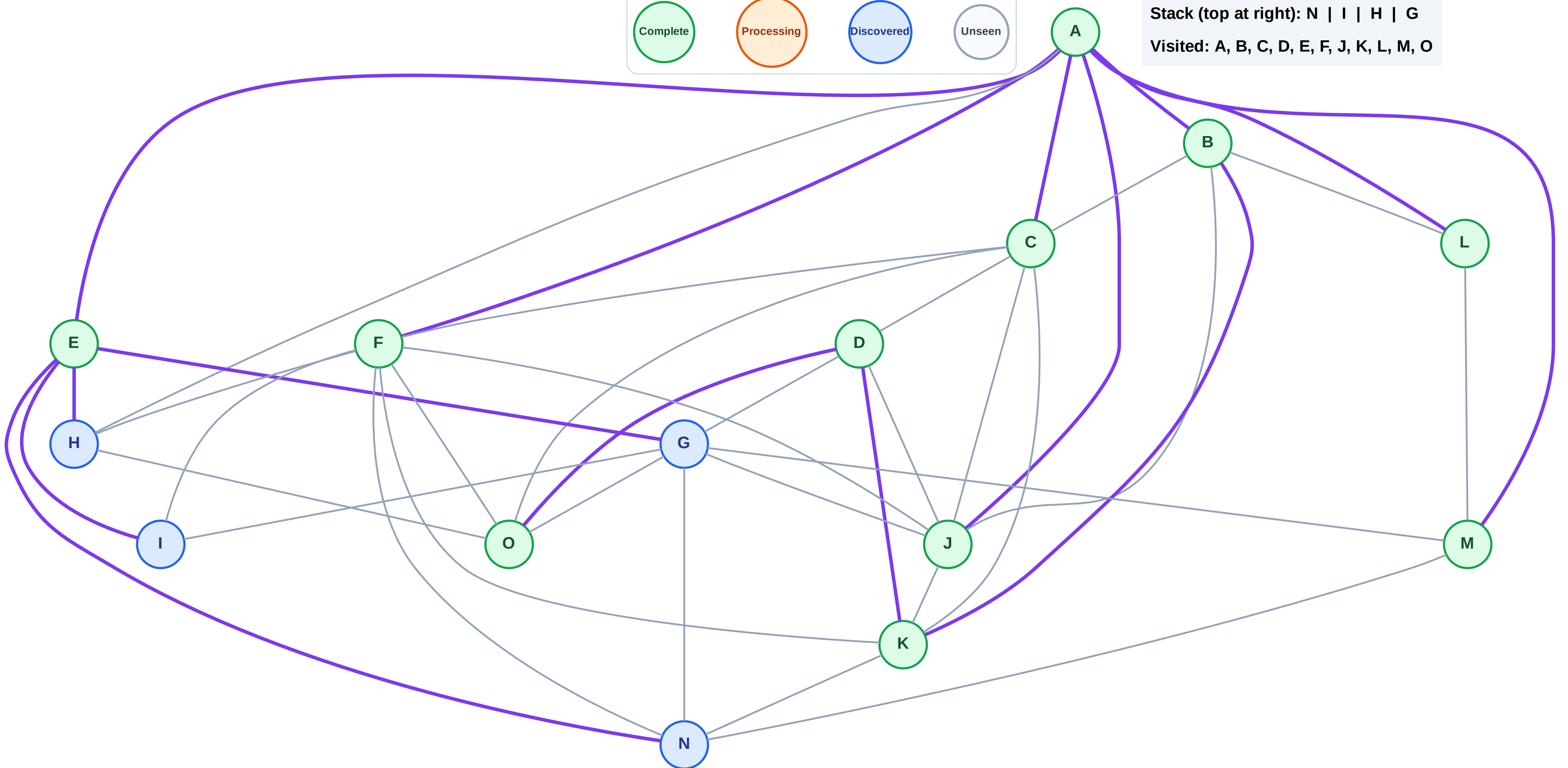
Step 23: No new neighbors from M

Legend



Stack (top at right): N | I | H | G

Visited: A, B, C, D, E, F, J, K, L, M, O



DFS Traversal

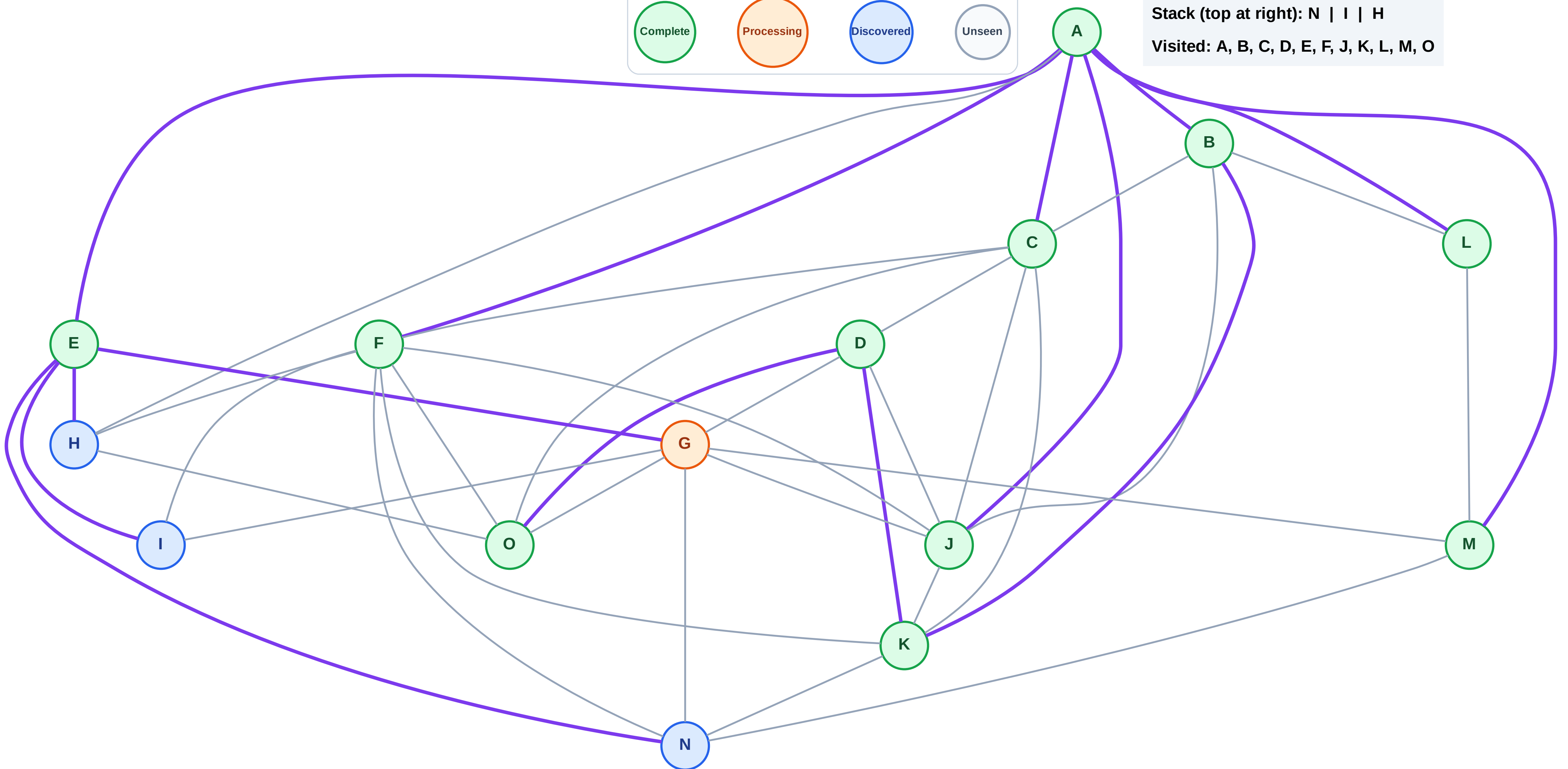
Step 24: Process node G

Legend



Stack (top at right): N | I | H

Visited: A, B, C, D, E, F, J, K, L, M, O



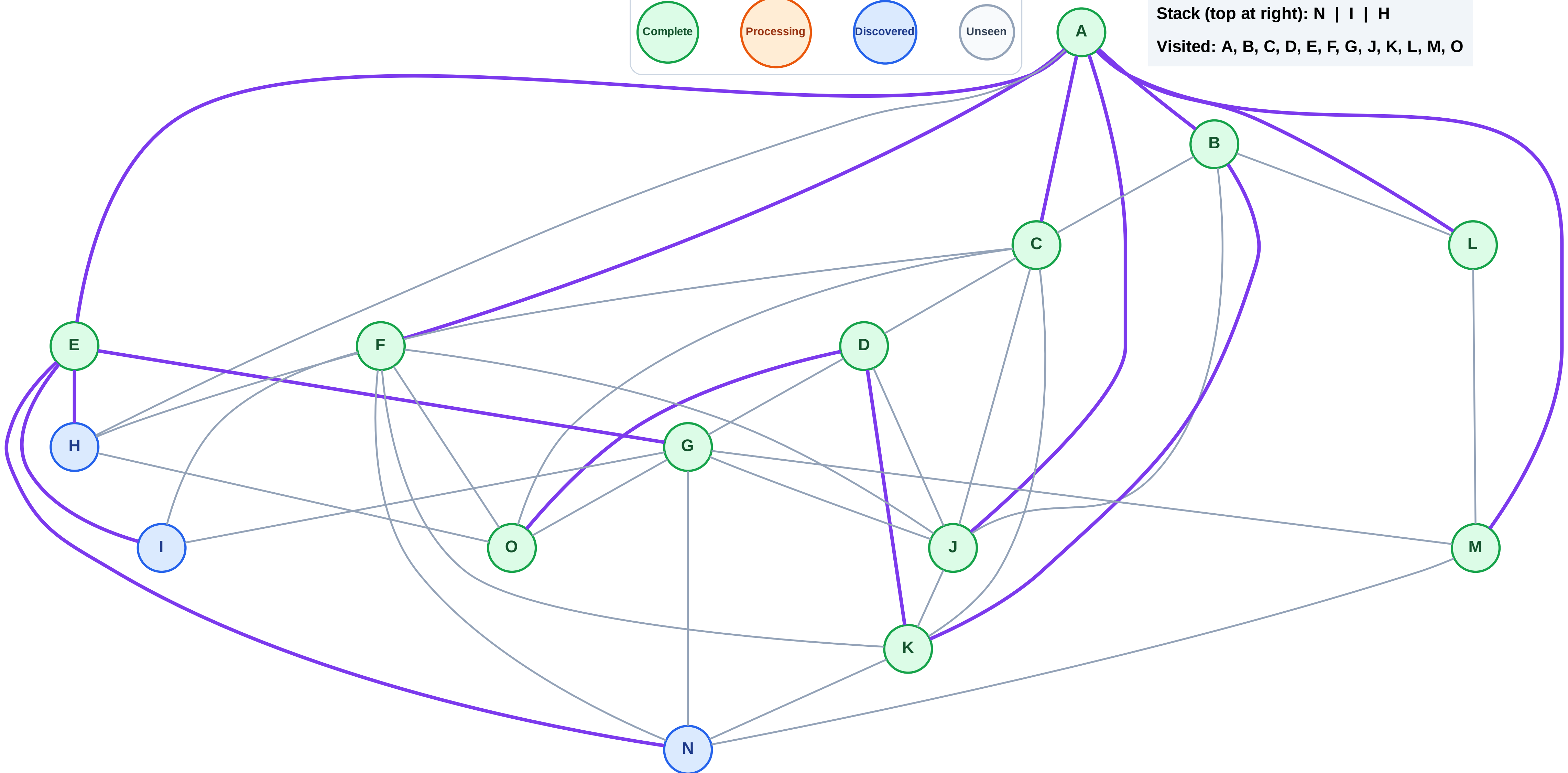
DFS Traversal

Step 25: No new neighbors from G

Legend



Stack (top at right): N | I | H
Visited: A, B, C, D, E, F, G, J, K, L, M, O



DFS Traversal

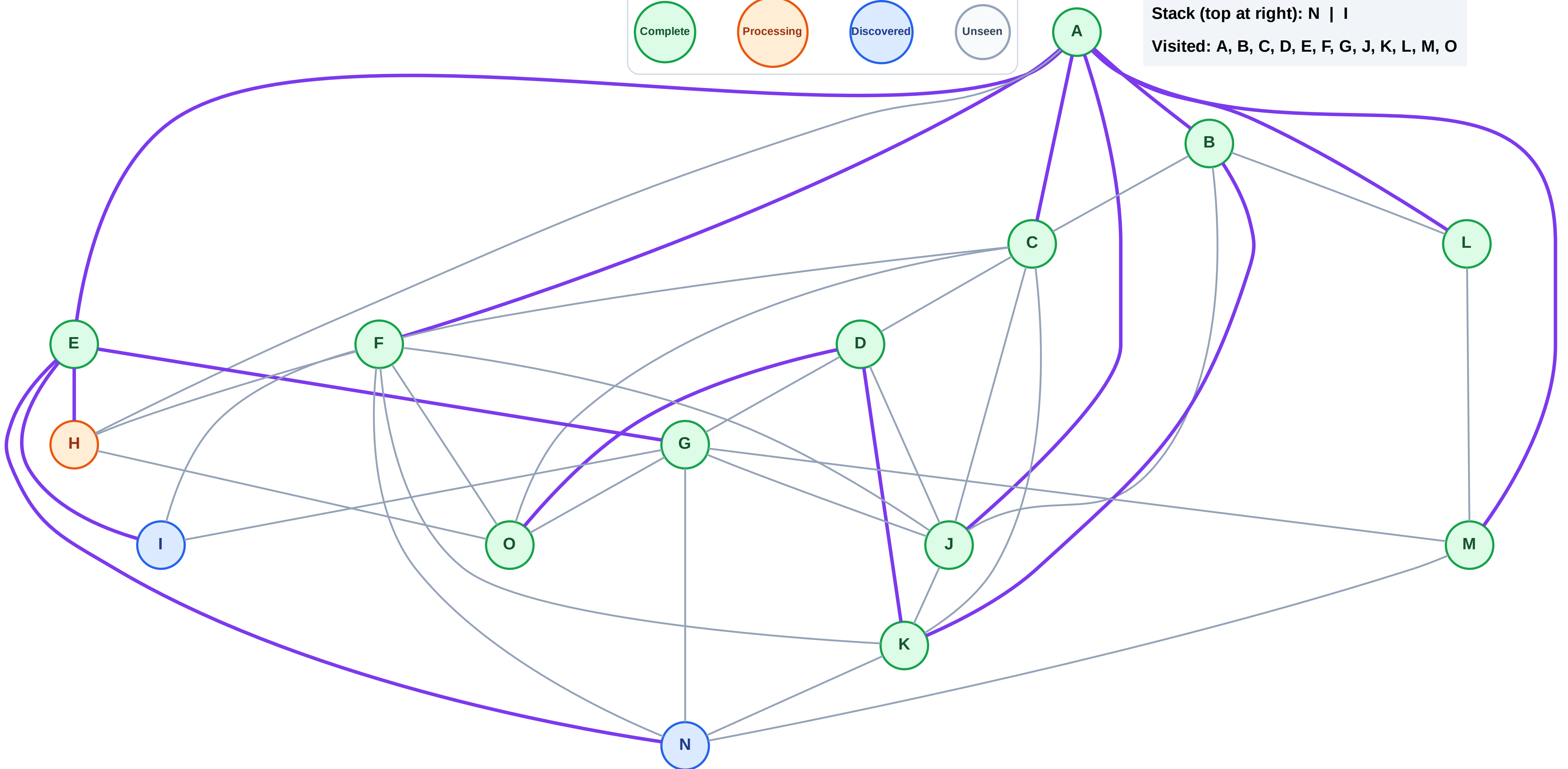
Step 26: Process node H

Legend



Stack (top at right): N | I

Visited: A, B, C, D, E, F, G, J, K, L, M, O



DFS Traversal

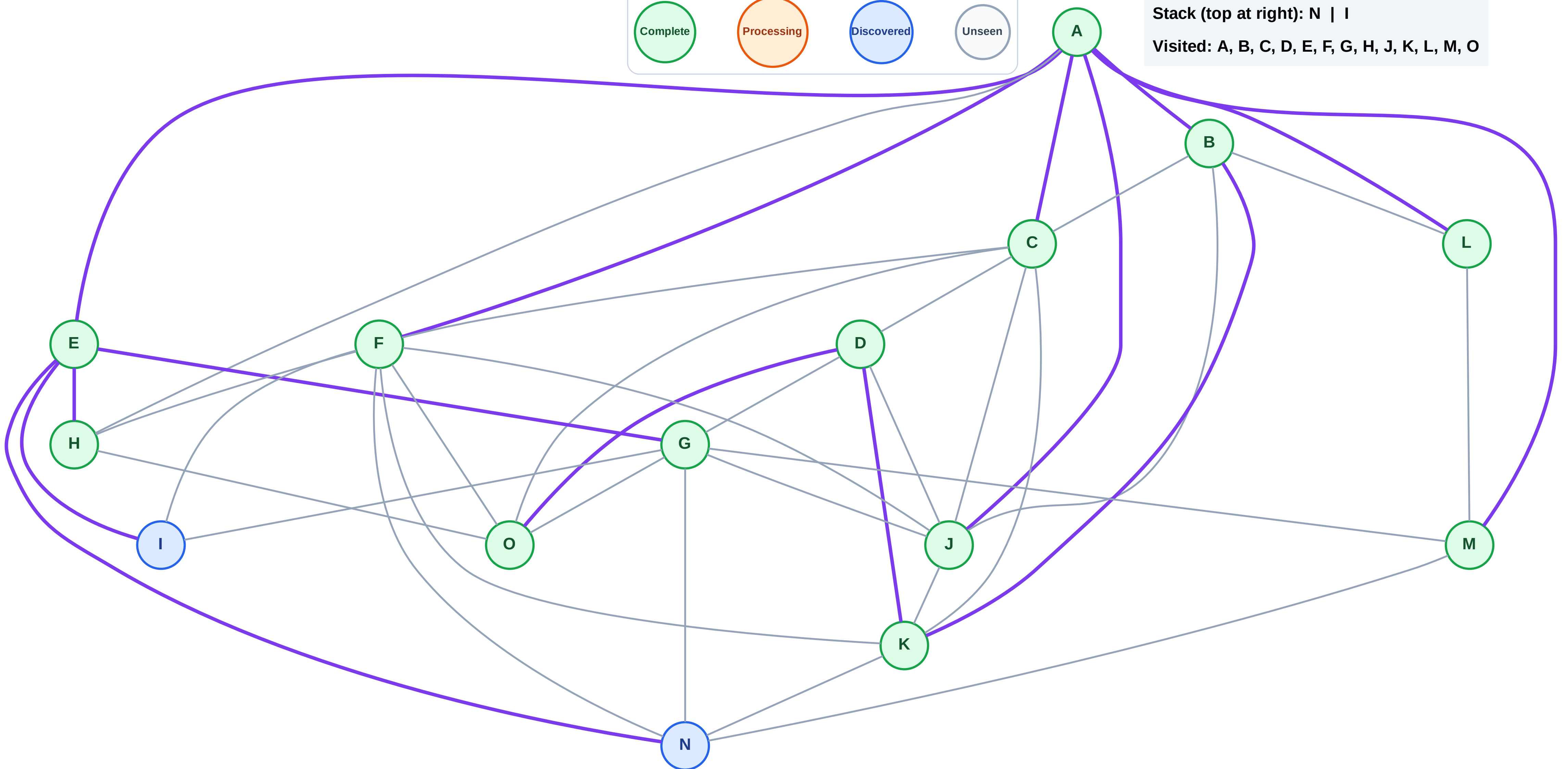
Step 27: No new neighbors from H

Legend



Stack (top at right): N | I

Visited: A, B, C, D, E, F, G, H, J, K, L, M, O



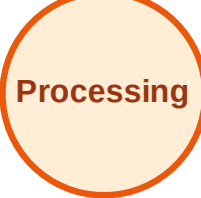
DFS Traversal

Step 28: Process node I

Legend



Complete



Processing

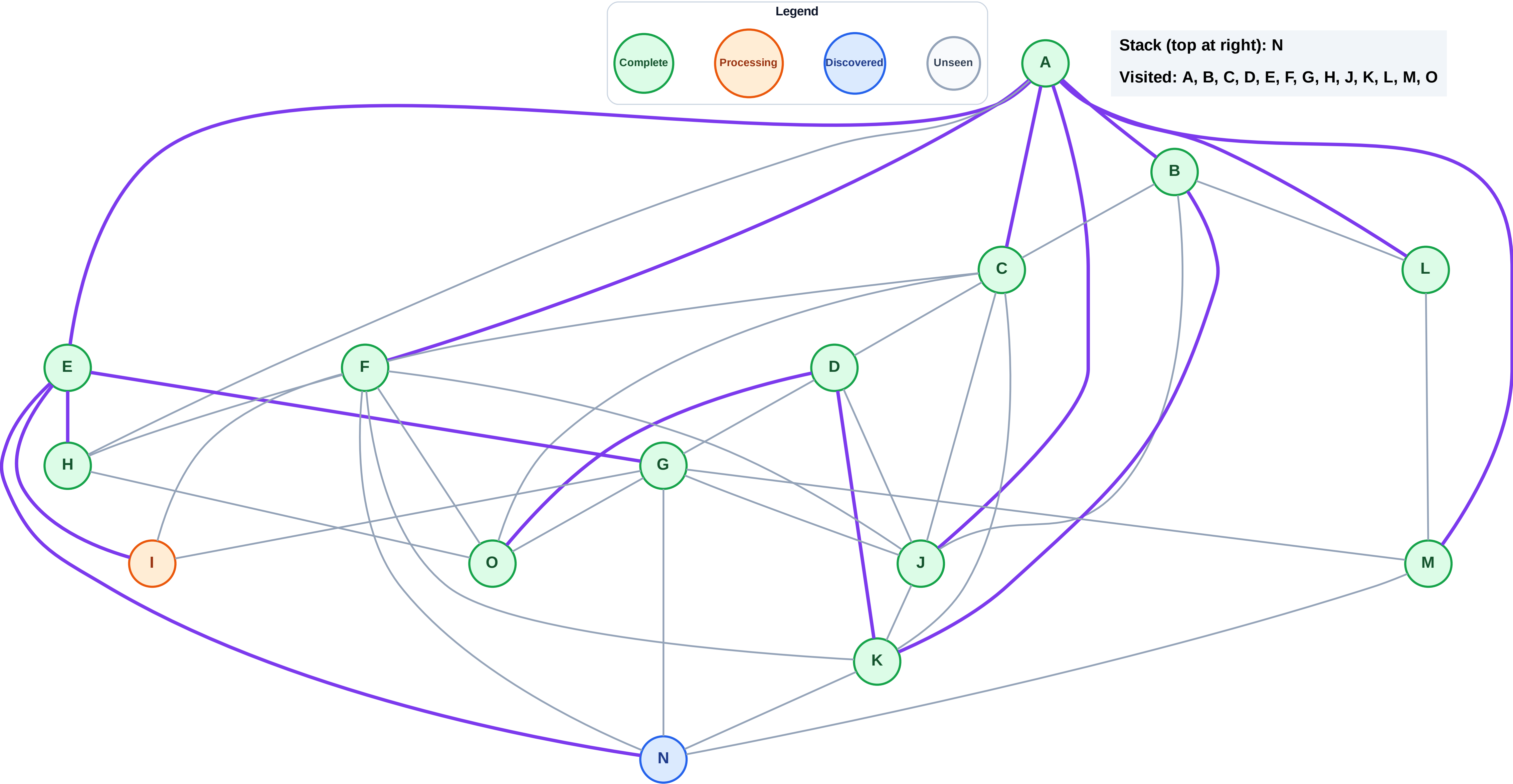


Discovered



Unseen

Stack (top at right): N
Visited: A, B, C, D, E, F, G, H, J, K, L, M, O



DFS Traversal

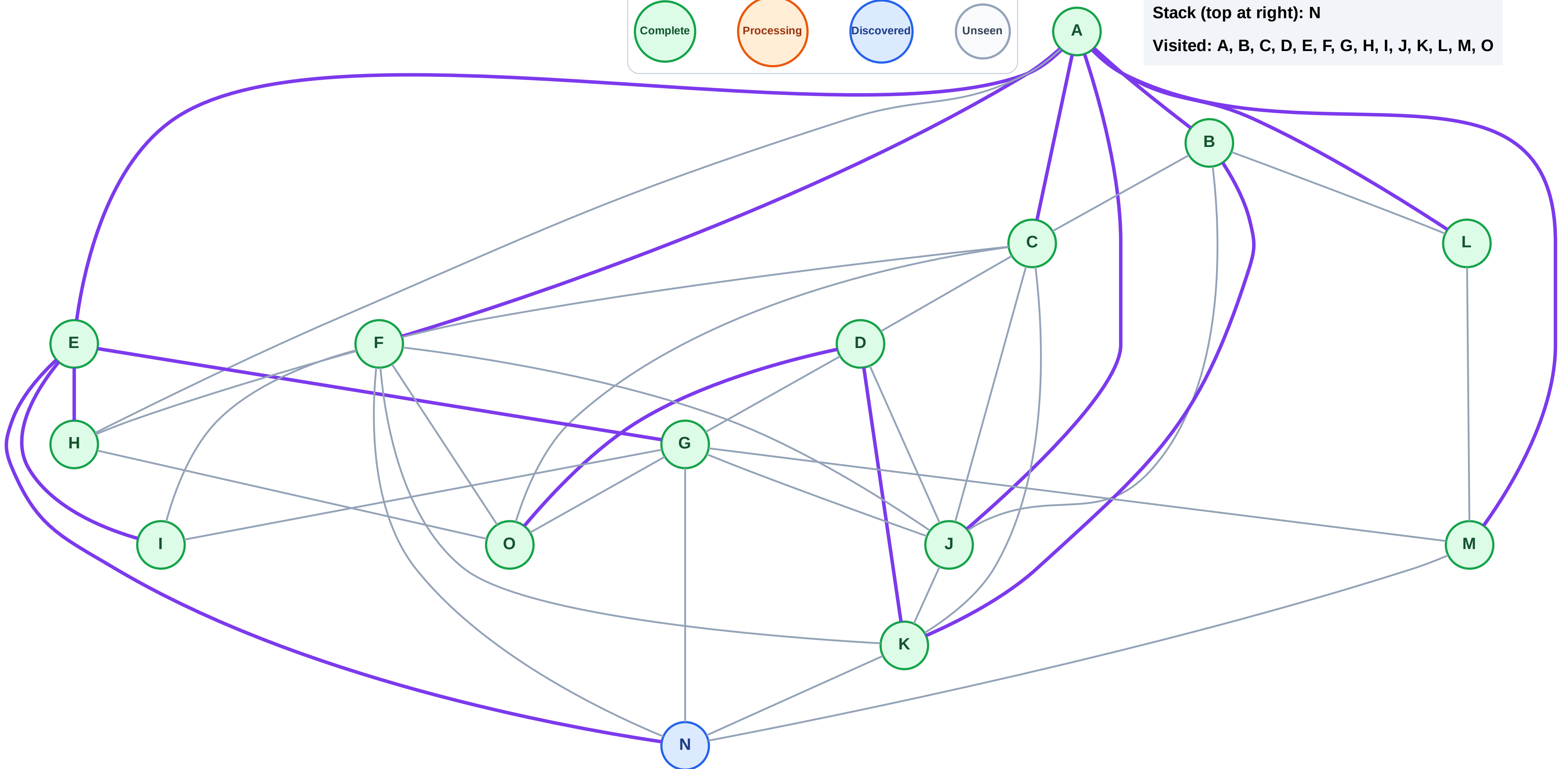
Step 29: No new neighbors from I

Legend



Stack (top at right): N

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, O



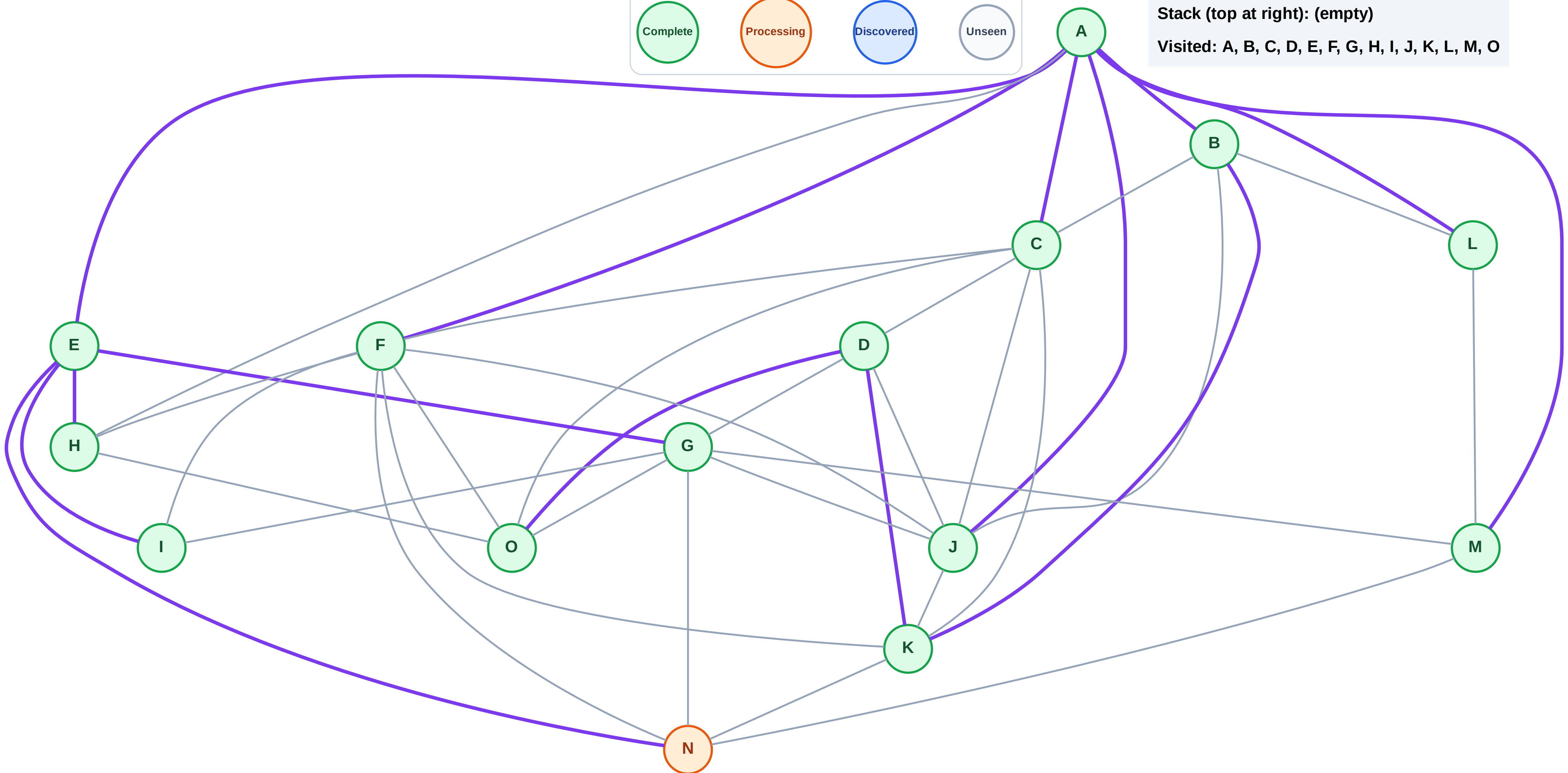
DFS Traversal

Step 30: Process node N

Legend



Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, O



DFS Traversal

Step 31: No new neighbors from N

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O

